The Energy Cost of a Phase Shift

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We prove that a passive causal system without zeros in the upper half-plane cannot shift every frequency of a band $[a, b]$ by the same phase $\theta > 0$ without loss: its mean attenuation over the band is at least $\theta\,(K(\kappa) - E(\kappa))/\kappa$ nepers, $\kappa = (b-a)/(b+a)$, and an explicit system attains this with the least attenuation at every frequency of the band at once. With zeros allowed, one explicit system with a single zero is optimal in the same pointwise sense. For a right angle it is algebraic, its mean attenuation is $\frac{\pi}{2}(\sqrt{b} - \sqrt{a})/(\sqrt{b} + \sqrt{a})$ nepers, and no passive system with this phase passes more than the fraction a/b of the power at $\omega = \sqrt{ab}$.

1. INTRODUCTION

Does it cost energy to shift the phase of a wave? At a single frequency the answer is no. A delay line of length τ multiplies a wave of angular frequency ω by $e^{i\omega\tau}$, and by choosing τ one obtains any phase at all, without loss.[1] The question becomes interesting only when the same phase shift is asked of every frequency in a band. A delay cannot do that, because its phase grows linearly with frequency. A system that does it moves the oscillations of a pulse under a fixed envelope, which is what one means by a phase shift of a signal rather than of a single sinusoid (Figure 2).

This paper answers the question for that case. It was raised in an earlier paper by one of us with S. N. Thakur [1, 2], where a 1° phase shift of a 5 MHz wave was assigned a delay of 555 ps and an "infinitesimal energy loss" $\Delta E = hf\Delta t$. The delay is right; the energy formula is not (Section 2). It has the units of an action, its numerical value in [1] is off by nine orders of magnitude, and a fixed phase shift changes neither the frequency nor the energy of a wave. What survives is the intuition behind it, that a phase shift has an energy price, and that intuition turns out to be correct in a precise and sharp form once the shift is required on a band.

1.1. The result

We work with causal, time-invariant, passive systems, described by a transfer function H that is analytic and bounded by 1 in the upper half-plane $\mathbb{C}_+$, with real impulse response. With the convention that a pure delay is $e^{i\omega\tau}$, the phase $\varphi(\omega)$ of H is a lag when it is positive, and $\alpha(\omega) = -\log|H(\omega)| \geq 0$ is the attenuation in nepers.[2] We call H *zero-free* when it has no zeros in $\mathbb{C}_+$; this class contains the minimum-phase systems, every system e^{ih} with h a passive response function, and the lossless resonances of Foster [11].[3] Our main theorem (Theorem 6.2) reads as follows.

> *Let H be zero-free and passive, continuous and nonzero near the band $[a, b]$, $0 < a < b$, and suppose that its phase on the band is $\varphi(\omega) = \theta + \tau\omega$ with $\theta \geq 0$ and any $\tau \in \mathbb{R}$. Then*
>
> $$\frac{1}{b-a}\int_a^b \alpha(\omega)\,\mathrm{d}\omega \;\geq\; \theta\,\Lambda, \qquad \Lambda = \frac{K(\kappa) - E(\kappa)}{\kappa}, \tag{1.1}$$
>
> *where $\kappa = (b-a)/(b+a)$ and K, E are the complete elliptic integrals. Equality holds if and only if the system has no response outside the band, in a sense made precise by its Herglotz measure, and it is attained by an explicit system $H_\star$.*

The constant is elementary to estimate. Its power series has positive coefficients and

$$\begin{gathered}\frac{\pi}{4}\,\kappa \;<\; \Lambda \;\leq\; \frac{\pi}{4}\,\frac{\kappa}{1-\kappa^2}, \\ \Lambda \sim \tfrac{1}{2}\log\frac{4b}{a} - 1 \quad (a/b \to 0),\end{gathered} \tag{1.2}$$

so for a narrow band of relative width $\Delta f/f$ a phase shift of one degree costs at least $1 - e^{-2\theta\Lambda}$, about

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[1] The factor $e^{i\omega\tau}$ is periodic in τ with period $2\pi/\omega$, so a delay of at most one period gives every phase; at 5 MHz one degree takes 555.6 ps (Section 2).

[2] An amplitude attenuation of one neper is $20/\ln 10 \approx 8.686$ dB, and the power transmission is $|H|^2 = e^{-2\alpha}$.

[3] Foster's reactance theorem: the impedance of a lossless one-port, as a function of $z = \omega$, is an odd Herglotz function, so its reactance increases with frequency between consecutive poles. The corresponding transfer factors e^{ih} have $|H| = 1$ on the real axis off the poles of h.

$(\pi^2/720)\,\Delta f/f \approx 0.0137\,\Delta f/f$, of the power at the worst frequency,[4] and the same fraction in the geometric mean over the band (Corollary 6.7 and Section 12.3). For an octave the bound is 0.0415 dB per degree, for a decade 0.143 dB per degree[5] (Table II).

Five features of the theorem deserve to be named at once.

Delay is free. The term $\tau\omega$ does not enter the bound.[6] A system may delay the envelope by any amount; what it pays for is the carrier–envelope offset θ (Proposition 3.3).

The bound is sharp. The extremal system $H_\star$ of Theorem 8.2 is given by an abelian integral,

$$H_\star(z) = \exp\Big(i\tau_\star \int_0^z \frac{s^2-m^2}{R(s)}\,\mathrm{d}s\Big),$$
$$R(s)^2 = (s^2-a^2)(s^2-b^2),$$

with $m^2 = b^2E(k)/K(k)$ and $\tau_\star = 2\theta K(k)/(\pi b)$, $k^2 = 1-a^2/b^2$. It is lossless outside the band, its attenuation vanishes at both band edges and peaks at $\omega = m$, and its phase is exactly θ across the band (Figure 8).

Lossless resonances make things worse. A reactive element resonating outside the band, the classical device for shaping phase, contributes a strictly positive term to the loss (Remark 6.4). A lossless system cannot have a constant phase on any band unless it is ± 1 (Corollary 4.6).

Phase leads obey the mirror bound. If $\theta < 0$ the same inequality holds for the mean of α weighted by ab/ω^2 (Theorem 6.6), so the worst frequency of a system whose phase is constant modulo 2π loses at least $\mathrm{dist}(\theta, \pi\mathbb{Z})\,\Lambda$ nepers (Corollary 6.7).

The extremal system wins at every frequency. Every zero-free system is the extremal function plus terms, one for each unit of measure off the band, that each raise the attenuation at every point of the band (Theorem 9.2). So for a phase $\theta+\tau\omega$ with $\theta \ge 0$, and for a lead $-s+\tau\omega$ with $\tau \le 0$, one explicit system has the least attenuation at every frequency at once (Theorems 9.4 and 9.6): for a lag it is $e^{i\tau z}H_\star$ as long as $\tau \ge -\tau_\star$, for a lead it is $H_\star(-ab/z)$. The sharp worst-frequency loss is $\alpha_\star(m)$, which for an octave is $1.273\,\theta\Lambda$, and the sharp plain mean for a lead is $|\theta|\,(E(\kappa)-\kappa'^2K(\kappa))/\kappa$ (Corollary 9.7).

Zeros help, by an amount we determine exactly. For $0 < \theta < a\pi/(a+b)$ a single zero on the imaginary axis already lowers the attainable mean attenuation strictly below the least value among all zero-free systems with the same argument (Proposition 10.2 and Corollary 10.3). The least loss over all passive systems, with any zeros, is the subject of our second theorem.

[4] For a narrow band $\kappa \approx \frac{1}{2}\Delta f/f$, so $\Lambda \approx \frac{\pi}{4}\kappa \approx \frac{\pi}{8}\frac{\Delta f}{f}$; one degree is $\pi/180$ radian, and $1-e^{-2\theta\Lambda} \approx 2\theta\Lambda = 2\cdot\frac{\pi}{180}\cdot\frac{\pi}{8}\frac{\Delta f}{f} = \frac{\pi^2}{720}\frac{\Delta f}{f}$ (Proposition 12.5(iv)).

[5] These are $\Lambda\cdot\frac{\pi}{180}\cdot\frac{20}{\ln 10}$ with $\Lambda = 0.27353$ for $b/a = 2$ and $\Lambda = 0.94478$ for $b/a = 10$.

[6] The reason is the vanishing first moment $\int_a^b \omega\,p(\omega)\,\mathrm{d}\omega = 0$ of the weight p in (1.3), Lemma 5.3(i).

1.2. Systems with zeros

Once zeros are allowed, $\log H$ is no longer defined and the sum rule does not apply. We pass instead to $\zeta = \log F$, where $F = e^{-i\theta}(H-e^{i\theta})/(H-e^{-i\theta})$ maps the values of H into the upper half-plane. Zeros of H become ordinary values of ζ, the phase condition becomes the condition that ζ lies on a fixed curve over the band, and a comparison argument in Hardy spaces gives a bound at every frequency (Theorem 11.1).

> *Let $0 < \theta < \pi$. There is a passive system $H_\theta^\sharp$, with one zero, on the imaginary axis, and lossless off the band, such that every passive system H that is continuous and nonzero near the band and has $\arg H \equiv \theta$ modulo 2π there satisfies*
>
> $$\alpha(\omega) \;\ge\; \alpha_\theta^\sharp(\omega) \qquad (a \le \omega \le b),$$
>
> *where $\alpha_\theta^\sharp$ is the attenuation of $H_\theta^\sharp$, with equality on a set of positive measure only for $H = H_\theta^\sharp$.*

The optimal system is $-B_{\eta_\star}e^{ih}$, a single Blaschke factor times a zero-free system, where the zero $i\eta_\star$ is fixed by the condition that the zero-free factor has no delay at high frequency. The minimal mean is $\theta\Lambda$ minus an explicit integral, it is positive, and this answers Problem 10.4. For a right angle the optimal system is algebraic (Theorem 11.2),

$$H_{\pi/2}^\sharp(z) = -\frac{z^2+ab}{R(z)+i(a+b)z},$$

its mean attenuation is $\frac{\pi}{2}(\sqrt{b}-\sqrt{a})/(\sqrt{b}+\sqrt{a})$, and its attenuation peaks at the geometric centre with the value $\frac{1}{2}\log(b/a)$. So no passive system whose phase is a right angle at every frequency of an octave transmits more than half of the power at $\omega = \sqrt{ab}$: the loss there is at least 3.01 dB, and for a decade at least 10 dB. For a narrow band the least mean loss with zeros is $\frac{\pi}{8}\sin\theta\,\Delta f/f$, against $\frac{\pi}{8}\theta\,\Delta f/f$ without (Corollary 11.9); small phase shifts gain almost nothing from zeros.

1.3. How the theorem is proved

The proof rests on a single identity. If $H = \pm e^{ih}$ with h a Herglotz function, then for every band

$$\int_a^b \alpha\,\mathrm{d}\omega = \frac{1}{2}\int_a^b \varphi(\omega)\,p(\omega)\,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W\,\mathrm{d}\mu, \qquad (1.3)$$

where $S = [-b,-a]\cup[a,b]$, μ is the Herglotz measure of h, and

$$p(\omega) = \frac{a^2+b^2-2\omega^2}{\sqrt{(\omega^2-a^2)(b^2-\omega^2)}},$$
$$W = \frac{(\rho-1)^2}{\rho}, \qquad \rho(z) = \sqrt{\frac{z^2-a^2}{z^2-b^2}}.$$

The function W is analytic in $\mathbb{C}_+$, its real part equals -2 on the band and is strictly positive on the rest of the real line, and its imaginary part is p on the band (Figure 3). The identity (1.3) then follows from the Kramers–Kronig relations [6–8] in their Herglotz form. Since $\int_a^b \omega p \, \mathrm{d}\omega = 0$, a delay drops out; since $\int_a^b p \, \mathrm{d}\omega > 0$ and the last term is nonnegative, a positive constant phase forces loss. The value of $\int_a^b p \, \mathrm{d}\omega$ is an elliptic integral, which a Landen transformation brings to the form (1.1).

Sum rules of this kind have a long history. Bode's gain–phase relations [10] bound the phase of a minimum-phase system by its gain; Fano [12] bounded the bandwidth of impedance matching; Rozanov [14] bounded the thickness of absorbers; and Gustafsson, Sjöberg, Bernland and Luger [15, 16] turned the Herglotz representation into a general machine for physical bounds. Miller [17] bounded the delay–bandwidth product of slow light. The inequality (1.1) is, to our knowledge, new: we did not find a bound on the loss forced by a frequency-independent phase shift in this literature, nor in the work on broadband phase shifters, where constant phase *differences* between two lossless channels are realised [13] and constant absolute phase is not attempted. The test function W, which is tailored to an affine phase, is what makes the bound sharp.

1.4. What the result is, and what it is not

The theorem is a sharp inequality of classical analysis with a direct physical reading: a carrier–envelope phase shift over a band has a minimal price in transmitted energy, proportional to the shift and, for narrow bands, to the relative bandwidth. It is not a statement about energy conservation: a constant phase factor is energy neutral (Proposition 3.1), and the price is paid because causality and passivity tie the phase of H on the band to its modulus everywhere. Nor does it rest on quantum mechanics; when the wave is quantised, the bound becomes a lower bound on the probability that a quantum in the band is not transmitted (Section 12.3). The energy hf of a quantum, which played the main part in [1], enters only through the kinematics of frequency shifts in Section 3; see also [3–5] for other uses of the photon energy by the first author and coworkers.

1.5. Plan

Section 2 checks the equations of [1]. Section 3 separates the three things a "phase shift" can mean: a constant factor, a delay, and a phase that changes in time. Section 4 sets up passive systems and their Herglotz functions, and Section 5 constructs the test function. The sum rule and the main theorem are in Section 6, the evaluation of the constant in Section 7, the extremal systems in Section 8, the pointwise bounds without zeros in Section 9, and the effect of a single zero in Section 10; Section 11 finds the least loss over all passive systems. Section 12 applies the results to waves in passive media and returns to the 5 MHz example. Section 13 lists open problems. Figure 1 shows how the results depend on each other. Appendix A derives, from the series alone, every fact on complete elliptic integrals that is used (Legendre's relation, the invariance of the arithmetic–geometric mean and Landen's transformations); Appendix B computes the integrals of the weight; Appendix C derives the formulas for the extremal profile; and Appendix D describes what has been checked by machine. Every statement is proved in the text. The finite and algebraic parts of the proofs (dimensional analysis, the coefficient recurrences, enclosures of Λ and the polynomial identities) are certified in Lean 4, and a script re-evaluates every numerical value.

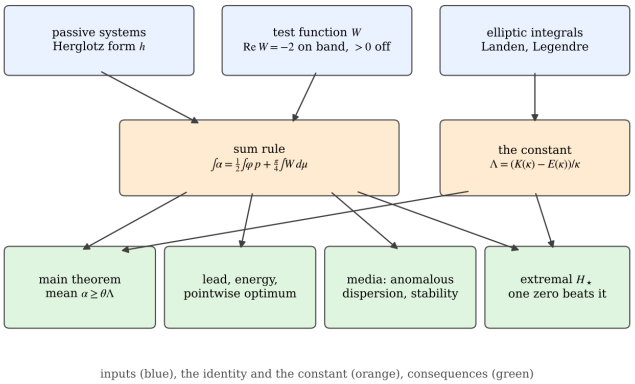


FIG. 1. How the results depend on each other. The sum rule (Theorem 6.1) is built from the test function W (Section 5); the constant Λ comes from the moments of the weight (Section 7); every later statement is a consequence of these two or of the extremal system of Theorem 8.2.

2. THE EQUATIONS OF THE EARLIER PAPER

We first go through the formulas of [1], since the rest of the paper replaces them. They fall into two groups: kinematic relations between phase, time and frequency, which are correct, and formulas for an energy loss ΔE, which are not.

2.1. Phase, time and frequency

For a sinusoid of frequency f, a delay Δt produces the phase

$$\Phi^\circ = 360^\circ f \, \Delta t, \tag{2.1}$$

and [1] rewrites this as $\Delta t = \Phi^\circ/(360^\circ f)$ and $f = \Phi^\circ/(360^\circ \Delta t)$. These are correct, with the understanding

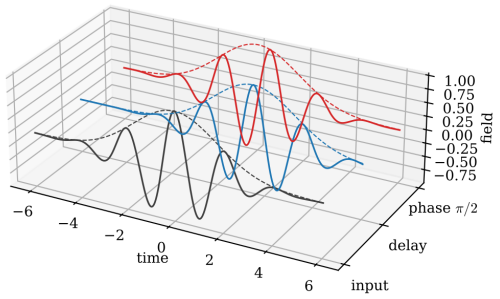


FIG. 2. A pulse (grey), the same pulse after a delay (blue), and after a constant phase shift of $\pi/2$ on its whole band (red). Dashed curves are the envelopes. The delay moves the envelope and the oscillations together; the constant phase shift moves the oscillations under a fixed envelope. Only the second operation costs energy in a passive system.

that the last one gives the frequency at which a *given* delay produces a *given* phase; it does not describe a change of frequency. The delay that produces one degree is

$$T(\text{deg}) = \frac{1}{360 f},$$

which at $f = 5\,\text{MHz}$ is 555.6 ps, as stated in [1]. Two of the phrasings there should be read with care. The equation "1° phase shift = $T/360$" equates an angle with a time; what is meant is that a delay of $T/360$ produces a phase of 1°, where $T = 1/f$ is the period. The name "time distortion" for Δt suggests something more than a delay; the quantity in (2.1) is just the delay. Finally, the statement that $T(\text{deg}) \propto 1/f$ is true of the delay needed for one degree, and it is the reason why a delay cannot produce the same phase at two frequencies.

2.2. The energy formulas

The paper [1] proposes three formulas for an "infinitesimal loss of wave energy",

$$\Delta E = h f \Delta t, \qquad \Delta E = \frac{2\pi h f_1}{360}\, T(\text{deg}), \qquad \Delta E = \frac{2\pi h}{360}\,\frac{T(\text{deg})}{\Delta t}, \tag{2.2}$$

with h Planck's constant, and evaluates the first one for $f = 4{,}986{,}111.11\,\text{Hz}$ and $\Delta t = 555\,\text{ps}$. Four things go wrong.

Units. Planck's constant is an action, measured in J s; hf is an energy and $hf\Delta t$ is again an action. The same holds for the other two formulas in (2.2), in which $T(\text{deg})/\Delta t$ is a pure number and $f_1 T(\text{deg})$ is the number $1/360$. None of the three expressions is an energy.[7]

Arithmetic. The product evaluated in [1] is

$$(6.626\times10^{-34})(4.986\times10^{6})(555\times10^{-12}) = 1.834\times10^{-36},$$

in J s, not 1.848×10^{-27} J. The stated value is larger by a factor of about 10^9, and it would amount to 56% of the energy $E_1 = hf_1 = 3.313\times10^{-27}$ J of a quantum at 5 MHz, which can hardly be called infinitesimal.

A fixed phase shift does not change the frequency. The second example of [1] assigns to a 1° phase shift the new frequency $f_2 = f_1 - f_1/360 = 4{,}986{,}111.11\,\text{Hz}$. Multiplying a wave by a constant $e^{i\theta}$, or delaying it, leaves its frequency unchanged (Proposition 3.1). A frequency change of $f_1/360$ occurs when the phase *slips* by one degree in every period, that is, when the phase grows at the rate $\dot\theta = 2\pi f_1/360$ (Proposition 3.2). In that case the energy of a quantum changes by

$$E_1 - E_2 = \frac{hf_1}{360} = 9.20\times10^{-30}\ \text{J},$$

and this is the only energy difference of the size suggested by the example. The paper also gives $E_2 = hf_2 \approx 3.313\times10^{-27}$ J; the correct value is 3.304×10^{-27} J, and the difference from E_1 is the number just computed, not 1.848×10^{-27} J.

What hf measures. The energy hf is that of one quantum. The energy of a classical wave is fixed by its amplitude, which the formulas (2.2) do not contain. A loss caused by a passive element is a fraction of the incident energy, whether the wave is classical or quantised, and the quantity to compute is therefore the transmission $|H|^2$, or, per quantum, the probability of transmission.

2.3. What remains

Table I lists the corrections. The kinematic part of [1] stands. The energy part asks a good question in the wrong variables: the phase shift that has a price is the one applied uniformly over a band, the price is a fraction of the energy rather than a multiple of h, and its size is governed by the bandwidth. Sections 4–12 compute it.

3. PHASE, DELAY AND FREQUENCY

3.1. Conventions

A signal is a real function $s(t)$ of finite energy, with Fourier transform $\hat s(\omega) = \int_{\mathbb{R}} s(t) e^{i\omega t}\,dt$, so that $\hat s(-\omega) =$

[7] If $T(\text{deg})$ is read as the angle $1° = \pi/180$ rather than as a time, the second formula has units of energy, but it then reads $\Delta E = (\pi/180)\, hf_1\, (2\pi/360)$, a fixed multiple of hf_1 without any connection to the delay.

TABLE I. The formulas of [1] and their status.

Formula in [1]	Status
$\Phi^\circ = 360^\circ f\Delta t$ and its inversions	correct
$T(\text{deg}) = 1/(360f)$, 555 ps at 5 MHz	correct
"1° phase shift = $T/360$"	angle equated with time; a delay $T/360$ gives 1°
$\Delta E = hf\Delta t$	an action, in J s; its value is 1.83×10^{-36}, not 1.848×10^{-27}
$\Delta E = (2\pi h f_1/360)T(\text{deg})$	an action; equals $2\pi h/360^2$
$\Delta E =$ $(2\pi h/360)\,T(\text{deg})/\Delta t$	an action
$f_2 = f_1 - f_1/360$ after a 1° shift	only if the phase slips 1° per period
$E_2 \approx 3.313 \times 10^{-27}$ J	3.304×10^{-27} J; $E_1 - E_2 = 9.20 \times 10^{-30}$ J

$\widehat{\hat{s}(\omega)}$. A linear time-invariant system acts by $\widehat{s_{\text{out}}} = H\hat{s}_{\text{in}}$. With this convention a delay by τ has $H(\omega) = e^{i\omega\tau}$, so a positive phase of H is a *lag*, and causality means that H extends analytically to the upper half-plane $\mathbb{C}_+ = \{\operatorname{Im} z > 0\}$ (Section 4). The analytic signal of s is

$$\psi(t) = \frac{1}{\pi}\int_0^\infty \hat{s}(\omega)e^{-i\omega t}\,\mathrm{d}\omega, \qquad s = \operatorname{Re}\psi.$$

For a narrowband pulse we write $\psi(t) = A(t)e^{-i(\omega_0 t+\phi)}$ with a slowly varying envelope $A \geq 0$, a carrier frequency ω_0 and a carrier–envelope phase ϕ. The phase ϕ locates the crests of the carrier relative to the peak of the envelope; it is the quantity controlled in frequency combs [21, 22].[8]

Three different operations go under the name of a phase shift. We treat them in turn.

3.2. A constant phase factor

The *ideal phase shifter* by θ multiplies every positive frequency by $e^{i\theta}$ and, to keep the signal real, every negative frequency by $e^{-i\theta}$:

$$H_\theta(\omega) = e^{i\theta\operatorname{sgn}\omega}, \qquad \psi_{\text{out}} = e^{i\theta}\psi. \tag{3.1}$$

It is not causal.[9]

Proposition 3.1. *The ideal phase shifter preserves the energy $\int s^2\,\mathrm{d}t$, the power spectrum $|\hat{s}|^2$ and hence every frequency present in the signal. For a single quantised mode of frequency ω the operator $U = e^{-i\theta N}$, N the number operator, commutes with the Hamiltonian $\hbar\omega(N + \frac{1}{2})$, so the probability distribution of the energy is unchanged. If $\sin\theta \neq 0$, the ideal phase shifter is not causal.*

Proof. The first statements follow from $|H_\theta| = 1$ and Parseval's identity, and from $[N, U] = 0$. For the last, $H_\theta = \cos\theta + i\sin\theta\operatorname{sgn}\omega$, and $i\operatorname{sgn}\omega$ is the transfer function of convolution with the Hilbert kernel $\pm 1/(\pi t)$, which does not vanish for $t < 0$. □

So a phase shift, as such, neither changes the frequency nor costs energy. What it costs is causality, and the price of restoring causality in a passive system is the subject of the rest of the paper.

3.3. A delay

A delay τ gives $\psi_{\text{out}}(t) = \psi(t - \tau)$ and the phase $\omega\tau$ at frequency ω, which is the relation (2.1). It is causal and lossless, and it realises any prescribed phase at one frequency. On a band it realises a phase that is proportional to frequency, never a constant one.

3.4. A phase that changes in time

Let a device multiply the analytic signal by $e^{i\theta(t)}$.

Proposition 3.2. *If $\theta(t) = \theta_0 + \Omega t$, then $\widehat{\psi_{\text{out}}}(\omega) = e^{i\theta_0}\hat{\psi}(\omega + \Omega)$: the spectrum is moved down by Ω, and a component of frequency ω leaves with frequency $\omega - \Omega$ and energy $\hbar(\omega - \Omega)$ per quantum. If the device conserves the number of quanta, the energy of the signal changes by the fraction Ω/ω at each frequency, the difference being exchanged with the agent that drives θ.*

Proof. The Fourier transform of $e^{i\Omega t}\psi(t)$ is $\hat{\psi}(\omega + \Omega)$ in our convention. The rest is the relation $E = \hbar\omega$ for each quantum. □

The conservation of the number of quanta is what the Manley–Rowe relations [19] give for a lossless parametric modulator, and the linear phase ramp is the serrodyne frequency translator [20]. A phase that slips by one degree per period has $\Omega = \omega/360$; at 5 MHz this is a shift of 13,888.89 Hz and an energy of $hf/360 = 9.20 \times 10^{-30}$ J per quantum, the numbers of Section 2. This is the only reading under which a "1° phase shift" changes frequency. It is an active operation, not a passive one.

3.5. Constant phase on a band

The operation that keeps the envelope and moves the carrier is the following.

[8] In a mode-locked laser ϕ advances by a fixed amount from one pulse to the next; the rate of this slip is the carrier–envelope offset frequency of the comb, which is measured and locked in [21, 22].

[9] With our Fourier convention the multiplier $i\operatorname{sgn}\omega$ has the kernel $1/(\pi t)$, since $\text{p.v.}\int \operatorname{sgn}\omega\, e^{-i\omega t}\,\mathrm{d}\omega = -2i/t$. Hence $H_\theta = \cos\theta + i\sin\theta\operatorname{sgn}\omega$ has the impulse response $\cos\theta\,\delta(t) + \sin\theta/(\pi t)$, which does not vanish for $t < 0$ unless $\sin\theta = 0$.

Proposition 3.3. *Let $0 < a < b$ and let H satisfy $H(-\omega) = \overline{H(\omega)}$ and*

$$H(\omega) = e^{-\alpha(\omega)} e^{i(\theta + \tau\omega)}, \qquad a \le \omega \le b. \tag{3.2}$$

If the spectrum of s lies in $[-b,-a] \cup [a,b]$ and $\alpha = 0$ on $[a,b]$, then $\psi_{\text{out}}(t) = e^{i\theta}\psi(t-\tau)$. In particular a pulse $A(t)e^{-i(\omega_0 t+\phi)}$ leaves as $A(t-\tau)e^{-i(\omega_0(t-\tau)+\phi-\theta)}$: the envelope is delayed by τ and the carrier–envelope phase is moved by $-\theta$. Conversely, if $\psi_{\text{out}} = e^{i\theta}\psi(\cdot - \tau)$ for every such signal, then (3.2) holds with $\alpha = 0$.

Proof. On positive frequencies $\widehat{\psi_{\text{out}}} = H\hat{\psi} = e^{i\theta}e^{i\tau\omega}\hat{\psi}$, which is the transform of $e^{i\theta}\psi(t-\tau)$. The converse follows by testing with signals whose spectra are concentrated near each $\omega \in [a,b]$. □

The phase of H on the band is then $\varphi(\omega) = \theta + \tau\omega$. Its slope τ is the group delay; its intercept

$$\theta = \varphi(\omega) - \omega\varphi'(\omega) = \omega\big(\tau_{\text{ph}} - \tau_{\text{gr}}\big), \qquad \tau_{\text{ph}} = \frac{\varphi}{\omega},\ \tau_{\text{gr}} = \varphi',$$

is the carrier–envelope offset, ω times the excess of phase delay over group delay. A delay has $\theta = 0$. The main theorem says that a passive zero-free system can have $\theta > 0$ only by attenuating, on average over the band, at least $\theta\Lambda$ nepers.

4. PASSIVE SYSTEMS AND HERGLOTZ FUNCTIONS

4.1. Passive systems

Definition 4.1. A *passive system* is a function H analytic in $\mathbb{C}_+$ with $|H(z)| \le 1$ there and with the real symmetry $H(-\bar{z}) = \overline{H(z)}$. Its frequency response is the boundary function $H(\omega) = \lim_{\varepsilon\downarrow 0} H(\omega + i\varepsilon)$, which exists for almost every real ω.[10]

The definition encodes three physical requirements. A time-invariant system that never puts out more energy than it receives acts on signals of finite energy by a multiplier of norm at most one; causality makes the multiplier the boundary value of a bounded analytic function in $\mathbb{C}_+$; and a real impulse response gives the symmetry. That these requirements are equivalent to Definition 4.1 is classical [8, 9, 23]. The transmission coefficient of a passive two-port, the reflection coefficient of a passive one-port and the propagation factor of a passive medium (Section 12) are passive systems.

The zeros of a passive system that is not identically zero satisfy the Blaschke condition, and $H = BG$ where B is the Blaschke product of the zeros and G is passive and has no zeros in $\mathbb{C}_+$ [23, Ch. II]. Since the zeros of H come in pairs $\zeta, -\bar{\zeta}$, B can be chosen with the real symmetry.[11] We call H *zero-free* when B is constant. This includes every minimum-phase system, and also the singular inner functions, which are not minimum-phase in the usual sense: the factor $\exp(i\gamma/(\omega_0 - z))$, $\gamma > 0$, is the lossless resonance of a reactance at ω_0 in the sense of Foster [11].

4.2. The Herglotz function of a zero-free system

A *Herglotz function* is an analytic map h from $\mathbb{C}_+$ to the closed upper half-plane. Every such function has the representation [9, 16, 23]

$$h(z) = c + \beta z + \int_{\mathbb{R}} \Big(\frac{1}{t-z} - \frac{t}{1+t^2}\Big)\,\mathrm{d}\mu(t), \tag{4.1}$$

with $c \in \mathbb{R}$, $\beta \ge 0$ and a positive measure μ with $\int \mathrm{d}\mu/(1+t^2) < \infty$. The measure is recovered by the Stieltjes inversion formula,

$$\mu\big((x_1,x_2)\big) + \tfrac{1}{2}\mu(\{x_1,x_2\}) = \lim_{y\downarrow 0} \frac{1}{\pi}\int_{x_1}^{x_2} \operatorname{Im} h(x+iy)\,\mathrm{d}x, \tag{4.2}$$

and $\beta = \lim_{y\to\infty} h(iy)/(iy)$.

Lemma 4.2. *Let H be a zero-free passive system, $H \not\equiv 0$. There are a sign $\sigma \in \{1,-1\}$ and a unique Herglotz function h with*

$$H = \sigma e^{ih}, \qquad h(iy) \in i\mathbb{R}\ (y>0), \qquad h(-\bar{z}) = -\overline{h(z)}. \tag{4.3}$$

In the representation (4.1) of h one has $c = 0$ and μ is even, so that

$$h(z) = \beta z - \frac{\mu(\{0\})}{z} + \int_{(0,\infty)} \frac{2z}{t^2 - z^2}\,\mathrm{d}\mu(t). \tag{4.4}$$

Proof. On the imaginary axis $H(iy) = \overline{H(iy)}$ is real and, having no zeros, of constant sign σ. Since $\mathbb{C}_+$ is simply connected, σH has an analytic logarithm, which we fix to be real on $i\mathbb{R}_+$. Put $h = -i\log(\sigma H)$. Then $\operatorname{Im} h = -\log|H| \ge 0$, so h is a Herglotz function, and $h(iy) \in i\mathbb{R}$. The function $g(z) = -\overline{h(-\bar{z})}$ is analytic, satisfies $e^{ig} = \overline{\sigma H(-\bar{z})} = \sigma H = e^{ih}$, and agrees with h on $i\mathbb{R}_+$; hence $g = h$. In (4.1), $c = \operatorname{Re} h(i) = 0$; the symmetry and (4.2) show that μ is even, and pairing t with $-t$ in (4.1) gives (4.4). Uniqueness holds because two such logarithms differ by a constant in $2\pi\mathbb{Z}$ that vanishes on $i\mathbb{R}_+$. □

[10] Fatou's theorem: a bounded analytic function in $\mathbb{C}_+$ has nontangential limits at almost every point of $\mathbb{R}$ [23, Ch. I].

[11] With $B_\zeta(z) = (z-\zeta)/(z-\bar{\zeta})$ one has $B_\zeta(-\bar{z}) = \overline{B_{-\bar{\zeta}}(z)}$, so the product $B_\zeta B_{-\bar{\zeta}}$ has the symmetry, and so does the Blaschke product over such pairs, with the usual unimodular convergence factors chosen in pairs.

We call

$$\varphi(\omega) = \operatorname{Re} h(\omega + i0),$$
$$\alpha(\omega) = \operatorname{Im} h(\omega + i0) = -\log|H(\omega)|$$

the *phase* and the *attenuation* of H. Then $\arg H(\omega) = \varphi(\omega) + \frac{1-\sigma}{2}\pi$ modulo 2π. The phase φ is the argument of σH continued from the imaginary axis, where it vanishes; when H is continuous and positive at $\omega = 0$ and nonzero on $[0, b]$, it is simply the argument of H unwrapped from $\varphi(0) = 0$. The representation (4.4) separates the three ingredients of a zero-free system: the delay βz, the lossless resonances, which are the atoms and the singular part of μ, and the losses, which are the absolutely continuous part of μ, with density α/π.

4.3. The band

Fix $0 < a < b$ and write $S = [-b, -a] \cup [a, b]$.

Definition 4.3. A zero-free passive system H is *regular on the band* if there is an open interval $I \supset [a, b]$, $I \subset (0, \infty)$, such that H extends continuously to $\mathbb{C}_+ \cup I$ and has no zeros on I.

Lemma 4.4. *If H is regular on the band, then h extends continuously to $\mathbb{C}_+ \cup I$, φ and α are continuous on I, and the restriction of μ to $I \cup (-I)$ is $\pi^{-1}\alpha(|t|)\,\mathrm{d}t$. In particular μ has no atoms at $\pm a, \pm b$.*

Proof. Near each point of I the function σH is continuous and nonzero, so its logarithm, normalised as in Lemma 4.2, is continuous up to I. The convergence in (4.2) is then uniform on compact subintervals of I, which gives $\mu = \pi^{-1}\alpha\,\mathrm{d}t$ on I; evenness gives the statement on $-I$. □

4.4. Lossless bands

The following is the form of Foster's reactance theorem [11] that we need.

Proposition 4.5. *Let $H = B\sigma e^{ih}$ be passive, with B a real-symmetric Blaschke product whose zeros stay away from I, and let σe^{ih} be regular on the band. If $|H| = 1$ on I, then the argument of H is real-analytic and nondecreasing on I, and strictly increasing unless H is constant.*

Proof. On I we have $\alpha = 0$, so by Lemma 4.4 μ vanishes on I and (4.4) continues analytically across I, with

$$\varphi'(x) = \beta + \frac{\mu(\{0\})}{x^2} + \int_{(0,\infty)\setminus I} \frac{2(t^2 + x^2)}{(t^2 - x^2)^2}\,\mathrm{d}\mu(t) \;\geq\; 0.$$

Each factor $(z - \zeta)/(z - \bar{\zeta})$ of B, $\operatorname{Im}\zeta > 0$, has argument with derivative $2\operatorname{Im}\zeta/|x - \zeta|^2 > 0$ on the real line, and the series for B converges with its derivatives on I. The sum is strictly positive unless $\beta = 0$, $\mu = 0$ and B is constant. □

Corollary 4.6. *Under the hypotheses of Proposition 4.5, a passive system that is lossless on I and has constant argument on $[a, b]$ is a constant, $H \equiv \pm 1$.*

So constant phase on a band is impossible without loss. The main theorem measures how much loss.

5. A TEST FUNCTION FOR THE BAND

5.1. The functions ρ and W

Let

$$\rho(z) = \Big(\frac{z-a}{z-b}\Big)^{1/2}\Big(\frac{z+a}{z+b}\Big)^{1/2}, \qquad z \in \mathbb{C}_+,$$

with principal square roots. The map $z \mapsto (z-a)/(z-b)$ is a real Möbius transformation of determinant $a - b < 0$ and sends $\mathbb{C}_+$ onto the lower half-plane; $z \mapsto (z+a)/(z+b)$ has determinant $b - a > 0$ and sends $\mathbb{C}_+$ onto itself. Neither meets the negative real axis, so ρ is analytic and zero-free in $\mathbb{C}_+$.[12] Put

$$W(z) = \rho(z) - 2 + \frac{1}{\rho(z)} = \frac{(\rho(z) - 1)^2}{\rho(z)}, \tag{5.1}$$

and, for $a < t < b$,

$$r(t) = \sqrt{(t^2 - a^2)(b^2 - t^2)},$$
$$p(t) = \frac{a^2 + b^2 - 2t^2}{r(t)} = \sqrt{\frac{b^2 - t^2}{t^2 - a^2}} - \sqrt{\frac{t^2 - a^2}{b^2 - t^2}}. \tag{5.2}$$

Lemma 5.1. *(i) $\rho(-\bar{z}) = \overline{\rho(z)}$, and $\rho(z) = 1 + \frac{b^2 - a^2}{2z^2} + O(z^{-4})$, so $W(z) = O(z^{-4})$ as $z \to \infty$ in $\overline{\mathbb{C}_+}$.*

(ii) ρ and W extend continuously to $\overline{\mathbb{C}_+}$ minus the four points $\pm a, \pm b$, near which $|W(z)| = O(|z \mp a|^{-1/2})$, $O(|z \mp b|^{-1/2})$.

(iii) On $\mathbb{R} \setminus S$, ρ is real and positive, and $W = (\rho - 1)^2/\rho$ is real and strictly positive.

(iv) For $a < t < b$, $\rho(t + i0) = -i\,s(t)$ with $s(t) = \sqrt{(t^2 - a^2)/(b^2 - t^2)}$, hence

$$W(t) = -2 + i\,p(t), \qquad W(-t) = -2 - i\,p(t).$$

Proof. (i) The symmetry holds for each factor. Since $\rho^2 = (z^2 - a^2)/(z^2 - b^2) = 1 + (b^2 - a^2)/(z^2 - b^2)$ and $\rho \to 1$, the expansion follows, and $W = (\rho - 1)^2/\rho$. (ii) Near b the factor $(z - a)/(z - b)$ behaves like a constant over $z - b$, so $\rho \sim c(z - b)^{-1/2}$ and $W \sim \rho$; near a, $\rho \sim c(z - a)^{1/2}$ and $W \sim 1/\rho$. The points $-a, -b$ follow by symmetry.

[12] The principal square root is analytic and zero-free on $\mathbb{C} \setminus (-\infty, 0]$, and an open half-plane bounded by the real axis does not meet $(-\infty, 0]$.

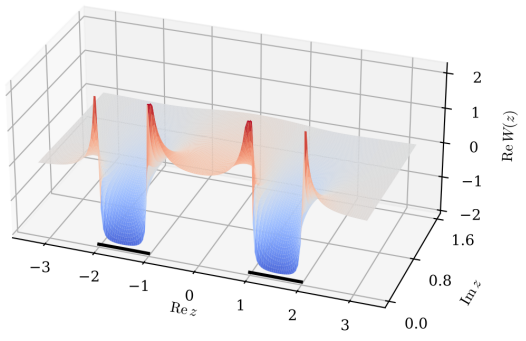


FIG. 3. The real part of the test function W over the upper half-plane for $a = 1$, $b = 2$ (values clipped at ± 2.2). On the real axis it equals -2 on the band (black segments) and is strictly positive elsewhere.

(iii) For real $t \notin S$ both quotients are positive. (iv) For $a < t < b$ and $\varepsilon \downarrow 0$ the number $(t-a)/(t-b+i\varepsilon)$ has negative real part and negative imaginary part, so its principal square root tends to $-i\sqrt{(t-a)/(b-t)}$, while $\sqrt{(t+a)/(t+b)}$ is positive. Hence $\rho = -is$ and $W = -is - 2 + i/s = -2 + i(1/s - s) = -2 + ip$. The value at $-t$ follows from (i). □

Figure 3 shows $\operatorname{Re} W$: a plateau at -2 over the band, positive values elsewhere, and integrable spikes at the band edges.

5.2. The kernel identity

Lemma 5.2. *For $s > 0$, $s \neq a, b$,*

$$\frac{1}{\pi}\,\mathrm{p.v.}\int_a^b p(x)\,\frac{2x}{x^2-s^2}\,\mathrm{d}x = \begin{cases} -2, & a < s < b, \\ W(s), & s \notin [a,b]. \end{cases} \tag{5.3}$$

Proof. Apply Cauchy's theorem to $W(z)/(z-s)$ on the boundary of the half-disc of radius R in $\mathbb{C}_+$, indented by semicircles of radius ε above $s, \pm a, \pm b$. By Lemma 5.1(ii) the small semicircles at $\pm a, \pm b$ contribute $O(\varepsilon^{1/2})$; by Lemma 5.1(i) the large one contributes $O(R^{-4})$; the one at s contributes $-\pi i W(s)$ in the limit, W being continuous at s. Therefore

$$\mathrm{p.v.}\int_{\mathbb{R}} \frac{W(x)}{x-s}\,\mathrm{d}x = \pi i\, W(s).$$

Taking imaginary parts and using that $\operatorname{Im} W$ vanishes off

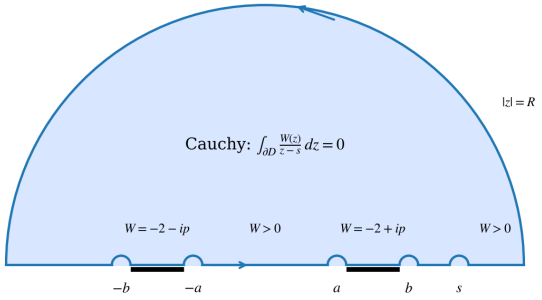


FIG. 4. The contour of the proof of Lemma 5.2: a half-disc of radius R in $\mathbb{C}_+$, indented above the branch points $\pm a, \pm b$ of W and above the pole s. The branch cuts of W are the two bands $\pm[a,b]$ (thick); the indentations at the branch points contribute nothing in the limit, the one at s contributes $-\pi i W(s)$.

S and equals $\pm p(|x|)$ on $\pm[a,b]$ (Lemma 5.1(iii),(iv)),

$$\begin{aligned} \pi \operatorname{Re} W(s) &= \mathrm{p.v.}\int_a^b p(x)\Big(\frac{1}{x-s} + \frac{1}{x+s}\Big)\,\mathrm{d}x \\ &= \mathrm{p.v.}\int_a^b p(x)\frac{2x}{x^2-s^2}\,\mathrm{d}x. \end{aligned}$$

By Lemma 5.1, $\operatorname{Re} W(s) = -2$ for $a < s < b$ and $\operatorname{Re} W(s) = W(s)$ otherwise. □

5.3. Moments of the weight

The weight p is positive on (a,c) and negative on (c,b), where $c^2 = (a^2+b^2)/2$. Under $\sigma = x^2$ it becomes antisymmetric, and this gives all the sign information we need.

Lemma 5.3. *Let $A = a^2$, $B = b^2$ and $\varpi(\sigma) = (A+B-2\sigma)/\sqrt{(\sigma-A)(B-\sigma)}$, so that $x\,p(x)\,\mathrm{d}x = \frac{1}{2}\varpi(x^2)\,\mathrm{d}(x^2)$.*

(i) $\varpi(A+B-\sigma) = -\varpi(\sigma)$, and $\int_a^b x\,p(x)\,\mathrm{d}x = 0$.

(ii) If f is continuous and nonincreasing on $[A,B]$, then $\int_a^b x\,p(x) f(x^2)\,\mathrm{d}x \geq 0$, with equality only for constant f.

(iii) $J := \int_a^b p(x)\,\mathrm{d}x > 0$ and $\int_a^b \frac{p(x)}{x}\,\mathrm{d}x = \frac{\pi(b-a)^2}{2ab}$.

(iv) $\int_a^b x^{2j+1} p(x)\,\mathrm{d}x < 0$ for $j \geq 1$, and $\int_a^b x^3 p(x)\,\mathrm{d}x = -\frac{\pi}{8}(b^2-a^2)^2$.

Proof. (i) The antisymmetry is visible, and it gives $\int_A^B \varpi = 0$. (ii) With $\sigma_m = (A+B)/2$,

$$\int_a^b x p f(x^2)\,\mathrm{d}x = \frac{1}{2}\int_A^{\sigma_m} \varpi(\sigma)\big(f(\sigma) - f(A+B-\sigma)\big)\,\mathrm{d}\sigma,$$

and both factors are nonnegative for $\sigma < \sigma_m$; the integral vanishes only if $f(\sigma) = f(A+B-\sigma)$ throughout, which for a monotone f means f constant. (iii) Apply (ii) to $f(\sigma) = \sigma^{-1/2}$ and $f(\sigma) = \sigma^{-1}$. The value follows from the elementary integrals

$$\int_A^B \frac{\mathrm{d}\sigma}{\sqrt{(\sigma-A)(B-\sigma)}} = \pi,$$
$$\int_A^B \frac{\mathrm{d}\sigma}{\sigma\sqrt{(\sigma-A)(B-\sigma)}} = \frac{\pi}{\sqrt{AB}},$$

which give $\frac{1}{2}\big((A+B)\pi/\sqrt{AB} - 2\pi\big) = \pi(b-a)^2/(2ab)$. (iv) Apply (ii) to $f(\sigma) = -\sigma^j$. For $j=1$ use also $\int_A^B \sigma\,\mathrm{d}\sigma/\sqrt{\cdot} = \pi(A+B)/2$ and $\int_A^B \sigma^2\,\mathrm{d}\sigma/\sqrt{\cdot} = \pi(3A^2 + 2AB + 3B^2)/8$. □

The first moment vanishing is the reason why delays are free; the positivity of J is the reason why a constant phase lag is not.

6. THE SUM RULE AND THE MAIN THEOREM

6.1. The sum rule

Theorem 6.1. *Let $H = \sigma e^{ih}$ be a zero-free passive system that is regular on the band, with phase φ, attenuation α and Herglotz measure μ. Then*

$$\int_a^b \alpha(\omega)\,\mathrm{d}\omega = \frac{1}{2}\int_a^b \varphi(\omega)\,p(\omega)\,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W(t)\,\mathrm{d}\mu(t), \tag{6.1}$$

and every term is finite. The last integral is nonnegative, and it vanishes if and only if $\mu(\mathbb{R}\setminus S) = 0$.

The proof uses one fact about the Hilbert transform $\mathcal{H}f(x) = \pi^{-1}\,\mathrm{p.v.}\int f(t)\,\mathrm{d}t/(x-t)$ on the line, the antisymmetry

$$\int_{\mathbb{R}} f\,\mathcal{H}g\,\mathrm{d}x = -\int_{\mathbb{R}} (\mathcal{H}f)\,g\,\mathrm{d}x, \qquad f \in L^q,\ g \in L^{q'}, \tag{6.2}$$

valid for $1 < q < \infty$ by M. Riesz's theorem [27, 28].[13]

Proof. Finiteness. φ is continuous on $[a,b]$ and p is integrable. On $I \setminus [a,b]$ the measure is $\pi^{-1}\alpha\,\mathrm{d}t$ with α bounded (Lemma 4.4), and W has integrable singularities at a, b. Away from $I \cup (-I)$, W is bounded and $O(t^{-4})$, while $\int \mathrm{d}\mu/(1+t^2) < \infty$.

Decomposition. Choose $a_1 < a$ and $b_1 > b$ with $[a_1,b_1] \subset I$, let $I_1 = (a_1,b_1)$, and split $\mu = \mu_1 + \mu_2$ with μ_1 the restriction of μ to $I_1 \cup (-I_1)$. By (4.4),

$$h = \beta z + h_1 + h_2,$$
$$h_j(z) = -\frac{\mu_j(\{0\})}{z} + \int_{(0,\infty)} \frac{2z}{t^2 - z^2}\,\mathrm{d}\mu_j(t),$$

and $\varphi = \beta x + \operatorname{Re} h_1 + \operatorname{Re} h_2$ on (a,b).

The delay. $\frac{1}{2}\int_a^b \beta x\,p(x)\,\mathrm{d}x = 0$ by Lemma 5.3(i).

The far part. The measure μ_2 has no mass in $I_1 \cup (-I_1)$, so h_2 is continuous up to (a,b) and the kernel $2x/(t^2-x^2)$ is bounded by a constant times $(1+t^2)^{-1}$ for $x \in [a,b]$ and $t \notin I_1$. Fubini's theorem and Lemmas 5.2 and 5.3(iii) give

$$\frac{1}{2}\int_a^b \operatorname{Re} h_2\,p\,\mathrm{d}x = -\frac{\mu(\{0\})}{2}\int_a^b \frac{p}{x}\,\mathrm{d}x - \frac{\pi}{2}\int_{(0,\infty)\setminus I_1} W\,\mathrm{d}\mu$$
$$= -\frac{\pi}{4}\int_{\mathbb{R}\setminus(I_1\cup -I_1)} W\,\mathrm{d}\mu,$$

where we used $W(0) = (b-a)^2/(ab)$,[14] the evenness of W on $\mathbb{R}\setminus S$ and of μ.

The near part. Here $\mu_1 = \psi(t)\,\mathrm{d}t$ with $\psi = \pi^{-1}\alpha(|t|)$ on $I_1 \cup (-I_1)$, a bounded function. For almost every $x \in (a,b)$,

$$\operatorname{Re} h_1(x) = \mathrm{p.v.}\int_{I_1} \psi(t)\frac{2x}{t^2-x^2}\,\mathrm{d}t$$
$$= \mathrm{p.v.}\int_{I_1} \psi(t)\Big(\frac{1}{t-x} - \frac{1}{t+x}\Big)\,\mathrm{d}t,$$

the boundary value of the real part of a Cauchy integral with bounded density [27, 29]. We claim that

$$\int_a^b p(x)\,\operatorname{Re} h_1(x)\,\mathrm{d}x$$
$$= \int_{I_1} \psi(t)\Big(\mathrm{p.v.}\int_a^b p(x)\frac{2x}{t^2-x^2}\,\mathrm{d}x\Big)\,\mathrm{d}t. \tag{6.3}$$

For the part $1/(t+x)$ of the kernel this is Fubini's theorem, the kernel being bounded. For the part $1/(t-x)$ it is (6.2) with $f = p\,\mathbf{1}_{[a,b]} \in L^{3/2}$ and $g = \psi\mathbf{1}_{I_1} \in L^3$; the function p lies in L^q for every $q < 2$ because its singularities are of order $|x-a|^{-1/2}$ and $|x-b|^{-1/2}$. By Lemma 5.2 the inner integral in (6.3) equals 2π for $t \in (a,b)$ and $-\pi W(t)$ for $t \in I_1 \setminus [a,b]$. Hence

$$\frac{1}{2}\int_a^b p\,\operatorname{Re} h_1\,\mathrm{d}x = \pi\int_a^b \psi\,\mathrm{d}t - \frac{\pi}{2}\int_{I_1\setminus[a,b]} W\,\mathrm{d}\mu$$
$$= \int_a^b \alpha\,\mathrm{d}t - \frac{\pi}{4}\int_{(I_1\cup -I_1)\setminus S} W\,\mathrm{d}\mu.$$

[13] Riesz's theorem is the boundedness of $\mathcal{H}$ on L^q, $1 < q < \infty$. For smooth compactly supported f, g the antisymmetry is Fubini's theorem for the truncated kernel $\mathbf{1}_{|x-t|>\varepsilon}/(x-t)$, which is odd under $x \leftrightarrow t$; it extends to $L^q \times L^{q'}$ by density and Riesz's bound.

[14] $\rho(0) = \sqrt{a/b}\sqrt{a/b} = a/b$, so $W(0) = a/b - 2 + b/a = (b-a)^2/(ab)$. With Lemma 5.3(iii), $\int_a^b p/x\,\mathrm{d}x = \pi(b-a)^2/(2ab) = \frac{\pi}{2}W(0)$, which turns the atom at 0 into $-\frac{\pi}{4}W(0)\mu(\{0\})$.

Adding the three parts gives (6.1). Since $W > 0$ on $\mathbb{R}\setminus S$ and μ has no atoms at $\pm a, \pm b$, the last term vanishes exactly when $\mu(\mathbb{R} \setminus S) = 0$. □

The identity (6.1) holds for every phase. Its content is that the band loss is a fixed linear functional of the band phase plus a nonnegative price for everything the system does outside the band: a lossless resonance at $\omega_0 \notin [a, b]$ with weight γ in μ costs $\frac{\pi}{2} W(\omega_0)\gamma$,[15] a pole at the origin costs $\frac{\pi}{4} W(0)\mu(\{0\})$, and attenuation outside the band costs its W-weighted total.

6.2. Constant phase lag

Put

$$\begin{aligned} \Lambda &= \frac{J}{2(b-a)} \\ &= \frac{1}{2(b-a)} \int_a^b \frac{a^2 + b^2 - 2\omega^2}{\sqrt{(\omega^2 - a^2)(b^2 - \omega^2)}}\, \mathrm{d}\omega > 0. \end{aligned} \tag{6.4}$$

Section 7 shows $\Lambda = (K(\kappa) - E(\kappa))/\kappa$.

Theorem 6.2 (Main theorem)**.** *Let H be a zero-free passive system, regular on the band $[a, b]$, whose phase on the band is*

$$\varphi(\omega) = \theta + \tau\omega, \qquad a < \omega < b,$$

with $\theta \geq 0$ and $\tau \in \mathbb{R}$. Then

$$\frac{1}{b-a} \int_a^b \alpha(\omega)\, \mathrm{d}\omega \;\geq\; \theta\, \Lambda. \tag{6.5}$$

Equality holds if and only if $\mu(\mathbb{R} \setminus S) = 0$; for every $\theta \geq 0$ it is attained by the system $H_\star$ of Theorem 8.2.

Proof. By Lemma 5.3(i), $\frac{1}{2}\int_a^b (\theta + \tau\omega) p\, \mathrm{d}\omega = \frac{1}{2}\theta J = (b-a)\theta\Lambda$, and Theorem 6.1 gives the inequality and the case of equality. □

Three remarks make the statement concrete.

Remark 6.3 (Constant argument). If $\arg H \equiv \theta$ modulo 2π on the band, then $\varphi = \theta - \frac{1-\sigma}{2}\pi$ modulo 2π, and since φ is continuous it is a constant θ' with $\theta' \equiv \theta \pmod{\pi}$. The theorem applies with θ' when $\theta' > 0$. Which representative occurs is not decided by the band: it depends on how much phase the system has accumulated from the imaginary axis, for instance from zero frequency.

Remark 6.4 (Lossless resonances). The equality case shows that every lossless resonance outside the band, and every atom of μ at zero frequency, raises the loss strictly above the bound. Shaping the phase with reactances, the classical tool of all-pass and phase-equaliser design, therefore works against a constant phase lag. Figure 5 shows the price of a unit of measure off the band, the factor $\frac{\pi}{4} W(t)$ of (6.1).

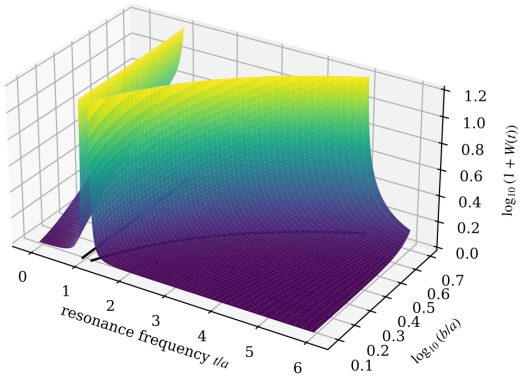


FIG. 5. The cost $\log_{10}(1 + W(t))$ of a lossless resonance at frequency t, as a function of t/a and of the band ratio b/a. It vanishes only on the band (the gap between the two sheets), grows without bound at the band edges, and decays like t^{-4} at high frequency (Lemma 5.1(i)); at $t = 0$ it is $W(0) = a/b + b/a - 2 > 0$.

Remark 6.5 (Passivity everywhere is needed). The constant-phase element $H(z) = (-iz/a)^{-\gamma}$, $0 < \gamma < 1$, familiar from dielectric relaxation [18], has phase $\theta = \gamma\pi/2$ at every positive frequency and attenuation $\alpha = \gamma \log(\omega/a) \geq 0$ on $[a, \infty)$. Over an octave its mean attenuation is $(2\theta/\pi)(2\log 2 - 1) = 0.2459\,\theta$, below the bound $0.2735\,\theta$. There is no contradiction: $|H| > 1$ for $\omega < a$, so H is not passive. The bound depends on passivity at all frequencies, not only on the band.

6.3. Constant phase lead

For a negative intercept the same identity, applied after an inversion of the frequency axis, gives a weighted bound.

Theorem 6.6. *Let H be as in Theorem 6.2 but with $\theta < 0$. Then*

$$\frac{1}{b-a} \int_a^b \alpha(\omega)\, \frac{ab}{\omega^2}\, \mathrm{d}\omega \;\geq\; |\theta|\, \Lambda - \frac{\pi}{4}\, \tau\, (b-a). \tag{6.6}$$

For $\tau = 0$ equality holds if and only if $\beta = 0$ and μ vanishes on $\mathbb{R} \setminus (S \cup \{0\})$; it is attained by $H_\star(-ab/z)$, where $H_\star$ is the extremal system for $|\theta|$.

Proof. The map $z \mapsto -ab/z$ preserves $\mathbb{C}_+$, so $\tilde{h}(z) = h(-ab/z)$ is a Herglotz function with the symmetry of (4.3), and $\tilde{H} = \sigma e^{i\tilde{h}}$ is zero-free, passive and regular on

[15] The measure is even, so the resonance puts mass γ at both $\pm\omega_0$, and $W(-\omega_0) = W(\omega_0)$ off the band; the two atoms give $\frac{\pi}{4} \cdot 2W(\omega_0)\gamma$.

the band, since $x \mapsto ab/x$ maps a neighbourhood of $[a,b]$ onto another. For $x \in (a,b)$ the point $-ab/(x+i0)$ lies above $-ab/x \in (-b,-a)$, so by the oddness of φ,

$$\tilde\varphi(x) = -\varphi(ab/x), \qquad \tilde\alpha(x) = \alpha(ab/x).$$

Substituting $x = ab/\omega$ and using $r(ab/\omega) = ab\,r(\omega)/\omega^2$,

$$\int_a^b f(ab/x)\,p(x)\,\mathrm{d}x = \int_a^b f(\omega)\,p^*(\omega)\,\mathrm{d}\omega,$$
$$p^*(\omega) = \frac{a^2+b^2-2a^2b^2/\omega^2}{r(\omega)},$$

for every integrable f. With $f=1$ and $f(\omega)=\omega$ this gives $\int_a^b p^* = J$ and $\int_a^b \omega p^* = ab\int_a^b p/x = \pi(b-a)^2/2$ (Lemma 5.3(iii)). Theorem 6.1 for $\tilde H$ now reads

$$\int_a^b \alpha(\omega)\frac{ab}{\omega^2}\,\mathrm{d}\omega = -\frac12\int_a^b(\theta+\tau\omega)p^*\,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W\,\mathrm{d}\tilde\mu,$$

which is (6.6) after division by $b-a$. The measure $\tilde\mu$ is the image of μ under $t\mapsto -ab/t$ with density ab/t^2 on $\mathbb{R}\setminus\{0\}$, plus an atom $ab\beta$ at 0 coming from the term βz; an atom of μ at 0 becomes a linear term of $\tilde h$ and costs nothing. This gives the case of equality, and $H_\star(-ab/z)$ has $\beta=0$ and its measure on $S\cup\{0\}$. □

6.4. The worst frequency

Corollary 6.7. *Let H be zero-free, passive and regular on the band, with $\arg H \equiv \theta$ modulo 2π on $[a,b]$. Then*

$$\max_{[a,b]}\alpha \;\geq\; \operatorname{dist}(\theta,\pi\mathbb{Z})\,\Lambda,$$

so at some frequency of the band the transmitted power is at most $e^{-2\operatorname{dist}(\theta,\pi\mathbb{Z})\Lambda}$. If the phase θ' of Remark 6.3 is positive, also

$$\exp\Big(\frac{1}{b-a}\int_a^b \log|H(\omega)|^2\,\mathrm{d}\omega\Big) \leq e^{-2\theta'\Lambda},$$

that is, the geometric mean of the power transmission over the band is at most $e^{-2\theta'\Lambda}$.

Proof. By Remark 6.3 the phase is a constant $\theta' \equiv \theta \pmod\pi$. If $\theta'>0$ then $\theta' \geq \operatorname{dist}(\theta,\pi\mathbb{Z})$ and Theorem 6.2 bounds the mean, hence the maximum, of α. If $\theta'<0$ then $|\theta'| \geq \operatorname{dist}(\theta,\pi\mathbb{Z})$, and since $\int_a^b ab\,\omega^{-2}\,\mathrm{d}\omega = b-a$ the left side of (6.6) with $\tau=0$ is at most $\max\alpha$. The second statement is (6.5) with $\log|H| = -\alpha$. □

The bound by the mean is not sharp for the maximum: Theorem 9.8 replaces Λ by the larger constant $A(m)$, which is attained.

Remark 6.8 (The lead extremal). For the system $H_\star(-ab/z)$ of Theorem 6.6, whose phase is $-|\theta|$ on the band, Theorem 6.1 applied directly gives the plain mean attenuation

$$\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega = |\theta|\;\frac{E(\kappa)-\kappa'^2K(\kappa)}{\kappa}, \qquad \kappa'^2 = 1-\kappa^2,$$

because its measure off S is the atom $ab\,\tau_\star$ at 0, of cost $\frac{\pi}{4}W(0)ab\tau_\star$, with $\tau_\star$ from Theorem 8.2 and the evaluation of Section 7. For an octave this is $0.2656\,|\theta|$, slightly below $\Lambda = 0.2735$. It is the least plain mean among systems with constant phase lead (Corollary 9.7(iii)).

7. THE CONSTANT

We use the complete elliptic integrals of modulus $k \in [0,1)$,

$$K(k) = \int_0^{\pi/2}\frac{\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}},$$
$$E(k) = \int_0^{\pi/2}\sqrt{1-k^2\sin^2\phi}\;\mathrm{d}\phi,$$

and the moduli attached to the band,

$$k = \sqrt{1-a^2/b^2}, \qquad k' = a/b, \qquad \kappa = \frac{b-a}{b+a} = \frac{1-k'}{1+k'}.$$

The facts on K and E that we use are collected in Appendix A.

Proposition 7.1. *With $I_0 = \int_a^b \mathrm{d}t/r(t)$ and $I_2 = \int_a^b t^2\,\mathrm{d}t/r(t)$,*

$$I_0 = \frac{K(k)}{b}, \qquad I_2 = b\,E(k),$$

and

$$\begin{aligned} J &= (a^2+b^2)I_0 - 2I_2 = b\big[(1+k'^2)K(k) - 2E(k)\big] \\ &= 2(a+b)\big[K(\kappa)-E(\kappa)\big]. \end{aligned} \tag{7.1}$$

Consequently

$$\Lambda = \frac{K(\kappa)-E(\kappa)}{\kappa}. \tag{7.2}$$

Proof. Substitute $t^2 = b^2-(b^2-a^2)\sin^2\phi$, $0\leq\phi\leq\pi/2$. Then $b^2-t^2 = (b^2-a^2)\sin^2\phi$, $t^2-a^2 = (b^2-a^2)\cos^2\phi$, $t\,\mathrm{d}t = -(b^2-a^2)\sin\phi\cos\phi\,\mathrm{d}\phi$, hence $\mathrm{d}t/r(t) = -\,\mathrm{d}\phi/t$ and $t = b\sqrt{1-k^2\sin^2\phi}$. This gives I_0 and I_2, and the second form of J. The Landen transformation (Appendix A) states

$$K(k) = (1+\kappa)K(\kappa), \qquad E(k) = (1+k')E(\kappa) - k'K(k).$$

Inserting these,[16] and using $b(1+\kappa) = 2b^2/(a+b)$ and $1+k' = (a+b)/b$,

$$\begin{aligned} J &= b(1+\kappa)(1+k')^2K(\kappa) - 2b(1+k')E(\kappa) \\ &= 2(a+b)\big[K(\kappa)-E(\kappa)\big]. \end{aligned}$$

[16] The coefficient of $K(\kappa)$ is $b(1+\kappa)\big[(1+k'^2)+2k'\big] = b(1+\kappa)(1+k')^2$, the term $2k'$ coming from $-2E(k)$.

Dividing by $2(b-a)=2\kappa(a+b)$ gives (7.2). □

Proposition 7.2. *(i) With* $c_n=\binom{2n}{n}4^{-n}$,

$$\Lambda=\frac{\pi}{2}\sum_{n\ge1}\frac{2n}{2n-1}\,c_n^2\,\kappa^{2n-1}=\frac{\pi}{4}\kappa\Big(1+\frac{3}{8}\kappa^2+\frac{15}{64}\kappa^4+\cdots\Big),$$

a series with positive coefficients;[17] *in particular* Λ *increases with* κ*, that is, with* b/a*.*

(ii) For all $0<a<b$*,*

$$\frac{\pi}{4}\,\frac{b-a}{b+a}\;<\;\Lambda\;\le\;\frac{\pi}{4}\,\frac{\kappa}{1-\kappa^2}=\frac{\pi}{16}\,\frac{b^2-a^2}{ab}.$$

(iii) As $a/b\to0$, $\Lambda=\log\frac{4}{\kappa'}-1+o(1)=\frac12\log\frac{4b}{a}-1+o(1)$*, with* $\kappa'=\sqrt{1-\kappa^2}=2\sqrt{ab}/(a+b)$*.*

Proof. (i) From $K=\frac{\pi}{2}\sum c_n^2\kappa^{2n}$ and $E=\frac{\pi}{2}\sum c_n^2\kappa^{2n}/(1-2n)$ one gets $K-E=\frac{\pi}{2}\sum_{n\ge1}c_n^2\kappa^{2n}\,2n/(2n-1)$. The coefficients for $n=1,2,3$ are $\frac12$, $\frac{3}{16}$ and $\frac{15}{128}$, which give the displayed expansion after division by κ. (ii) The lower bound is the first term. For the upper bound, the ratio of consecutive coefficients $d_n=c_n^2\,2n/(2n-1)$ is

$$\frac{d_{n+1}}{d_n}=\frac{(2n+1)(2n-1)}{(2n+2)\,2n}<1,$$

so $d_n\le d_1=\frac12$ and $K-E\le\frac{\pi}{4}\sum_{n\ge1}\kappa^{2n}=\frac{\pi}{4}\kappa^2/(1-\kappa^2)$. Finally $\kappa/(1-\kappa^2)=(b^2-a^2)/(4ab)$. (iii) As $\kappa\to1$, $K(\kappa)=\log(4/\kappa')+o(1)$ and $E(\kappa)\to1$ (Appendix A); and $\log(4/\kappa')=\log\big(2(a+b)/\sqrt{ab}\big)=\frac12\log(4b/a)+o(1)$. □

For a narrow band with centre $\bar\omega=(a+b)/2$ and width $\Delta\omega=b-a$, $\kappa=\Delta\omega/(2\bar\omega)$, so

$$\Lambda=\frac{\pi}{8}\,\frac{\Delta\omega}{\bar\omega}\Big(1+O\big((\Delta\omega/\bar\omega)^2\big)\Big).\tag{7.3}$$

The bound is linear in the relative bandwidth, which is what one expects: on an infinitely narrow band a delay achieves any phase without loss. For wide bands it grows only logarithmically in b/a. Table II gives values, also in decibels per degree of phase, which is Λ times $(20/\ln10)(\pi/180)=0.15160$, and Figure 6 shows the bound as a surface.

TABLE II. The constant Λ of (7.2), the least mean attenuation per radian of constant phase lag, for several band ratios. The last column is the least worst-frequency power loss for $\theta=1^\circ$.

b/a	κ	Λ (Np/rad)	dB/deg	$1-e^{-2\Lambda\pi/180}$
1.01	0.004975	0.003907	0.000592	1.36×10^{-4}
1.1	0.04762	0.03743	0.00567	1.31×10^{-3}
1.5	0.2	0.1595	0.0242	5.55×10^{-3}
2	0.3333	0.2735	0.0415	9.50×10^{-3}
3	0.5	0.4366	0.0662	1.51×10^{-2}
10	0.8182	0.9448	0.143	3.24×10^{-2}
100	0.9802	2.016	0.306	6.80×10^{-2}
1000	0.9980	3.150	0.478	1.04×10^{-1}

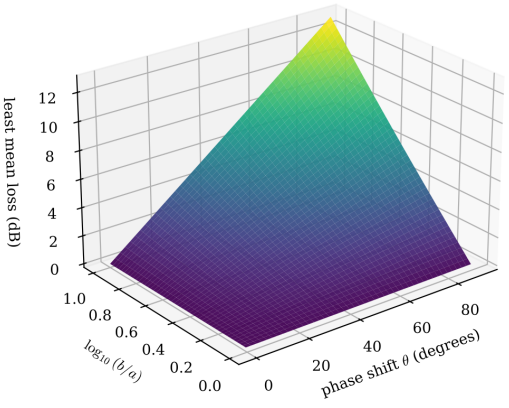


FIG. 6. The least mean loss $\theta\Lambda$, in decibels, over the phase lag θ and the band ratio b/a. The surface is linear in θ and increasing in b/a.

8. SHARPNESS: THE EXTREMAL SYSTEMS

8.1. The square root R

Let R be the branch of $\sqrt{(z^2-a^2)(z^2-b^2)}$ that is analytic in $\mathbb{C}\setminus S$ and satisfies $R(z)=z^2\big(1+O(z^{-2})\big)$ at infinity. It exists because a loop around one component of S encircles two branch points.[18]

Lemma 8.1. *$R(-z)=R(z)$ and $R(\bar z)=\overline{R(z)}$. On the real axis, $R(t)=\sqrt{(t^2-a^2)(t^2-b^2)}>0$ for $|t|>b$, $R(t)=-\sqrt{(a^2-t^2)(b^2-t^2)}<0$ for $|t|<a$, and*

$$R(t+i0)=i\,r(t),\qquad R(-t+i0)=-i\,r(t)\qquad(a<t<b).$$

Proof. The symmetries hold because $R(-z)$ and $\overline{R(\bar z)}$ are branches of the same root with the same behaviour at infinity. Near $z=b$, $R(z)=(z-b)^{1/2}G(z)$ with G analytic and $G(b)=\sqrt{2b(b^2-a^2)}>0$, the root being positive for $z>b$. Passing from $t>b$ to $t<b$ through $\mathbb{C}_+$ turns $(z-b)^{1/2}$ into $i\sqrt{b-t}$, so $R=ir$ on (a,b). Near $z=a$ write $R=(z-a)^{1/2}G_1(z)$ with $(z-a)^{1/2}>0$ for $t>a$; then G_1 is positive imaginary on (a,b), and passing to $t<a$ through $\mathbb{C}_+$ gives $R=i\sqrt{a-t}\,G_1<0$. The values on the negative axis follow from the symmetries. □

[17] By (A.1), $K-E=\frac{\pi}{2}\sum_{n\ge1}c_n^2\big(1-\frac{1}{1-2n}\big)\kappa^{2n}$ and $1-\frac{1}{1-2n}=\frac{2n}{2n-1}$; dividing by κ gives (i).

[18] Each of the four factors $(z\mp a)^{1/2}$, $(z\mp b)^{1/2}$ changes sign along a loop around its branch point; a loop around $[a,b]$ changes the sign twice, so the product returns to its value.

8.2. The extremal system for constant phase

Put

$$m^2 = b^2\,\frac{E(k)}{K(k)}, \qquad \tau_\star = \frac{2\theta K(k)}{\pi b} = \frac{4\theta K(\kappa)}{\pi(a+b)}, \tag{8.1}$$

so that $a^2 < m^2 < b^2$, because $m^2 = I_2/I_0$ is an average of t^2 over (a,b) (Proposition 7.1), and define

$$h_\star(z) = \tau_\star \int_0^z \frac{s^2 - m^2}{R(s)}\,\mathrm{d}s, \qquad H_\star = e^{ih_\star}. \tag{8.2}$$

Theorem 8.2. *Let $\theta > 0$.*

(i) The integral (8.2) does not depend on the path in $\mathbb{C}\setminus S$; $h_\star$ is analytic there, odd, real on $\mathbb{R}\setminus S$, and $h_\star(z) = \tau_\star z + O(z^{-1})$ at infinity.

(ii) On the band, $h_\star(x+i0) = \theta + i\alpha_\star(x)$ with

$$\alpha_\star(x) = \tau_\star \int_a^x \frac{m^2 - s^2}{r(s)}\,\mathrm{d}s, \tag{8.3}$$

which is positive on (a,b), vanishes at a and b, and is largest at $x = m$.

(iii) $h_\star$ is a Herglotz function, and $H_\star$ is a zero-free passive system, regular on the band, lossless on $\mathbb{R}\setminus S$, with constant phase θ on $[a,b]$ and

$$\frac{1}{b-a}\int_a^b \alpha_\star(\omega)\,\mathrm{d}\omega = \theta\Lambda.$$

Proof. (i) The integrand is even, analytic in $\mathbb{C}\setminus S$, and integrable at the four branch points. A loop around $[a,b]$ can be shrunk onto the two sides of the cut, where $R = \pm ir$; its integral is $\pm 2i\int_a^b (s^2-m^2)\,\mathrm{d}s/r(s) = \pm 2i(I_2 - m^2 I_0) = 0$. By evenness the same holds around $[-b,-a]$. Hence $h_\star$ is single-valued, and odd. On $(-a,a)$ and on $|t| > b$ the integrand is real, so $h_\star$ is real there; it is also real along the path from 0 to $\pm a$, and the vanishing of the periods shows that it returns to real values beyond $\pm b$. At infinity, $(s^2-m^2)/R(s) = 1 + O(s^{-2})$, so $h_\star(z) - \tau_\star z$ tends to a constant; it is odd and analytic for $|z| > b$, so its Laurent series has only odd powers and the constant is 0. (ii) For $0 < x < a$, $h_\star(x) = \tau_\star \int_0^x (m^2 - s^2)\,\mathrm{d}s/\sqrt{(a^2-s^2)(b^2-s^2)}$ by Lemma 8.1, and at $x = a$ this is

$$\begin{aligned}
&\tau_\star\Big(\frac{m^2}{b}K(k') - b\big(K(k') - E(k')\big)\Big)\\
&\quad = \frac{\tau_\star b}{K(k)}\big(E(k)K(k') + E(k')K(k) - K(k)K(k')\big)\\
&\quad = \frac{\pi b\,\tau_\star}{2K(k)} = \theta,
\end{aligned}$$

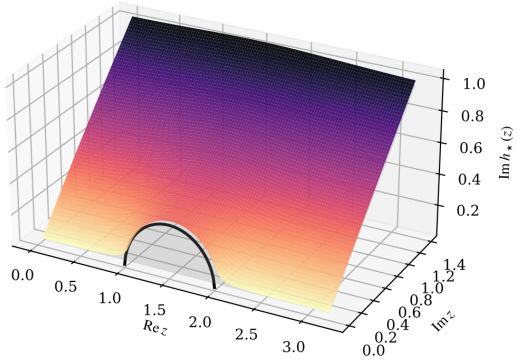


FIG. 7. The surface $\mathrm{Im}\,h_\star(x+iy)$ over the upper half-plane for an octave and $\theta = 1$. It is positive in $\mathbb{C}_+$, as a Herglotz function must be; on the real axis it vanishes off the band, and over the band its boundary values are the attenuation $\alpha_\star$ (black curve). Far from the band it approaches the plane $\tau_\star y$ (grey) of the delay.

by the substitution $s = a\sin\phi$[19] and Legendre's relation (Appendix A). Continuing from a along the upper side of the cut, where $R = ir$,

$$h_\star(x+i0) = \theta + \tau_\star\int_a^x \frac{s^2-m^2}{ir(s)}\,\mathrm{d}s = \theta + i\alpha_\star(x).$$

The integrand of (8.3) is positive for $s < m$ and negative for $s > m$, and its integral over (a,b) is $I_2 - m^2 I_0 = 0$; so $\alpha_\star$ rises from 0 at a to its maximum at m and falls back to 0 at b. (iii) The function $v = \mathrm{Im}(h_\star - \tau_\star z)$ is harmonic in $\mathbb{C}_+$, continuous on $\overline{\mathbb{C}_+}$, tends to 0 at infinity by (i), vanishes on $\mathbb{R}\setminus S$, and equals $\alpha_\star \geq 0$ on (a,b) and, by the symmetry $h_\star(-\bar z) = -\overline{h_\star(z)}$, also on $(-b,-a)$. By the minimum principle $v \geq 0$, so $\mathrm{Im}\,h_\star \geq \tau_\star\,\mathrm{Im}\,z \geq 0$. Thus $h_\star$ is a Herglotz function with $h_\star(iy) \in i\mathbb{R}$, $H_\star$ is passive and zero-free, and it is regular on the band since $h_\star$ is continuous up to $(0,\infty)$. As $\mathrm{Im}\,h_\star = 0$ on $\mathbb{R}\setminus S$, (4.2) shows $\mu_\star(\mathbb{R}\setminus S) = 0$, and Theorem 6.2 holds with equality. □

Remark 8.3. The extremal system is a delay $\tau_\star$ at high frequency, as its asymptotics shows, yet its group delay on the band is zero, its phase being constant there. All of its attenuation is in the band, and none of its response outside the band is reactive: in the language of (4.4) it has $\beta = \tau_\star$ and a measure carried by S. For an octave and $\theta = 1$, $\tau_\star = 0.6864/a$, $m = 1.4988\,a$, and the peak

[19] It gives $\mathrm{d}s/\sqrt{(a^2-s^2)(b^2-s^2)} = \mathrm{d}\phi/(b\sqrt{1-k'^2\sin^2\phi})$, and $\int_0^{\pi/2}\sin^2\phi\,\mathrm{d}\phi/\sqrt{1-k'^2\sin^2\phi} = (K(k') - E(k'))/k'^2$ with $a^2/k'^2 = b^2$.

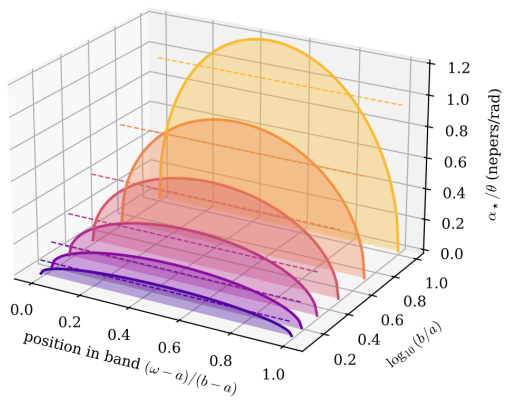


FIG. 8. The attenuation $\alpha_\star/\theta$ of the extremal system across the band, for $b/a = 1.25, 1.5, 2, 3, 5, 10$. Dashed lines mark the mean, Λ. The profile vanishes at both edges.

attenuation is $\alpha_\star(m) = 0.3482\,\mathrm{Np}$, against the mean 0.2735. Figure 8 shows $\alpha_\star$ for several band ratios, and Figure 7 the function $\operatorname{Im} h_\star$ over $\mathbb{C}_+$.

8.3. Extremal systems for other phases

The construction extends to phases that are not constant, and this is what we need for systems with zeros.

Proposition 8.4. *Let g be continuously differentiable on $[a,b]$ and put*

$$h_g(z) = \frac{2zR(z)}{\pi}\int_a^b \frac{g(t)}{r(t)\,(z^2-t^2)}\,\mathrm{d}t,$$
$$\beta_g = \frac{2}{\pi}\int_a^b \frac{g(t)}{r(t)}\,\mathrm{d}t.$$

(i) h_g is analytic in $\mathbb{C}\setminus S$, odd, real on $\mathbb{R}\setminus S$, continuous up to S from both sides, and $h_g(z) = \beta_g z + O(z^{-1})$ at infinity.

(ii) For $a < x < b$, $\operatorname{Re} h_g(x+i0) = g(x)$ and

$$\operatorname{Im} h_g(x+i0) = v_g(x) = \frac{2x\,r(x)}{\pi}\,\mathrm{p.v.}\int_a^b \frac{g(t)}{r(t)\,(x^2-t^2)}\,\mathrm{d}t.$$

(iii) If $v_g \ge 0$ on (a,b) and $\beta_g \ge 0$, then e^{ih_g} is a zero-free passive system, regular on the band, with phase g and attenuation v_g on the band and no measure outside S, so that $\int_a^b v_g = \frac{1}{2}\int_a^b g\,p$.

(iv) For constant $g = \theta$, $h_g = h_\star$.

Proof. (i) The Cauchy integral $q(z) = \int_a^b g(t)\,(r(t)(z^2-t^2))^{-1}\,\mathrm{d}t$ is analytic off S and even; $zR(z)q(z)$ is odd and real on $\mathbb{R}\setminus S$ because R is real there (Lemma 8.1). The density g/r has singularities of order $|t-a|^{-1/2}$, $|t-b|^{-1/2}$ at the ends, so q is $O(|z-a|^{-1/2})$, $O(|z-b|^{-1/2})$ there and Hölder continuous up to the open cut [29, Ch. 4]; multiplied by R, which vanishes like the square root, the product is continuous up to S. At infinity $q(z) = z^{-2}\int_a^b g/r + O(z^{-4})$ and $zR(z) = z^3(1+O(z^{-2}))$, which gives the expansion. (ii) Write $2z/(z^2-t^2) = 1/(z-t)+1/(z+t)$. By the Plemelj formula, at $x+i0$ with $a<x<b$ the first part of q contributes $\mathrm{p.v.}\int g\,(r(t)(x-t))^{-1}\,\mathrm{d}t - i\pi g(x)/r(x)$, so

$$\frac{2}{\pi}xq(x+i0) = \frac{2x}{\pi}\,\mathrm{p.v.}\int_a^b \frac{g(t)\,\mathrm{d}t}{r(t)(x^2-t^2)} - \frac{ig(x)}{r(x)}.$$

Multiplying by $R(x+i0) = ir(x)$ gives the real part $g(x)$ and the imaginary part $v_g(x)$. (iii) As in the proof of Theorem 8.2(iii), $\operatorname{Im}(h_g - \beta_g z)$ is harmonic, continuous up to the axis, zero on $\mathbb{R}\setminus S$, equal to $v_g \ge 0$ on S, and tends to 0 at infinity; so $\operatorname{Im} h_g \ge \beta_g \operatorname{Im} z \ge 0$. The last identity is Theorem 6.1. (iv) Let $d = h_g - h_\star$. Both functions have real part θ on the upper side of (a,b), are real on $\mathbb{R}\setminus S$ and satisfy $h(\bar z) = \overline{h(z)}$, so on the lower side the real part is again θ and $d(x\pm i0) = \pm i\,e(x)$ with e real and continuous. Since $R(x\pm i0) = \pm ir(x)$, the function d/R has the same boundary values e/r from both sides of (a,b), and likewise on $(-b,-a)$. It is therefore analytic across the open cuts, by Morera's theorem, and its singularities at $\pm a, \pm b$ are $O(|z-c|^{-1/2})$, hence removable. So d/R is entire. Both functions have the linear term $\tau_\star z$, because $\beta_\theta = 2\theta I_0/\pi = \tau_\star$, so $d = O(z^{-1})$ and $d/R \to 0$ at infinity. Thus $d = 0$. □

Lemma 8.5. *If $g(t)/t$ is nonincreasing on $[a,b]$, then $v_g \ge 0$. In particular this holds when $g > 0$ and g is nonincreasing, and then also $\beta_g > 0$.*

Proof. Substitute $\sigma = t^2$ and write $G(\sigma) = g(\sqrt\sigma)/\sqrt\sigma$, which is nonincreasing. Then

$$\frac{g(t)\,\mathrm{d}t}{r(t)} = \frac{G(\sigma)\,\mathrm{d}\sigma}{2\sqrt{(\sigma-a^2)(b^2-\sigma)}},$$

and with $\sigma_0 = x^2$,

$$v_g(x) = \frac{x\,r(x)}{\pi} \times \mathrm{p.v.}\int_{a^2}^{b^2} \frac{G(\sigma)}{(\sigma_0-\sigma)\sqrt{(\sigma-a^2)(b^2-\sigma)}}\,\mathrm{d}\sigma.$$

Glauert's integral [27],

$$\mathrm{p.v.}\int_A^B \frac{\mathrm{d}\sigma}{(\sigma_0-\sigma)\sqrt{(\sigma-A)(B-\sigma)}} = 0$$

for $A < \sigma_0 < B$, which follows from the substitution $\sigma = \frac{A+B}{2} - \frac{B-A}{2}\cos\phi$ and $\mathrm{p.v.}\int_0^\pi \mathrm{d}\phi/(\cos\phi - \cos\phi_0) = 0$,

allows us to subtract $G(\sigma_0)$:

$$v_g(x) = \frac{x\,r(x)}{\pi}\int_{a^2}^{b^2} \frac{G(\sigma)-G(\sigma_0)}{\sigma_0-\sigma}\,\frac{\mathrm{d}\sigma}{\sqrt{(\sigma-a^2)(b^2-\sigma)}}.$$

For a nonincreasing G the difference quotient is nonnegative, so $v_g \geq 0$. If $g > 0$ is nonincreasing, g/t is decreasing, and $\beta_g > 0$ is clear. □

9. POINTWISE BOUNDS WITHOUT ZEROS

The sum rule bounds a mean. For zero-free systems with an affine phase on the band there is a stronger statement: one explicit system has the least attenuation at every frequency of the band at once. The proof writes every admissible h as the function h_g of Proposition 8.4 plus terms, one for each unit of measure off the band, each of which raises the attenuation at every point of the band.[20]

Throughout, $\Sigma = (0,a)\cup(b,\infty)$ is the part of the positive axis off the band, and $\mu_0 = \mu(\{0\})$. Put

$$\tau_\circ = \frac{2K(k)}{\pi b} = \frac{\tau_\star}{\theta}, \qquad h_\circ = \frac{h_\star}{\theta}, \qquad A = \frac{\alpha_\star}{\theta},$$

so that $h_\circ$ is the extremal function of Theorem 8.2 for $\theta = 1$, with linear coefficient $\tau_\circ$ and attenuation $A(x) = \tau_\circ \int_a^x (m^2 - s^2)\,\mathrm{d}s/r(s)$ on the band.

9.1. One unit of measure off the band

For $t \in \Sigma$ and $a < x < b$ put

$$K_0(z) = \frac{R(z)}{abz}, \qquad K_t(z) = \frac{2zR(z)}{R(t)\,(t^2-z^2)},$$
$$k_0(x) = \frac{r(x)}{abx}, \qquad k_t(x) = \frac{2x\,r(x)}{R(t)\,(t^2-x^2)}.$$

By Lemma 8.1, $R(t)$ and $t^2 - x^2$ have the same sign for $t \in \Sigma$, so $k_0 > 0$ and $k_t > 0$ on (a,b).

Lemma 9.1. *K_0 and K_t, $t \in \Sigma$, are odd, analytic in $\mathbb{C}\setminus(S\cup\{0,\pm t\})$, real on $\mathbb{R}\setminus S$ away from their poles, and*

$$K_0(z) = \frac{z}{ab} - \frac{1}{z} + \frac{1}{\pi}\int_a^b \frac{2z\,k_0(s)}{s^2-z^2}\,\mathrm{d}s,$$
$$K_t(z) = -\frac{2z}{R(t)} + \frac{2z}{t^2-z^2} + \frac{1}{\pi}\int_a^b \frac{2z\,k_t(s)}{s^2-z^2}\,\mathrm{d}s.$$

On the band $\operatorname{Re} K_0(x+i0) = \operatorname{Re} K_t(x+i0) = 0$, $\operatorname{Im} K_0(x+i0) = k_0(x)$ *and* $\operatorname{Im} K_t(x+i0) = k_t(x)$.

Proof. The parity and reality follow from Lemma 8.1, and the boundary values on the band from $R(x+i0) = ir(x)$. Call D the difference of the two sides of either identity. At infinity $R(z) = z^2 + O(1)$, so $K_0 = z/(ab) + O(z^{-1})$ and $K_t = -2z/R(t) + O(z^{-1})$, and the Cauchy integrals are $O(z^{-1})$; hence $D = O(z^{-1})$. At $z = 0$, $R(0) = -ab$, so $K_0 = -1/z + O(z)$. At $z = \pm t$ the residue of K_t is -1, as is that of $2z/(t^2-z^2)$. So D is analytic off S. The Cauchy integral of the density k/π, $k = k_0$ or k_t, which is Hölder continuous inside the band and of order $|s-c|^{1/2}$ at its ends c, has the boundary imaginary part k on (a,b) by the Plemelj formula, as in the proof of Proposition 8.4(ii), and is continuous up to S [29, Ch. 4]. Both sides of each identity have imaginary part k on the upper side of the band, so D is real there, and by the symmetry $D(\bar z) = \overline{D(z)}$ it takes the same values from below. So D has no jump across the open band, and by oddness none across $(-b,-a)$; it is bounded near $\pm a, \pm b$, hence entire, and it tends to 0. Thus $D = 0$. □

9.2. The reconstruction

Theorem 9.2. *Let $H = \sigma e^{ih}$ be a zero-free passive system, regular on the band, whose phase $g = \varphi$ is continuously differentiable on $[a,b]$. Then $\int_\Sigma \mathrm{d}\mu(t)/|R(t)| < \infty$ and*

$$h = h_g + \mu_0 K_0 + \int_\Sigma K_t\,\mathrm{d}\mu(t). \tag{9.1}$$

Consequently, for $a < x < b$,

$$\alpha(x) = v_g(x) + \mu_0\,k_0(x) + \int_\Sigma k_t(x)\,\mathrm{d}\mu(t), \tag{9.2}$$
$$\beta = \beta_g + \frac{\mu_0}{ab} - \int_\Sigma \frac{2\,\mathrm{d}\mu(t)}{R(t)}. \tag{9.3}$$

The first identity says that the attenuation on the band is the attenuation v_g forced by the phase alone, plus a strictly positive contribution of every unit of measure off the band. The second shows the price of that measure in another currency: mass at frequencies below the band, where $R < 0$, raises the linear coefficient, and mass above the band lowers it.

Proof. Finiteness. For $t \to \infty$, $|R(t)| \sim t^2$, and $\int \mathrm{d}\mu/(1+t^2) < \infty$. Near a and b the measure has a bounded density by Lemma 4.4, and $|R|^{-1}$ has singularities of order $|t-a|^{-1/2}$, $|t-b|^{-1/2}$. Near 0, $|R| \geq ab/2$.

The right side. Call it $\hat h$. Since $|K_t(z)|$ is bounded by a constant times $|R(t)|^{-1}$ for $t \in \Sigma$ and z in a compact set at positive distance from $S\cup\{0\}\cup\operatorname{supp}\mu$, and for x in a compact subset of (a,b) the same holds near $x + i0$, the integral converges locally uniformly there; $\hat h$ is analytic off $S\cup\{0\}\cup\operatorname{supp}\mu\cup -\operatorname{supp}\mu$ and continuous up to the open band, where by Proposition 8.4(ii) and Lemma 9.1 its real part is g. Integrating the identities of Lemma 9.1 against

[20] The positive reconstruction of Theorem 9.2 and the pointwise comparisons Theorems 9.4, 9.6 and 9.8 were found with OpenAI's ChatGPT, starting from the sum rule of Section 6. The uniqueness step below, which reuses the argument of Proposition 8.4(iv), is ours.

μ, by Fubini's theorem since every term is absolutely integrable,

$$\hat{h}(z) = \hat{\beta} z - \frac{\mu_0}{z} + \int_\Sigma \frac{2z}{t^2 - z^2}\,\mathrm{d}\mu(t) + \frac{1}{\pi}\int_a^b \frac{2z\,\hat{v}(s)}{s^2 - z^2}\,\mathrm{d}s,$$

with $\hat{\beta}$ the right side of (9.3) and $\hat{v}$ that of (9.2). Here we used that $h_g = \beta_g z + \frac{1}{\pi}\int_a^b 2z v_g(s)\,\mathrm{d}s/(s^2 - z^2)$: the difference of the two sides is odd, analytic off S, $O(z^{-1})$ at infinity, continuous up to S, and has no jump across the band, so it is entire and vanishes.

$\hat{v}$ is bounded. v_g is continuous on $[a, b]$ by Proposition 8.4(i). In $\hat{v}(x)$ only the mass of μ near the band edges can produce a singularity. For $x = a + \varepsilon$ and $t = a - u$ with $0 < u < \delta$, $k_t(x)$ is at most a constant times $\sqrt{\varepsilon}/(\sqrt{u}\,(u + \varepsilon))$, and the density of μ is bounded there, while

$$\sqrt{\varepsilon}\int_0^\delta \frac{\mathrm{d}u}{\sqrt{u}\,(u + \varepsilon)} \le \pi;$$

the edge b is the same. The rest of μ contributes a bounded amount by the finiteness above.

Uniqueness. By (4.4) and Lemma 4.4, $h = \beta z - \mu_0/z + \int_\Sigma 2z\,\mathrm{d}\mu/(t^2 - z^2) + \frac{1}{\pi}\int_a^b 2z\alpha(s)\,\mathrm{d}s/(s^2 - z^2)$. So

$$q = h - \hat{h} = (\beta - \hat{\beta})z + \frac{1}{\pi}\int_a^b \frac{2z\,(\alpha - \hat{v})(s)}{s^2 - z^2}\,\mathrm{d}s,$$

which is analytic off S, odd, $O(z)$ at infinity, and $O(\log|z - c|)$ near each end c of S, because $\alpha - \hat{v}$ is bounded. Both h and $\hat{h}$ are continuous up to the open band with real part g, so, exactly as in the proof of Proposition 8.4(iv), q/R has the same boundary values from both sides of (a, b) and of $(-b, -a)$. It is therefore analytic across the open cuts, its singularities at $\pm a, \pm b$ are $o(|z - c|^{-1})$ and hence removable, and q/R is entire. At infinity $q/R = O(z^{-1})$. Hence $q = 0$, which is (9.1); comparing linear coefficients and densities gives $\beta = \hat{\beta}$ and $\alpha = \hat{v}$ almost everywhere, hence everywhere by continuity. □

Corollary 9.3. *Under the hypotheses of Theorem 9.2, for $a < x < b$,*

$$\alpha(x) \;\ge\; v_g(x) + \frac{r(x)}{x}\,\max\{0, -\beta_g\}.$$

Proof. For $0 < t < a < x$, $x/(x^2 - t^2) > 1/x$, so $k_t(x) > 2r(x)/(x\,|R(t)|)$. Put

$$C_{\mathrm{in}} = \frac{\mu_0}{ab} + \int_{(0,a)} \frac{2\,\mathrm{d}\mu(t)}{|R(t)|}, \qquad C_{\mathrm{out}} = \int_{(b,\infty)} \frac{2\,\mathrm{d}\mu(t)}{R(t)},$$

both nonnegative. By (9.2), $\alpha(x) \ge v_g(x) + \frac{r(x)}{x}C_{\mathrm{in}} + \int_{(b,\infty)} k_t(x)\,\mathrm{d}\mu(t)$, and by (9.3) and $\beta \ge 0$, $C_{\mathrm{in}} \ge C_{\mathrm{out}} - \beta_g \ge -\beta_g$. □

9.3. Affine phase

For $g(t) = t$, Proposition 8.4 gives $\beta_g = \frac{2}{\pi}\int_a^b t\,\mathrm{d}t/r(t) = 1$ and $v_g = 0$, by the substitution $\sigma = t^2$ and Glauert's integral (proof of Lemma 8.5); hence $h_g = z$. By linearity and Proposition 8.4(iv), the phase $\theta + \tau\omega$ has

$$h_g = \theta h_\circ + \tau z, \qquad v_g = \theta A, \qquad \beta_g = \theta\tau_\circ + \tau. \tag{9.4}$$

The function $R(z)/z = abK_0(z)$ has, by Lemma 9.1, linear coefficient 1, an atom ab at the origin and band attenuation $r(x)/x$.

Theorem 9.4. *Let $\theta \ge 0$, $\tau \in \mathbb{R}$, and $C = \max\{0, -\theta\tau_\circ - \tau\}$. Every zero-free passive system, regular on the band, with phase $\theta + \tau\omega$ on (a, b) satisfies*

$$\alpha(x) \;\ge\; \theta A(x) + C\,\frac{r(x)}{x} \qquad (a \le x \le b). \tag{9.5}$$

Equality holds throughout the band for

$$h_{\mathrm{opt}}(z) = \theta h_\circ(z) + \tau z + C\,\frac{R(z)}{z},$$

and equality at a single point of (a, b) forces $h = h_{\mathrm{opt}}$.

Proof. The bound is Corollary 9.3 with (9.4), extended to the ends by continuity. By Lemma 9.1 and (9.4), h_{opt} has the form (4.4) with linear coefficient $\theta\tau_\circ + \tau + C \ge 0$, an atom $Cab \ge 0$ at the origin, and the measure $\pi^{-1}(\theta A + Cr/x)\,\mathrm{d}x \ge 0$ on the band and nothing else. So it is a Herglotz function with the symmetry of (4.3), continuous up to a neighbourhood of $[a, b]$, and $e^{ih_{\mathrm{opt}}}$ is admissible with phase $\theta + \tau\omega$ and attenuation $\theta A + Cr/x$.

Suppose equality at $x_0 \in (a, b)$. In the proof of Corollary 9.3 each inequality is then an equality at x_0. Since $k_t(x_0) > 0$ for $t > b$, μ vanishes on (b, ∞); since the comparison for $0 < t < a$ is strict, μ vanishes on $(0, a)$; and $C_{\mathrm{in}} = \mu_0/(ab)$ equals C. By (9.1), $h = \theta h_\circ + \tau z + CabK_0 = h_{\mathrm{opt}}$. □

For $\tau \ge -\theta\tau_\circ$, in particular for a pure phase lag, the optimum is $e^{i\tau z}H_\star$, and an additional delay costs nothing. A negative slope beyond $-\theta\tau_\circ = -\tau_\star$ would make the linear coefficient negative; the cheapest compensation, at every frequency of the band at once, is then a pole of h at zero frequency.

9.4. Constant phase lead

Lemma 9.5. *$h_\circ(-ab/z) = -h_\circ(z) + \tau_\circ R(z)/z$. Consequently*

$$A(ab/x) = \tau_\circ\frac{r(x)}{x} - A(x) > 0 \qquad (a < x < b).$$

Proof. The function $f(z) = h_\circ(-ab/z)$ is a Herglotz function with the symmetry of (4.3), regular on the band, as in the proof of Theorem 6.6; its phase on the band

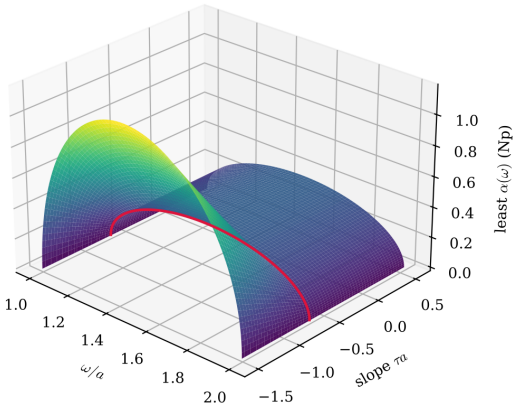


FIG. 9. The least attenuation $A(\omega)+\max\{0,-\tau_\circ-\tau\}\,r(\omega)/\omega$ of Theorem 9.4 at each frequency of an octave, for the phase $1+\tau\omega$, as a function of the slope τ. For $\tau\ge-\tau_\circ$ (red curve at $\tau=-\tau_\circ=-0.6864/a$) it does not depend on τ; below it the pole at zero frequency adds $r(\omega)/\omega$ per unit of slope.

is -1 and its attenuation is $A(ab/x)$. Since $h_\circ$ is analytic at 0 with $h_\circ(0)=0$, f has linear coefficient 0; since $h_\circ(w)=\tau_\circ w+O(w^{-1})$, $f(z)=-\tau_\circ ab/z+O(z)$ near 0, an atom $\tau_\circ ab$; and f is real on Σ, so μ has no other mass there. Theorem 9.2 with $g=-1$, for which $h_g=-h_\circ$, gives $f=-h_\circ+\tau_\circ abK_0$. Taking imaginary parts on the band gives the second identity; $A(ab/x)>0$ by Theorem 8.2(ii). □

Theorem 9.6. *Let $s>0$ and $\tau\le0$. Every zero-free passive system, regular on the band, with phase $-s+\tau\omega$ on (a,b) satisfies*

$$\alpha(x)\ \ge\ s\,A(ab/x)-\tau\,\frac{r(x)}{x}\qquad(a\le x\le b),$$

with equality throughout the band for $h=-sh_\circ+\tau z+(s\tau_\circ-\tau)R/z$, and equality at one point of (a,b) forces this h. For $\tau=0$ the optimum is $H_\star(-ab/z)$ with $H_\star$ the extremal system for $\theta=s$.

Proof. Here $v_g=-sA$ and $\beta_g=-s\tau_\circ+\tau<0$, so Corollary 9.3 and Lemma 9.5 give $\alpha\ge-sA+(s\tau_\circ-\tau)r/x=sA(ab/x)-\tau r/x$. The stated h has linear coefficient 0, an atom $(s\tau_\circ-\tau)ab>0$ at the origin and band density $\pi^{-1}(sA(ab/x)-\tau r/x)>0$, so it is admissible. Equality at one point forces h as in the proof of Theorem 9.4. For $\tau=0$, Lemma 9.5 identifies h with $sh_\circ(-ab/z)$. □

9.5. Consequences

Corollary 9.7. *(i) For phase $\theta+\tau\omega$ with $\theta\ge0$, $\tau\in\mathbb{R}$, or with $\theta<0$, $\tau\le0$, the optimal system of Theorems 9.4 and 9.6 minimises $\int_a^b w\,\alpha$ for every weight $w\ge0$, every L^p norm of α on the band, $1\le p\le\infty$, and every functional of α that increases with α; for $\int w\alpha$ with $w\ne0$ it is the only minimiser.*

(ii) For phase $\theta\ge0$, the least value of $\max_{[a,b]}\alpha$ is $\theta A(m)=\alpha_\star(m)$; for phase $-s<0$ it is $sA(m)$, attained at $x=ab/m$.

(iii) For phase $-s<0$ the least mean attenuation over the band is $s\Lambda_-$, with

$$\Lambda_-=\frac{E(\kappa)-\kappa'^2K(\kappa)}{\kappa},\qquad\kappa'^2=1-\kappa^2.$$

(iv) For phase $\theta+\tau\omega$ with $\theta\ge0$ and $\tau<-\tau_\star$, the least mean attenuation is

$$\theta\Lambda+\frac{\pi}{4}\,(b-a)\,(-\tau-\tau_\star).$$

Proof. (i) follows from the pointwise bounds and their equality cases; a weight $w\ge0$ that is not almost everywhere zero is positive on a set of positive measure, where any other admissible system has strictly larger attenuation, by the equality cases and continuity. (ii) A is largest at m (Theorem 8.2(ii)). (iii) is the value computed in Remark 6.8 for $H_\star(-ab/z)$. (iv) Integrate (9.5) with $C=-\tau-\tau_\star$, using

$$\int_a^b\frac{r(x)}{x}\,\mathrm{d}x=\frac{\pi}{4}(b-a)^2,$$

which follows from $\sigma=x^2$, $A=a^2$, $B=b^2$ and the substitution $\sigma=c-d\cos\phi$, $c=\frac{A+B}{2}$, $d=\frac{B-A}{2}$:

$$\begin{aligned}\int_A^B\frac{\sqrt{(\sigma-A)(B-\sigma)}}{\sigma}\,\mathrm{d}\sigma&=d^2\int_0^\pi\frac{\sin^2\phi\,\mathrm{d}\phi}{c-d\cos\phi}\\&=\pi\big(c-\sqrt{c^2-d^2}\big)\\&=\frac{\pi}{2}\big(\sqrt{B}-\sqrt{A}\big)^2,\end{aligned}$$

since $c^2-d^2=AB$. □

For an octave $A(m)=0.34824$, so the least worst-frequency attenuation for a lag θ is $1.2731\,\theta\Lambda$, against the mean $\theta\Lambda=0.27353\,\theta$; and $\Lambda_-=0.26560$. So the sharp worst-frequency bound for a lag and the sharp plain mean for a lead are both attained by the extremal system or by its inversion.

9.6. A prescribed argument

Theorem 9.8. *Let $0<\vartheta<\pi$. Among zero-free passive systems regular on the band with $\arg H\equiv\vartheta$ modulo 2π on $[a,b]$, the least mean attenuation is $\min\{\vartheta\Lambda,(\pi-\vartheta)\Lambda_-\}$ and the least value of $\max_{[a,b]}\alpha$ is $\min\{\vartheta,\pi-\vartheta\}A(m)$.*

Proof. By Remark 6.3 the phase is a constant $\theta'\equiv\vartheta\pmod\pi$. If $\theta'>0$ then $\theta'\ge\vartheta$, and by Theorem 6.2 and Corollary 9.7(ii) both costs are at least $\vartheta\Lambda$ and $\vartheta A(m)$, with equality for $H_\star$ with $\theta=\vartheta$. If $\theta'<0$ then $|\theta'|\ge\pi-\vartheta$, and Corollary 9.7(ii), (iii) give $(\pi-\vartheta)\Lambda_-$ and $(\pi-\vartheta)A(m)$, attained by $-H_\star(-ab/z)$ with $\theta=\pi-\vartheta$. □

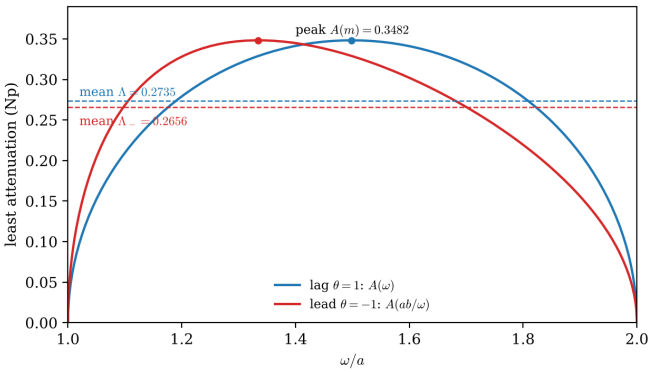


FIG. 10. The least attenuation at each frequency of an octave for a unit lag (Theorem 9.4) and a unit lead (Theorem 9.6). The lead profile is the lag profile reflected by $\omega \mapsto ab/\omega$; the two have the same peak $A(m)$, at m and at ab/m, and different means.

So the mean is least on the lagging branch for $\vartheta < \vartheta_c = \pi\Lambda_-/(\Lambda + \Lambda_-)$ and on the leading branch beyond it, while the worst frequency changes branch at $\pi/2$. For an octave $\vartheta_c = 1.5477$, just below $\pi/2$: the two costs prefer different systems for $\vartheta_c < \vartheta < \pi/2$. Theorem 9.8 sharpens Corollary 6.7, whose constant Λ becomes $A(m)$.

10. SYSTEMS WITH ZEROS

The main theorem concerns zero-free systems. A passive system may also have zeros in $\mathbb{C}_+$, and a zero contributes phase without contributing loss on the real axis. This section shows that zeros do help, and by how much one zero helps.

10.1. The bound with an inner factor

Proposition 10.1. *Let $H = BG$ be passive, where B is a real-symmetric Blaschke product whose zeros stay away from $I \cup (-I)$ and G is zero-free and regular on the band. Let φ be a continuous argument of H on $[a,b]$ and let ψ be the continuous argument of B on $[a,b]$ for which $\varphi - \psi$ is the phase of G in the sense of Lemma 4.2. Then $\alpha = \alpha_G$ on the band, and*

$$\begin{aligned}\int_a^b \alpha \,\mathrm{d}\omega &= \frac{1}{2}\int_a^b (\varphi - \psi)\, p \,\mathrm{d}\omega + \frac{\pi}{4}\int_{\mathbb{R}\setminus S} W \,\mathrm{d}\mu_G \\ &\geq \frac{1}{2}\int_a^b (\varphi - \psi)\, p \,\mathrm{d}\omega.\end{aligned} \tag{10.1}$$

The function ψ is real-analytic and strictly increasing on $[a,b]$.

Proof. The Blaschke product is analytic across I and unimodular there, so $\log|H| = \log|G|$ on the band, and $\arg H = \arg B + \arg G$ modulo 2π. The continuous arguments on $[a,b]$ therefore differ by a constant in $2\pi\mathbb{Z}$, which fixes ψ. Theorem 6.1 applied to G gives (10.1). The monotonicity of ψ is the computation in the proof of Proposition 4.5.[21] □

An increasing ψ behaves like a delay distributed over frequency. A pure delay is free (Lemma 5.3(i)), but a phase that rises steeply at low frequencies and flattens on the band is not a delay: its weighted integral $\int \psi p$ is positive, and (10.1) then allows less loss than (6.5). The simplest instance is a single zero on the imaginary axis.

10.2. One zero

For $\eta > 0$ let

$$B_\eta(z) = \frac{z - i\eta}{z + i\eta}.$$

It is passive, unimodular on $\mathbb{R}$, has the real symmetry, and the argument of $-B_\eta$ on $(0,\infty)$ is

$$\arg\big(-B_\eta(\omega)\big) = \pi - 2\arctan\frac{\eta}{\omega} = 2\arctan\frac{\omega}{\eta} \in (0,\pi). \tag{10.2}$$

For $0 < \theta < \pi$ put $\eta_\theta = b\cot(\theta/2)$, so that $2\arctan(b/\eta_\theta) = \theta$.

Proposition 10.2. *Let $0 < \theta < \pi$ and $\eta \geq \eta_\theta$, and put*

$$g_\eta(\omega) = \theta - 2\arctan\frac{\omega}{\eta}, \qquad H_\eta = -B_\eta\, e^{ih_{g_\eta}},$$

with h_g as in Proposition 8.4. Then H_η is passive and regular on the band, has exactly one zero in $\mathbb{C}_+$, at $i\eta$, has argument θ on $[a,b]$, and

$$\begin{aligned}\frac{1}{b-a}\int_a^b \alpha_\eta \,\mathrm{d}\omega &= \theta\Lambda - \delta(\eta), \\ \delta(\eta) &= \frac{1}{b-a}\int_a^b \arctan\frac{\omega}{\eta}\, p(\omega)\,\mathrm{d}\omega.\end{aligned} \tag{10.3}$$

The reduction δ is strictly positive and strictly decreasing in η, and

$$\delta(\eta) = \frac{\pi(b^2-a^2)^2}{24(b-a)\eta^3} + O(\eta^{-5}) \qquad (\eta \to \infty). \tag{10.4}$$

Proof. On $[a,b]$ the function g_η is decreasing, and $g_\eta(b) \geq 0$ because $\eta \geq \eta_\theta$. Hence $g_\eta \geq 0$, $g_\eta > 0$ on $[a,b)$, and $g_\eta(t)/t$ is decreasing. By Lemma 8.5, $v_{g_\eta} \geq 0$ and $\beta_{g_\eta} > 0$, so by Proposition 8.4(iii) $e^{ih_{g_\eta}}$ is a zero-free passive system, regular on the band, with phase g_η and attenuation v_{g_η}, and with no measure outside S. Multiplying by $-B_\eta$

[21] For a single factor $(z-\zeta)/(z-\bar\zeta)$ with $\zeta = \xi + i\eta$, $\eta > 0$, the argument on the real line is $\arg(x-\zeta) - \arg(x-\bar\zeta)$, and $\frac{\mathrm{d}}{\mathrm{d}x}\arg(x - \xi \mp i\eta) = \pm\eta/((x-\xi)^2 + \eta^2)$; the difference is $2\eta/|x-\zeta|^2$.

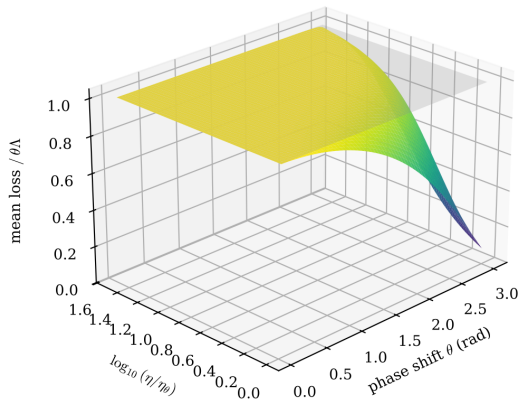


FIG. 11. The mean attenuation of the one-zero system H_η of Proposition 10.2, divided by the zero-free bound $\theta\Lambda$, for $a = 1$, $b = 2$, over the phase θ and $\log_{10}(\eta/\eta_\theta)$. The grey plane is the level 1. The ratio is below 1 everywhere and tends to 1 as the zero moves to infinity.

keeps passivity and the real symmetry, adds the single zero $i\eta$, leaves the modulus on $\mathbb{R}$ unchanged, and adds the argument (10.2). The argument of H_η on the band is therefore $g_\eta + 2\arctan(\omega/\eta) = \theta$, and Proposition 8.4(iii) gives

$$\int_a^b \alpha_\eta = \frac{1}{2}\int_a^b g_\eta\, p = \frac{\theta}{2}J - \int_a^b \arctan\frac{\omega}{\eta}\, p\,\mathrm{d}\omega,$$

which is (10.3) since $J = 2(b-a)\Lambda$.

For the sign, write $\arctan(x/\eta) = x\, f(x^2)$ with $f(\sigma) = \arctan(\sqrt{\sigma}/\eta)/\sqrt{\sigma}$. Since $u \mapsto \arctan(u)/u$ is strictly decreasing on $(0,\infty)$,[22] f is strictly decreasing, and Lemma 5.3(ii) gives $\delta(\eta) > 0$. Similarly

$$\delta'(\eta) = -\frac{1}{b-a}\int_a^b \frac{x\,p(x)}{\eta^2 + x^2}\,\mathrm{d}x < 0,$$

by Lemma 5.3(ii) with the strictly decreasing $f(\sigma) = 1/(\eta^2+\sigma)$. Finally $\arctan u = u - u^3/3 + O(u^5)$ uniformly on $[0, b/\eta]$, and Lemma 5.3(i),(iv) give $\int_a^b x\,p = 0$ and $\int_a^b x^3 p = -\frac{\pi}{8}(b^2-a^2)^2$, which yields (10.4). □

Since δ decreases, the best choice within the family is $\eta = \eta_\theta$, where the phase g_η of the zero-free factor reaches zero at the upper band edge. Figure 11 shows the mean loss of H_η, relative to $\theta\Lambda$, over θ and η/η_θ.

[22] Its derivative is $\big(u/(1+u^2) - \arctan u\big)/u^2$, and $\arctan u > u/(1+u^2)$ for $u > 0$ because both vanish at 0 and $\frac{\mathrm{d}}{\mathrm{d}u}\arctan u = \frac{1}{1+u^2} > \frac{1-u^2}{(1+u^2)^2} = \frac{\mathrm{d}}{\mathrm{d}u}\frac{u}{1+u^2}$.

TABLE III. Mean attenuation over the octave $[1,2]$ of three systems with argument θ modulo 2π: the lag extremal $H_\star$ (the least value among zero-free systems with phase θ), the lead extremal $-H_\star(-ab/z)$ built for $\pi - \theta$, and the one-zero system H_{η_θ}. For $\theta < \pi/3$ the first column is the least value among all zero-free systems (Corollary 10.3).

θ	$\theta\Lambda$	$(\pi-\theta)\,0.26560$	H_{η_θ}
0.5	0.13677	0.70160	0.13443
1	0.27353	0.56880	0.25394
$\pi/2$	0.42966	0.41720	0.34659
2	0.54707	0.30320	0.35977
2.5	0.68383	0.17040	0.27677

10.3. One zero beats every zero-free system

The statement of Theorem 6.2 fixes the continuous phase φ. A measurement only gives the argument modulo 2π, and a zero-free system with $\arg H \equiv \theta$ may realise θ through a lead $\theta - \pi$, by Remark 6.3. The next corollary takes this into account.

Corollary 10.3. *Let $0 < \theta < a\pi/(a+b)$. Among zero-free passive systems regular on the band with $\arg H \equiv \theta$ modulo 2π on $[a,b]$, the least mean attenuation over the band is $\theta\Lambda$, attained by $H_\star$. The system H_{η_θ} of Proposition 10.2, which has one zero, has the same argument and a strictly smaller mean attenuation.*

Proof. By Remark 6.3 the phase of such a system is a constant θ' with $\theta' \equiv \theta \pmod{\pi}$. If $\theta' \geq \theta$, Theorem 6.2 gives mean attenuation at least $\theta'\Lambda \geq \theta\Lambda$. Otherwise $\theta' \leq \theta - \pi < 0$, and Theorem 6.6 with $\tau = 0$ bounds the ab/ω^2-weighted mean by $|\theta'|\,\Lambda \geq (\pi - \theta)\Lambda$. Since $ab/\omega^2 \leq b/a$ on $[a,b]$, the plain mean is at least $(a/b)(\pi-\theta)\Lambda$, which exceeds $\theta\Lambda$ exactly when $\theta < a\pi/(a+b)$. The value $\theta\Lambda$ is attained by $H_\star$ (Theorem 8.2), and Proposition 10.2 gives the second statement. □

For larger θ the lead branch competes. The system $-H_\star(-ab/z)$, built from the extremal for $\pi - \theta$, is zero-free, has $\sigma = -1$ and phase $\theta - \pi$, hence argument θ modulo 2π, and by Remark 6.8 its mean attenuation is $(\pi-\theta)\,(E(\kappa) - \kappa'^2 K(\kappa))/\kappa$. Table III compares the three systems for an octave.

The one-zero system is the best of the three for $\theta \leq \pi/2$; by $\theta = 2$ the lead branch is better. For $\theta = 0.5$ the zero lowers the loss by 1.7%, for $\theta = 1$ by 7.2%.

10.4. The problem

Problem 10.4. Determine

$$\Lambda_{\mathrm{all}}(\theta) = \inf\Big\{\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega\Big\},$$

the infimum over all passive systems H, with or without zeros, that are continuous and nonzero near the band and

satisfy $\arg H \equiv \theta$ modulo 2π on $[a,b]$. Is $\Lambda_{\mathrm{all}}(\theta) > 0$ for $\theta \notin \pi\mathbb{Z}$?

Corollary 4.6 shows that the infimum is not attained by a lossless system, but it does not exclude that it is zero. By (10.1), the question is how large $\int_a^b \psi\, p$ can be made by the argument ψ of a Blaschke product, while the remaining phase $\theta - \psi$ is still the phase of a zero-free passive system. The difficulty is the second condition: by Proposition 8.4 it requires $v_g \geq 0$ and $\beta_g \geq 0$ for $g = \theta - \psi$, and $\beta_g \geq 0$ fails as soon as g is too negative on the band.[23] The next section solves the problem. The infimum is positive and attained, the minimiser has exactly one zero, and it is the least lossy system at every frequency of the band at once.

11. THE LEAST LOSS WITH ZEROS

This section solves Problem 10.4. The answer is stronger than the problem asks. For every phase $\theta \notin \pi\mathbb{Z}$ there is one passive system whose attenuation is the least possible at every frequency of the band simultaneously, among all passive systems with that argument, with or without zeros. It has exactly one zero, on the imaginary axis, and it is lossless off the band. For $\theta = \pi/2$ it is algebraic, and every quantity is elementary.

11.1. Statement

We call a passive system H *admissible for* θ if it is regular on the band in the sense of Definition 4.3 (the definition makes sense for any passive system) and $\arg H \equiv \theta \pmod{2\pi}$ on $[a,b]$. The admissible systems are exactly those of Problem 10.4. Since $-H$ is passive whenever H is, and $\arg(-H) = \arg H + \pi$, it suffices to treat $0 < \theta < \pi$. We use r, h_g, v_g, β_g, B_η and δ as in Sections 8 and 10.

Theorem 11.1. *Let $0 < \theta < \pi$. The equation*

$$\int_a^b \frac{2\arctan(t/\eta)}{r(t)}\,\mathrm{d}t = \theta \int_a^b \frac{\mathrm{d}t}{r(t)} \tag{11.1}$$

has exactly one root $\eta_\star \in (0, \eta_\theta)$. Put

$$g_\sharp(\omega) = \theta - 2\arctan\frac{\omega}{\eta_\star}, \qquad H_\theta^\sharp = -B_{\eta_\star} e^{ih_{g_\sharp}},$$
$$\alpha_\theta^\sharp = v_{g_\sharp} \quad \text{on } [a,b].$$

(i) $H_\theta^\sharp$ is admissible for θ. It has exactly one zero in $\mathbb{C}_+$, at $i\eta_\star$, it is lossless on $\mathbb{R}\setminus S$, it satisfies $H_\theta^\sharp(0) = 1$ and $H_\theta^\sharp(iy) \to -1$ as $y \to \infty$, and its attenuation $\alpha_\theta^\sharp$ is positive on (a,b) and vanishes at a and b.

(ii) Every system H admissible for θ satisfies

$$\alpha(\omega) \geq \alpha_\theta^\sharp(\omega) \qquad (a \leq \omega \leq b). \tag{11.2}$$

If equality holds on a set of positive measure, then $H = H_\theta^\sharp$.

(iii) Consequently

$$\Lambda_{\mathrm{all}}(\theta) = \frac{1}{b-a}\int_a^b \alpha_\theta^\sharp\,\mathrm{d}\omega = \theta\Lambda - \delta(\eta_\star) > 0, \tag{11.3}$$

the infimum is attained only by $H_\theta^\sharp$, and it is strictly smaller than the mean attenuation of every system of Proposition 10.2. For $\pi < \theta < 2\pi$, $\Lambda_{\mathrm{all}}(\theta) = \Lambda_{\mathrm{all}}(\theta - \pi)$, attained only by $-H_{\theta-\pi}^\sharp$.

The inequality (11.2) is pointwise. A single system is optimal for the mean, for every weighted mean, for the worst frequency and for the geometric mean at the same time (Corollary 11.8). The zero-free theory has the same feature for an affine phase (Theorem 9.4), although its sum rule bounds only the mean; there the constraint $\beta \geq 0$ ties the frequencies of the band together, and zeros remove it (Remark 11.5).

For a right angle everything is explicit.

Theorem 11.2. *Let $\theta = \pi/2$. Then $\eta_\star = \sqrt{ab}$, and*

$$H_{\pi/2}^\sharp(z) = -\frac{z^2 + ab}{R(z) + i(a+b)z}, \tag{11.4}$$

with R as in Lemma 8.1. Its attenuation is

$$\begin{aligned}\alpha_{\pi/2}^\sharp(\omega) &= \log\frac{(a+b)\,\omega + r(\omega)}{\omega^2 + ab} \\ &= 2\,\mathrm{artanh}\sqrt{\frac{(\omega-a)(b-\omega)}{(\omega+a)(\omega+b)}},\end{aligned} \tag{11.5}$$

and

$$\int_a^b \alpha_{\pi/2}^\sharp\,\mathrm{d}\omega = \frac{\pi}{2}\bigl(\sqrt{b}-\sqrt{a}\bigr)^2, \quad \Lambda_{\mathrm{all}}\Bigl(\frac{\pi}{2}\Bigr) = \frac{\pi}{2}\,\frac{\sqrt{b}-\sqrt{a}}{\sqrt{b}+\sqrt{a}}. \tag{11.6}$$

The profile $\alpha_{\pi/2}^\sharp$ is invariant under $\omega \mapsto ab/\omega$ and has its only maximum at the geometric centre $\omega_0 = \sqrt{ab}$, where $\alpha_{\pi/2}^\sharp(\omega_0) = \frac{1}{2}\log(b/a)$. Hence every passive system with $\arg H \equiv \pi/2 \pmod{\pi}$ on $[a,b]$ that is regular on the band satisfies

$$|H(\omega)| \leq \frac{\omega^2 + ab}{(a+b)\,\omega + r(\omega)}, \qquad \bigl|H(\sqrt{ab})\bigr|^2 \leq \frac{a}{b}. \tag{11.7}$$

For an octave, a passive system whose phase is a right angle at every frequency of the band transmits at most half of the power at the geometric centre: its loss there is at least 3.01 dB, and for a decade at least 10 dB.[24] The

[23] For instance, two zeros chosen so that $g < 0$ on the whole band give $\beta_g = \frac{2}{\pi}\int g/r < 0$; the resulting function e^{ih_g} grows like $e^{-\beta_g y}$ along $i\mathbb{R}_+$ and is not passive.

[24] The power transmission is $|H|^2 \leq a/b$, so the loss is at least $10\log_{10}(b/a)$ dB: $10\log_{10} 2 = 3.0103$ and $10\log_{10} 10 = 10$.

TABLE IV. The least mean attenuation over the octave $[1, 2]$ of passive systems with argument θ modulo 2π, with or without zeros (Theorem 11.1), the position of the zero of the optimal system relative to $\eta_\theta = b\cot(\theta/2)$ of Proposition 10.2, the ratio to the zero-free value $\theta\Lambda$, and the least possible worst-frequency attenuation $\max \alpha^\sharp_\theta$.

θ	$\eta_\star/\eta_\theta$	$\Lambda_{\mathrm{all}}(\theta)$	$\Lambda_{\mathrm{all}}/\theta\Lambda$	$\max \alpha^\sharp_\theta$
0.5	0.72576	0.13090	0.9571	0.16677
1	0.71853	0.22860	0.8357	0.29204
$\pi/2$	0.70711	0.26951	0.6272	0.34657
2	0.69843	0.24357	0.4452	0.31540
2.5	0.69050	0.15941	0.2331	0.20805

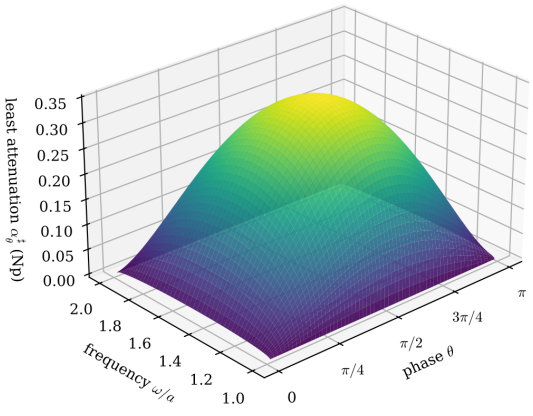


FIG. 12. The least attenuation $\alpha^\sharp_\theta(\omega)$ at each frequency of the octave, over the phase $\theta \in (0, \pi)$ (Theorem 11.1). No passive system with argument θ on the band lies below the surface at any frequency. The black curve is the closed form (11.5) at $\theta = \pi/2$, with its peak $\frac{1}{2}\log 2$ at $\omega = \sqrt{2}$.

sum rule of the zero-free theory gives no bound of this pointwise kind. The mean attenuation of $H^\sharp_{\pi/2}$ over an octave is $\frac{\pi}{2}(3 - 2\sqrt{2}) = 0.26951$ Np, which is 62.7% of the zero-free value $\frac{\pi}{2}\Lambda = 0.42966$. For a narrow band the ratio of the two minimal means tends to $\sin\theta/\theta$ (Corollary 11.9), which is $2/\pi$ at a right angle.

Table IV gives the octave values, and Figure 12 the optimal profiles. Comparing with Table III: the optimal system beats the one-zero system H_{η_θ} for every θ, by 2.6% at $\theta = 0.5$ and 22% at $\theta = \pi/2$, and for $\theta \geq 2$ it also beats the lead branch. Its zero sits closer to the origin than η_θ, so that the phase $g_\sharp$ of its zero-free factor is negative on part of the band, which Lemma 8.5 does not allow. At $\theta = \pi/2$ the ratio $\eta_\star/\eta_\theta$ is $\sqrt{a/b}$, by Theorem 11.2.

11.2. A change of variables

The phase condition $\arg H \equiv \theta$ is not linear in $\log H$ once zeros are allowed, since $\log H$ is then not defined. A

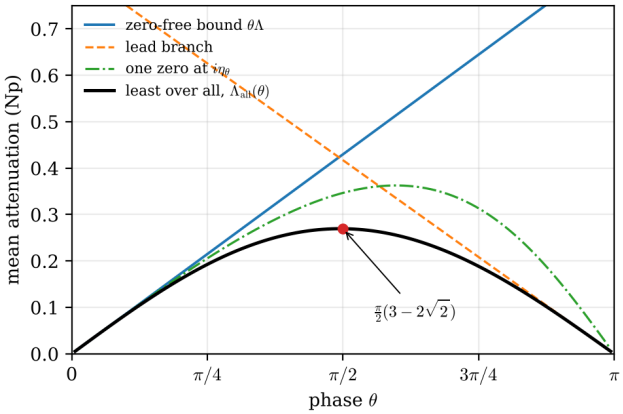


FIG. 13. Mean attenuation over the octave against the phase θ: the zero-free bound $\theta\Lambda$ (Theorem 6.2), the lead branch of Table III, the one-zero system H_{η_θ} (Proposition 10.2), and the exact minimum $\Lambda_{\mathrm{all}}(\theta)$ over all passive systems (Theorem 11.1). The dot marks the closed form $\frac{\pi}{2}(3 - 2\sqrt{2})$ at $\theta = \pi/2$.

Möbius transformation of the values removes this difficulty.

Lemma 11.3. *Let $0 < \theta < \pi$, let H be a passive system that is not constant, and put*

$$F = e^{-i\theta}\,\frac{H - e^{i\theta}}{H - e^{-i\theta}}, \qquad H = \frac{F - e^{i\theta}}{e^{i\theta}F - 1}. \tag{11.8}$$

(i) F is analytic in $\mathbb{C}_+$ with values in $\mathbb{C}_+$, and $F(-\bar z) = 1/\overline{F(z)}$. The function $\zeta = \log F$, with $0 < \operatorname{Im}\zeta < \pi$, satisfies $\zeta(-\bar z) = -\overline{\zeta(z)}$, and

$$\zeta(z) = \frac{1}{\pi}\int_0^\infty g(t)\,\frac{2z}{t^2 - z^2}\,\mathrm{d}t, \tag{11.9}$$

where $g(t) = \operatorname{Im}\zeta(t + i0) \in [0, \pi]$ for almost every $t > 0$. In particular $\operatorname{Re}\zeta = 0$ on $i\mathbb{R}_+$.

(ii) Conversely, if F is analytic in $\mathbb{C}_+$ with values in $\mathbb{C}_+$ and $F(-\bar z) = 1/\overline{F(z)}$, then (11.8) defines a passive system.

(iii) $|H| = 1$ at a point $t \in \mathbb{R}$ where H is continuous if and only if $F(t)$ is real there, and then $g(t) \in \{0, \pi\}$. The zeros of H are the points where $F = e^{i\theta}$.

(iv) Let $H(\omega) = e^{i\theta - \alpha}$ with $\alpha > 0$ at a point ω of the band, and write $\zeta = u + iv$ there. (If $\alpha = 0$, then $F = 0$.) Then

$$\frac{1}{F} = \cos\theta - i\sin\theta\coth\frac{\alpha}{2}, \qquad e^u\cos\theta = \cos v, \tag{11.10}$$

$u < 0$, v lies strictly between θ and $\pi/2$ (or $v = \pi/2$ if $\theta = \pi/2$), and

$$\alpha = A_\theta(u), \qquad A_\theta(u) = 2\operatorname{artanh}\frac{e^u\sin\theta}{\sqrt{1 - e^{2u}\cos^2\theta}}, \tag{11.11}$$

where A_θ is a strictly increasing bijection of $(-\infty,0)$ onto $(0,\infty)$. Conversely, a value $1/F$ of the form (11.10) with $\alpha>0$ corresponds to $H=e^{i\theta-\alpha}$.

Proof. (i) Since H is not constant, $|H|<1$ in $\mathbb{C}_+$ by the maximum principle, so $H\neq e^{\pm i\theta}$ there. The Möbius map $w\mapsto e^{-i\theta}(w-e^{i\theta})/(w-e^{-i\theta})$ sends the unit circle into the real line[25] and 0 to $e^{i\theta}\in\mathbb{C}_+$, so it maps the open disc onto $\mathbb{C}_+$. The symmetry $H(-\bar z)=\overline{H(z)}$ gives $F(-\bar z)=e^{-i\theta}(\bar H-e^{i\theta})/(\bar H-e^{-i\theta})=1/\overline{F(z)}$, hence the symmetry of ζ. The function ζ is a Herglotz function with bounded imaginary part, so in (4.1) $\beta=0$ and $\mathrm{d}\mu=\pi^{-1}g\,\mathrm{d}t$ with g the boundary values of $\operatorname{Im}\zeta$ [23, Ch. I]. On $i\mathbb{R}_+$ the symmetry reads $\zeta=-\bar\zeta$, so $\operatorname{Re}\zeta=0$ there and $c=\operatorname{Re}\zeta(i)=0$. The symmetry also makes g even, and folding the integral over $\mathbb{R}$ onto $(0,\infty)$ with $\frac{1}{t-z}-\frac{1}{-t-z}=\frac{2z}{t^2-z^2}$ gives (11.9); the terms $t/(1+t^2)$ cancel. (ii) The inverse Möbius map in (11.8) sends $\mathbb{C}_+$ into the open unit disc, and the symmetry of H follows by reversing the computation in (i). (iii) The Möbius map sends the unit circle onto the real line, and $H=0$ exactly when $F=e^{-i\theta}(-e^{i\theta})/(-e^{-i\theta})=e^{i\theta}$. (iv) Substituting $H=e^{i\theta-\alpha}$ in $1/F=e^{i\theta}(H-e^{-i\theta})/(H-e^{i\theta})$ gives

$$\frac{1}{F}=\frac{(e^{-\alpha}-1)\cos\theta+i(e^{-\alpha}+1)\sin\theta}{e^{-\alpha}-1},$$

which is the first identity of (11.10). Since $1/F=e^{-u}(\cos v-i\sin v)$, the real parts give $e^u\cos\theta=\cos v$, and the imaginary parts $\tanh(\alpha/2)=e^u\sin\theta/\sin v$. As $\coth(\alpha/2)>1$, $|1/F|>1$, so $u<0$, and then $|\cos v|<|\cos\theta|$ with the same sign, which places v between θ and $\pi/2$. Finally $\sin v=\sqrt{1-e^{2u}\cos^2\theta}$ gives (11.11). The quotient $e^u\sin\theta/\sqrt{1-e^{2u}\cos^2\theta}$ increases with u, tends to 0 as $u\to-\infty$ and to 1 as $u\to0$. □

So the band condition places the boundary value of ζ on the curve

$$\Gamma_\theta=\{u+iv:\ u<0,\ e^u\cos\theta=\cos v\},$$

with v between θ and $\pi/2$, and the attenuation is an increasing function of u alone. Off the band, g is free in $[0,\pi]$; a lossless stretch has $g\in\{0,\pi\}$. In the variable ζ the zeros of H have disappeared.

11.3. The comparison theorem

We use two facts about Hardy spaces of the unit disc $\mathbb{D}$ [24, Chs. 3, 4]. (a) If f is analytic in $\mathbb{D}$ and $|\arg f|\le\lambda<\pi/2$, then $f\in H^q$ for every $q<\pi/(2\lambda)$; this follows from the case $\lambda=\pi/2$, $q<1$, applied to $f^{\pi/(2\lambda)}$. (b) If $\operatorname{Im}f$ is bounded, then $f\in H^p$ for every $p<\infty$, by the theorem of M. Riesz. A function in H^1 is the Poisson integral of its boundary values, and so is its imaginary part. Through the Cayley map $w=(z-i)/(z+i)$ these statements apply to functions in $\mathbb{C}_+$, and for Y in H^1 of the disc they give

$$Y(z)=\operatorname{Re}Y(i)+\frac{1}{\pi}\int_{\mathbb{R}}\operatorname{Im}Y(t)\Big(\frac{1}{t-z}-\frac{t}{1+t^2}\Big)\,\mathrm{d}t,\tag{11.12}$$

with $\int|Y(t)|\,\mathrm{d}t/(1+t^2)<\infty$.[26]

Proposition 11.4. *Let $0<\theta<\pi$ and let $H^\sharp$ be admissible for θ, with*

$$g_\sharp=\pi\ \text{ on }(0,a),\qquad g_\sharp=0\ \text{ on }(b,\infty)\tag{11.13}$$

for the function $g_\sharp$ of Lemma 11.3 belonging to $H^\sharp$. Then every H admissible for θ satisfies $\alpha\ge\alpha^\sharp$ on $[a,b]$, where $\alpha^\sharp$ is the attenuation of $H^\sharp$. If $\alpha=\alpha^\sharp$ on a set of positive measure, then $H=H^\sharp$. In particular at most one admissible system satisfies (11.13).

Proof. Let $\zeta=u+iv$ and $\zeta_\sharp=u_\sharp+iv_\sharp$ be the functions of Lemma 11.3 for H and $H^\sharp$, and $d=\zeta-\zeta_\sharp$. The function F of H is continuous up to (a,b), so $u=\log|F|$ is continuous there with values in $[-\infty,0)$, and $u=-\infty$ exactly on the closed set Z where $H=e^{i\theta}$. Since $\operatorname{Im}\zeta$ is bounded, ζ has finite boundary values almost everywhere (fact (b) above), so Z has measure zero. The attenuation $\alpha^\sharp$ of $H^\sharp$ is positive on (a,b), since $u_\sharp$ is finite there, $H^\sharp$ being regular on the band. By Lemma 11.3(iv) it suffices to show $u\ge u_\sharp$ almost everywhere on (a,b): then, by continuity, everywhere, so that Z is empty and $\alpha=A_\theta(u)\ge A_\theta(u_\sharp)=\alpha^\sharp$.

Step 1: the chord. The curve Γ_θ is the graph of $V(u)=\arccos(e^u\cos\theta)$, and $V'(u)=-\cot V(u)$, where $\cot V$ has the sign $\epsilon=\operatorname{sign}\cos\theta$ and $|\cot V|\le|\cot\theta|$ because V lies between θ and $\pi/2$. By the mean value theorem, at each $x\in(a,b)\setminus Z$

$$d(x)=\big(u(x)-u_\sharp(x)\big)\big(1-i\,m(x)\big)$$

with $m(x)=\cot V(\xi)$ for some ξ between $u(x)$ and $u_\sharp(x)$ (and $m=0$ where $u=u_\sharp$). Put $\psi=\arctan m$. Then $|\psi|\le\lambda$, with $\lambda=|\pi/2-\theta|<\pi/2$, ψ is measurable, and $1-im=\sqrt{1+m^2}\,e^{-i\psi}$.

Step 2: a rotating factor. Define

$$\Xi(z)=\frac{1}{\pi}\int_a^b\psi(t)\,\frac{2t}{t^2-z^2}\,\mathrm{d}t,\qquad X=e^{\Xi}.$$

The density $\frac{1}{\pi}\psi(|t|)\operatorname{sign}t$ on S is bounded, so $\operatorname{Im}\Xi$ is its Poisson integral: $|\operatorname{Im}\Xi|\le\lambda$ in $\mathbb{C}_+$, $\operatorname{Im}\Xi=\psi$ almost everywhere on (a,b), $\operatorname{Im}\Xi=0$ on $\mathbb{R}\setminus S$, and $\Xi(-\bar z)=\overline{\Xi(z)}$.

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[25] For $w=e^{i\phi}$, $e^{i\phi}-e^{i\theta}=2ie^{i(\phi+\theta)/2}\sin\frac{\phi-\theta}{2}$ and $e^{i\phi}-e^{-i\theta}=2ie^{i(\phi-\theta)/2}\sin\frac{\phi+\theta}{2}$, so the image is $\sin\frac{\phi-\theta}{2}/\sin\frac{\phi+\theta}{2}\in\mathbb{R}$.

[26] At $z=i$ the integral is $\frac{i}{\pi}\int\operatorname{Im}Y(t)\,\mathrm{d}t/(1+t^2)$, which is purely imaginary, so the real constant is $\operatorname{Re}Y(i)$. The imaginary part of the kernel is the Poisson kernel $y/((t-x)^2+y^2)$.

Hence X is real and positive on $i\mathbb{R}_+$ and on $\mathbb{R}\setminus S$, $X\to 1$ at infinity, and $|\arg X|\le\lambda$. If $\theta=\pi/2$, then $X\equiv 1$.

Step 3: the product. Let $Y=Xd$. By (a), $X\in H^q$ for some $q>1$; by (b), $d\in H^p$ for every $p<\infty$, since $|\operatorname{Im} d|\le\pi$. By Hölder's inequality $Y\in H^1$. At almost every $x\in(a,b)$,

$$Y(x)=|X(x)|\sqrt{1+m(x)^2}\,\big(u(x)-u_\sharp(x)\big)\in\mathbb{R},\quad(11.14)$$

since the factor $e^{i\psi}$ of X cancels $e^{-i\psi}$. On $\mathbb{R}\setminus S$, $\operatorname{Im} Y=X\,(g-g_\sharp)$, which by (11.13) and $0\le g\le\pi$ is ≤ 0 on $(0,a)$ and ≥ 0 on (b,∞). Both X and d have the symmetries $X(-\bar z)=\overline{X(z)}$, $d(-\bar z)=-\overline{d(z)}$, so $Y(-\bar z)=-\overline{Y(z)}$; thus $\operatorname{Im} Y$ is even on $\mathbb{R}$ and Y is purely imaginary on $i\mathbb{R}_+$, so that $\operatorname{Re} Y(i)=0$.

Step 4: the sign. By (11.12), folded onto $(0,\infty)$ as in Lemma 11.3, and since $\operatorname{Im} Y=0$ on (a,b),

$$Y(z)=\frac{1}{\pi}\int_{(0,a)\cup(b,\infty)}\operatorname{Im} Y(t)\,\frac{2z}{t^2-z^2}\,\mathrm{d}t.$$

The right side is analytic across (a,b), so for $a<x<b$

$$Y(x)=\frac{1}{\pi}\int_0^a\big(-\operatorname{Im} Y(t)\big)\frac{2x}{x^2-t^2}\,\mathrm{d}t+\frac{1}{\pi}\int_b^\infty\operatorname{Im} Y(t)\,\frac{2x}{t^2-x^2}\,\mathrm{d}t\;\ge\;0,\quad(11.15)$$

both integrands being nonnegative. With (11.14) this gives $u\ge u_\sharp$ almost everywhere on (a,b).

Equality. If $\alpha=\alpha^\sharp$ on a set of positive measure, pick a point of it at which (11.14) holds. There $Y=0$, so both integrals in (11.15) vanish, and since $X>0$ and the kernels do not vanish, $g=g_\sharp$ almost everywhere off the band. Then $\operatorname{Im} Y=0$ almost everywhere on $\mathbb{R}$, so $Y\equiv 0$ by (11.12), hence $d\equiv 0$ and $H=H^\sharp$. □

Remark 11.5. For zero-free systems the analogue of ζ is ih, and the constraint $\operatorname{Re} h=\theta$ on the band is linear; there the coefficient $\beta\ge 0$ of the linear term of h acts as a single global constraint, which Corollary 9.3 turns into a pointwise price $r(x)/x$ per unit of $-\beta_g$. In the variable ζ the linear term is absent, because $\operatorname{Im}\zeta$ is bounded, and the band condition is a curve rather than a line. The proof above replaces linearity by the observation that the chords of Γ_θ have slopes in a fixed sector, which the factor X rotates to the real axis. The constraint $\beta\ge 0$ is what zeros remove: in the factorisation $H^\sharp_\theta=-B_{\eta_\star}e^{ih_{g_\sharp}}$ the zero-free factor has $\beta_{g_\sharp}=0$ (Lemma 11.7).

11.4. The right angle

Proof of Theorem 11.2. Let $Q(z)=(z+a)(z+b)$ and

$$\zeta_0(z)=\tfrac12\big(\log(z-a)+\log(z-b)-\log(z+a)-\log(z+b)\big)$$

with the arguments in $(0,\pi)$. For $z\in\mathbb{C}_+$ the angle $\arg(z-c)-\arg(z+c)$ lies in $(0,\pi)$ for $c=a,b$,[27] so $0<\operatorname{Im}\zeta_0<\pi$. On $(0,a)$, $\operatorname{Im}\zeta_0=\frac12(\pi+\pi)=\pi$; on (b,∞) it is 0; on (a,b) it is $\pi/2$, and $\zeta_0(-\bar z)=-\overline{\zeta_0(z)}$. So ζ_0 has the form (11.9), with $g_0=\pi$ on $(0,a)$, $\pi/2$ on (a,b), 0 on (b,∞). Since $e^{2\zeta_0}=(z-a)(z-b)/Q(z)=R^2/Q^2$ and both e^{ζ_0} and R/Q tend to 1 at infinity, $F_0=e^{\zeta_0}=R/Q$. By Lemma 11.3(ii), (11.8) with $\theta=\pi/2$ defines a passive system

$$K=\frac{F_0-i}{iF_0-1}=\frac{R-iQ}{iR-Q}=-\frac{z^2+ab}{R+i(a+b)z},$$

where the last step uses $R^2=(z^2-a^2)(z^2-b^2)$ and $(z^2-a^2)(z^2-b^2)+(a+b)^2z^2=(z^2+ab)^2$.[28] On the band $F_0=ir/Q$ is purely imaginary, so $\operatorname{Re}(1/F_0)=0=\cos(\pi/2)$ and, by (11.10), $K=e^{i\pi/2-\alpha}$ with $\tanh(\alpha/2)=e^u=r/Q$, which is the second form of (11.5). The first form is $\log\frac{1+s}{1-s}$ with $s=r/Q$, because $(Q+r)(\omega^2+ab)=(Q-r)\big((a+b)\omega+r\big)$ on the band.[29] The system K is regular on the band, with $K\to e^{i\pi/2}$ at the band edges, and its function g_0 satisfies (11.13). By Proposition 11.4, K is the unique admissible system with this property. It has exactly one zero, a simple one at $i\sqrt{ab}$, where $z^2+ab=0$ and $R+i(a+b)z=-2(a+b)\sqrt{ab}\ne 0$;[30] moreover $K(0)=-ab/R(0)=1$ and $K(iy)\to -1$. So Lemma 11.7 below, applied to K, gives $K=H^\sharp_{\pi/2}$ and $\eta_\star=\sqrt{ab}$. (Directly: the substitution $t=ab/s$ maps $\mathrm{d}t/r(t)$ to $\mathrm{d}s/r(s)$ and $\arctan(t/\sqrt{ab})$ to $\frac\pi2-\arctan(s/\sqrt{ab})$, so the left side of (11.1) at $\eta=\sqrt{ab}$ is $\frac\pi2\int\mathrm{d}t/r$.)

For (11.6), write $K=-B_{\sqrt{ab}}\,e^{ih}$, where $h=h_{g_\sharp}$. By Proposition 8.4(iii) and (4.4),

$$h(z)=\frac{1}{\pi}\int_a^b\alpha^\sharp_{\pi/2}(t)\,\frac{2z}{t^2-z^2}\,\mathrm{d}t,$$

so

$$\lim_{y\to\infty} iy\,h(iy)=-\frac{2}{\pi}\int_a^b\alpha^\sharp_{\pi/2}\,\mathrm{d}t.$$

On the other hand $R+i(a+b)z=z^2\big(1+i(a+b)/z+O(z^{-2})\big)$ and $-B_\eta(z)=-1+2i\eta/z+O(z^{-2})$, so

$$e^{ih}=\frac{K}{-B_{\sqrt{ab}}}=1-\frac{i\,(a+b-2\sqrt{ab})}{z}+O(z^{-2}),$$

---

[27] It is the angle at z of the triangle with vertices $-c$, c, z.

[28] Cross-multiplying, $(R-iQ)(R+i(a+b)z)=R^2+(a+b)zQ-iR(z^2+ab)$, since $(a+b)z-Q=-(z^2+ab)$, while $-(z^2+ab)(iR-Q)=(z^2+ab)Q-iR(z^2+ab)$, and $R^2=Q\,(z-a)(z-b)=Q(z^2+ab-(a+b)z)$.

[29] Write $c=(a+b)\omega$ and $m=\omega^2+ab$, so $Q=m+c$. The two sides are $m^2+cm+rm$ and $mc+mr+c^2-r^2$, and $c^2-m^2=(a^2+b^2)\omega^2-\omega^4-a^2b^2=r^2$.

[30] On $i\mathbb{R}_+$, $R(iy)=-\sqrt{(y^2+a^2)(y^2+b^2)}$, being real, nonvanishing and $\sim -y^2$.

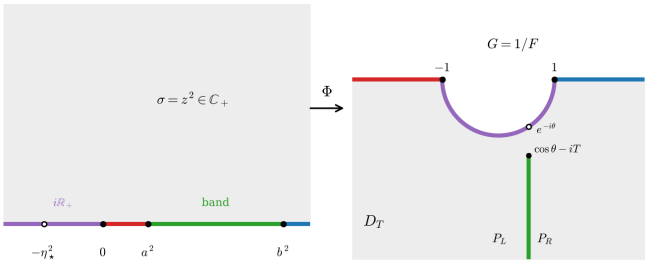


FIG. 14. The construction of Proposition 11.6. Left: the plane of $\sigma = z^2$, with the boundary arcs coloured. Right: the domain D_T of $G = 1/F$, the part of the lower half-plane outside the unit disc, cut along the vertical ray $\operatorname{Re} G = \cos\theta$ below the tip $\cos\theta - iT$. The map Φ sends each arc on the left to the arc of the same colour; the two ends of the band go to the two ends of the slit at infinity. The zero of the optimal system is the preimage of $e^{-i\theta}$.

whence $zh(z) \to -(\sqrt{b} - \sqrt{a})^2$, which is (11.6).

Finally, $q(\omega) = (\omega - a)(b - \omega)/((\omega + a)(\omega + b))$ satisfies $q(ab/\omega) = q(\omega)$, and $\log q$ is strictly concave on (a, b), since $(\log q)'' = \frac{1}{(\omega+a)^2} - \frac{1}{(\omega-a)^2} + \frac{1}{(\omega+b)^2} - \frac{1}{(b-\omega)^2} < 0$. So its only maximum is at the fixed point $\sqrt{ab}$, where $\sqrt{q} = (\sqrt{b} - \sqrt{a})/(\sqrt{b} + \sqrt{a})$ and $2\operatorname{artanh}\sqrt{q} = \log(\sqrt{b}/\sqrt{a})$. The bound (11.7) is $|H| = e^{-\alpha} \le e^{-\alpha^\sharp_{\pi/2}}$, from Theorem 11.1(ii), applied to H or $-H$. □

11.5. Existence for every phase

For $\theta \neq \pi/2$ the optimal system is not algebraic, and we construct it by a conformal map. In the variable $\sigma = z^2$, the first quadrant becomes $\mathbb{C}_+$, and the boundary conditions of Lemma 11.3 for a system with (11.13) become conditions on $G = 1/F$: G is real and below -1 on $(0, a^2)$, real and above 1 on (b^2, ∞), unimodular on $(-\infty, 0)$ (the imaginary axis of z, where $\operatorname{Re}\zeta = 0$), and $\operatorname{Re} G = \cos\theta$ on (a^2, b^2), by (11.10). This describes a slit domain (Figure 14).

Proposition 11.6. *Let $0 < \theta < \pi$. There is a system $H^\sharp$, admissible for θ, with (11.13), lossless on $\mathbb{R} \setminus S$, with exactly one zero in $\mathbb{C}_+$, which is simple and lies on $i\mathbb{R}_+$, and with $H^\sharp(0) = 1$, $H^\sharp(iy) \to -1$ as $y \to \infty$, and attenuation positive on (a, b) and zero at a and b.*

Proof. For $T > \sin\theta$ let $\Sigma_T = \{\cos\theta - is : \ s \ge T\}$ and

$$D_T = \{G : \ \operatorname{Im} G < 0, \ |G| > 1\} \setminus \Sigma_T.$$

The ray Σ_T does not meet the closed unit disc, since $|\cos\theta - is| > 1$ for $s > \sin\theta$. The domain D_T is simply connected and its boundary is locally connected, so a conformal map Φ of $\mathbb{C}_+$ onto D_T extends to a homeomorphism of $\mathbb{R} \cup \{\infty\}$ onto the prime ends of D_T [25, Ch. 2]. Traversed with D_T on the left, the boundary consists of the lower unit semicircle C from 1 to -1, the ray $(-\infty, -1]$, the left side of Σ_T up to the tip and its right side down again, and the ray $[1, \infty)$. It has two prime ends at infinity: P_L, between $(-\infty, -1]$ and the left side of the slit, and P_R, between the right side and $[1, \infty)$. We normalise Φ by $\Phi(\infty) = 1$, $\Phi(0) = -1$ and $\Phi(a^2) = P_L$, and let $s(T) > a^2$ be the preimage of P_R.

Choice of T. Let $\lambda(T)$ be the extremal distance in D_T between C and Σ_T [26, Ch. 4]. By conformal invariance it equals the extremal distance in $\mathbb{C}_+$ between $(-\infty, 0)$ and $[a^2, s(T)]$, which is $L(s(T)/a^2)$ for a continuous strictly decreasing bijection L of $(1, \infty)$ onto $(0, \infty)$.[31] The function λ is continuous in T: for T' near T a diffeomorphism $G \mapsto G - i(T' - T)\chi(G)$, with χ a smooth bump equal to 1 near the tip and supported in a disc around it that avoids the unit circle and the real axis, maps D_T onto $D_{T'}$ arc by arc, with dilatation $1 + O(|T' - T|)$, and extremal distances change at most by that factor.[32] As $T \to \infty$, every curve in D_T from C to Σ_T crosses the annulus $1 < |G| < T$, so $\lambda(T) \ge \frac{1}{2\pi}\log T \to \infty$. As $T \downarrow \sin\theta$, the tip $\cos\theta - iT$ lies at the distance $\delta = T - \sin\theta$ below the point $e^{-i\theta}$ of C, and for $\delta < \rho < \rho_0$ an arc of the circle $\left|G - e^{-i\theta}\right| = \rho$ of length less than $2\pi\rho$ joins C to Σ_T inside D_T; this gives $\lambda(T) \le 2\pi/\log(\rho_0/\delta) \to 0$. Hence $s(T)/a^2$ is continuous and takes every value in $(1, \infty)$, and we fix T with $s(T) = b^2$.

The system. On the open first quadrant put $F(z) = 1/\Phi(z^2)$, and on the open second quadrant $F(z) = 1/\overline{F(-\bar z)}$. On $i\mathbb{R}_+$, $z^2 < 0$ is mapped to C, where $|\Phi| = 1$, so the two definitions have the same continuous boundary values there, and F is analytic in $\mathbb{C}_+$ by Morera's theorem. F maps $\mathbb{C}_+$ into $\mathbb{C}_+$, since $\operatorname{Im}\Phi < 0$, and $F(-\bar z) = 1/\overline{F(z)}$. Define $H^\sharp$ from F by (11.8). By Lemma 11.3(ii) it is a passive system. On $(0, a)$, $\Phi(t^2) < -1$, so $F \in (-1, 0)$ and $g_\sharp = \pi$; on (b, ∞), $F \in (0, 1)$ and $g_\sharp = 0$; in both cases $\left|H^\sharp\right| = 1$. On (a, b), $\Phi(t^2) = \cos\theta - is$ with $s \ge T > \sin\theta$, which by (11.10) is $H = e^{i\theta - \alpha}$ with $\coth(\alpha/2) = s/\sin\theta$, so $\alpha > 0$, and $\alpha \to 0$ at the band edges, where $s \to \infty$, $F \to 0$ and $H^\sharp \to e^{i\theta}$. Hence $H^\sharp$ is continuous and nonzero on a neighbourhood of $[a, b]$, that is, regular on the band, with argument θ there. By Lemma 11.3(iii) its zeros are the points with $1/F = e^{-i\theta}$, a point of C; Φ takes this boundary value at exactly one point of $(-\infty, 0)$, and analytically,

[31] By scaling, $L(\rho)$ is the extremal distance between $(-\infty, 0)$ and $[1, \rho]$. It decreases because the curve family grows with ρ; it tends to ∞ as $\rho \to 1$, since every curve crosses the half-annulus $\rho - 1 < |\sigma - 1| < 1$, of extremal distance $\frac{1}{\pi}\log\frac{1}{\rho-1}$; and to 0 as $\rho \to \infty$, since the half-circles $|\sigma| = t$, $1 < t < \rho$, join the two sets and have length πt, which gives modulus at least $\frac{1}{\pi}\log\rho$. Strictness and continuity follow from the explicit value as a ratio of complete elliptic integrals.

[32] It moves the slit along itself and fixes the rest of the boundary; it is injective once $|T' - T|\,|\nabla\chi| < 1$. A diffeomorphism whose differential has ratio of singular values at most K changes an extremal distance by a factor in $[1/K, K]$, by transporting the admissible metrics [26, Ch. 4].

since C is an analytic arc, with nonzero derivative. So $H^\sharp$ has exactly one zero in $\mathbb{C}_+$, a simple one on $i\mathbb{R}_+$. Finally $\Phi(0) = -1$ gives $F(0) = -1$ and $H^\sharp(0) = 1$, and $\Phi(\sigma) \to 1$ as $\sigma \to -\infty$ gives $H^\sharp(iy) \to -1$. □

Lemma 11.7. *Let $H^\sharp$ be admissible for θ, with (11.13), lossless on $\mathbb{R} \setminus S$, with exactly one zero in $\mathbb{C}_+$, a simple one at $i\eta$, and with $H^\sharp(0) = 1$ and $H^\sharp(iy) \to -1$ as $y \to \infty$; for instance the system of Proposition 11.6. Then $H^\sharp = -B_\eta e^{ih_g}$ with $g(\omega) = \theta - 2\arctan(\omega/\eta)$, $\beta_g = 0$, and the attenuation of $H^\sharp$ on the band is v_g. Consequently η is the root $\eta_\star$ of (11.1).*

Proof. The quotient $E = H^\sharp/(-B_\eta)$ is analytic and zero-free in $\mathbb{C}_+$, unimodular where $H^\sharp$ is, and bounded, hence bounded by 1 by the maximum principle, since $|B_\eta| \to 1$ at $\mathbb{R} \cup \{\infty\}$. It has the real symmetry and $E(0) = H^\sharp(0)/(-B_\eta(0)) = 1$, so by Lemma 4.2 $E = e^{ih}$ with an odd Herglotz function h, $h(0) = 0$. On $\mathbb{R} \setminus S$, $|E| = 1$, so $\operatorname{Im} h = 0$ there: the measure of h is carried by S, and by (4.1), h being odd, $h(z) = \beta z + O(z^{-1})$ at infinity. Along $[0, a]$ the function F of $H^\sharp$ is negative, by (11.13), with $F(0) = -1$ and $F(a) = 0$, so the argument of $H^\sharp$ moves continuously from 0 to θ inside $(-\theta, \theta]$,[33] and that of $-B_\eta$ from 0 to $2\arctan(a/\eta)$ by (10.2). So on the band $\operatorname{Re} h = \theta - 2\arctan(\omega/\eta) = g$, with no multiple of 2π. The argument of Proposition 8.4(iv) now applies to $h - h_g$: it is real on $\mathbb{R} \setminus S$, has real part 0 on the band from above, is continuous up to S (since E is continuous and nonzero near the band), so $(h - h_g)/R$ is entire, and it is $O(z^{-1})$ at infinity. Hence $h = h_g$ and $\beta = \beta_g$. Finally $E(iy) \to (-1)/(-1) = 1$, so $h(iy) \to 0$ and $\beta = 0$. The attenuation is $\operatorname{Im} h_g = v_g$ on the band. The left side of (11.1) decreases strictly in η from $\pi \int \mathrm{d}t/r$ to 0, so it has exactly one root, and $\beta_g = 0$ says that η is this root. □

Proof of Theorem 11.1. The system $H_\theta^\sharp = -B_{\eta_\star} e^{ih_{g_\sharp}}$ is the system of Proposition 11.6, by Lemma 11.7. This gives (i), with $\alpha_\theta^\sharp = v_{g_\sharp} > 0$ on the open band. Since $g_\sharp \geq 0$ and decreasing for $\eta \geq \eta_\theta$, $\beta_{g_\eta} > 0$ there (Lemma 8.5); and β_{g_η} increases with η; so $\eta_\star < \eta_\theta$. Part (ii) is Proposition 11.4. For (iii), by Proposition 8.4(iii) with $v_{g_\sharp} \geq 0$ and $\beta_{g_\sharp} = 0$, $\int_a^b \alpha_\theta^\sharp = \frac{1}{2}\int_a^b g_\sharp\, p$, and the computation in the proof of Proposition 10.2 gives $\theta\Lambda - \delta(\eta_\star)$. It is positive since $\alpha_\theta^\sharp > 0$. The systems H_η of Proposition 10.2 are admissible and differ from $H_\theta^\sharp$, since $\beta_{g_\eta} > 0$, so by (ii) their mean is strictly larger. The case $\pi < \theta < 2\pi$ follows by passing to $-H$. □

11.6. Consequences

Corollary 11.8. *Let $0 < \theta < \pi$ and let H be admissible for θ.*

(i) For every measurable $w \geq 0$ on $[a, b]$, $\int w\alpha \geq \int w\alpha_\theta^\sharp$; in particular $\max_{[a,b]} \alpha \geq \max_{[a,b]} \alpha_\theta^\sharp$. The system $H_\theta^\sharp$ attains all of these bounds at once.

(ii) A phase lead costs at least as much as no delay: if H is passive, regular on the band, and $\arg H \equiv \theta - \tau\omega \pmod{2\pi}$ on $[a, b]$ for some $\tau \geq 0$, then $\alpha \geq \alpha_\theta^\sharp$ on $[a, b]$.

(iii) Every zero-free system with phase θ on the band, in the sense of Theorem 6.2, satisfies $\alpha \geq \alpha_\theta^\sharp$ pointwise, in addition to $\frac{1}{b-a}\int \alpha \geq \theta\Lambda$.

Proof. (i) is (11.2). (ii) The system $e^{i\tau z}H$ is passive, has the same modulus on $\mathbb{R}$, and has argument θ on the band. (iii) is a special case of (ii) with $\tau = 0$. □

Part (i) answers, for systems with zeros allowed, the worst-frequency question of the zero-free case (Section 13, problem (3)). Part (ii) leaves open the case of a delay, $\arg H = \theta + \tau\omega$ with $\tau > 0$: the system $e^{i\tau z}H_\theta^\sharp$ shows that the bound $\alpha_\theta^\sharp$ can still be attained, but we do not know whether a delay together with zeros can do better.

Corollary 11.9. *Fix $0 < \theta < \pi$, and let $f = (a + b)/2$, $\Delta f = b - a$. As $\Delta f/f \to 0$,*

$$\Lambda_{\mathrm{all}}(\theta) = \frac{\pi}{8}\sin\theta\,\frac{\Delta f}{f} + O\Big(\Big(\frac{\Delta f}{f}\Big)^2\Big),$$

so that $\Lambda_{\mathrm{all}}(\theta)/\theta\Lambda \to \sin\theta/\theta$.

Proof. Write $\epsilon = \Delta f/f$. By (1.2), $\Lambda = \frac{\pi}{8}\epsilon + O(\epsilon^3)$. By the proof of Proposition 10.2, $\int_a^b p = J = 2(b - a)\Lambda$, and by Lemma 5.3(i), $\int_a^b \omega p = 0$, so $\frac{1}{b-a}\int_a^b (\omega - f)p = -2f\Lambda$. Taylor's formula for $\arctan(\omega/\eta)$ at $\omega = f$, with a remainder $O((\omega - f)^2/\eta^2)$ that is integrated against $|p|$, whose mean over the band is $O(1)$,[34] gives

$$\delta(\eta) = 2\Lambda\arctan\frac{f}{\eta} - 2\Lambda\frac{f\eta}{f^2 + \eta^2} + O\Big(\frac{\epsilon^2 f^2}{\eta^2}\Big).$$

The root of (11.1) satisfies $2\arctan(f/\eta_\star) = \theta + O(\epsilon)$, since the weight $\mathrm{d}t/r$ is carried by an interval of length ϵf around f. With $\phi = 2\arctan(f/\eta_\star)$ and $2f\eta/(f^2 + \eta^2) = \sin\phi$,

$$\begin{aligned}\Lambda_{\mathrm{all}} &= \theta\Lambda - \delta(\eta_\star) = \Lambda\,(\theta - \phi + \sin\phi) + O(\epsilon^2)\\ &= \Lambda\sin\theta + O(\epsilon^2),\end{aligned}$$

because $\Lambda = O(\epsilon)$ and $\theta - \phi = O(\epsilon)$; also $f/\eta_\star$ stays bounded. □

[33] The inverse Möbius map of (11.8) sends $(-\infty, 0]$ to the arc of the unit circle from $e^{-i\theta}$ to $e^{i\theta}$ through 1, with $-1 \mapsto 1$ and $0 \mapsto e^{i\theta}$.

[34] In $\omega = f + \frac{\Delta f}{2}s$, $p\,\mathrm{d}\omega/(b - a) = -s\,\mathrm{d}s/\sqrt{1 - s^2}\,(1 + O(\epsilon))$.

In the computations (Appendix D) the relative error is of the order of $(\Delta f/f)^2$: the ratio of Λ_{all} to the leading term is 1.0021 for $b/a = 1.2$ and 1.000149 for $b/a = 1.05$ at $\theta = \pi/2$. For small phases zeros hardly help: the gain is a factor $\sin\theta/\theta = 1 - \theta^2/6 + \dots$. For the 1° shift at 5 MHz of Section 12, the least mean attenuation with zeros allowed is 1.37588×10^{-3} Np, against 1.37595×10^{-3} Np without zeros.

For general θ the profile $\alpha_\theta^\sharp$ is not elementary, but a closed-form bound with the right shape survives, for the equilibrium weight of the band

$$w_0(\omega) = \frac{1}{I_0\, r(\omega)}, \qquad I_0 = \int_a^b \frac{\mathrm{d}t}{r(t)} = \frac{K(k)}{b},$$

$k^2 = 1 - a^2/b^2$.

Corollary 11.10. *Let $0 < \theta < \pi$ and let H be admissible for θ. Then*

$$\int_a^b w_0 \log\tanh\frac{\alpha}{2}\,\mathrm{d}\omega \;\geq\; \log\sin\theta - \pi\frac{K(a/b)}{K(k)}, \quad (11.16)$$

that is, the w_0-geometric mean of $\tanh(\alpha/2)$ is at least $q\sin\theta$, where $q = e^{-\pi K(a/b)/K(k)}$ is the nome of the modulus k. Equality holds if and only if $\theta = \pi/2$ and $H = H^\sharp_{\pi/2}$. For an octave $q = 0.085796$, for a decade $q = 0.26220$.

Proof. By Lemma 11.3(iv), $\tanh(\alpha/2) = e^u \sin\theta/\sin v \geq e^u \sin\theta$, with equality only where $v = \pi/2$, which requires $\theta = \pi/2$. Let ζ_0 be the function in the proof of Theorem 11.2 and $d = \zeta - \zeta_0$, so that $\operatorname{Im} d = g - g_0$, which is ≤ 0 on $(0, a)$ and ≥ 0 on (b, ∞). Both functions have the form (11.9), so $\operatorname{Re} d(x) = \frac{1}{\pi}\,\mathrm{p.v.}\int_0^\infty \operatorname{Im} d(t)\,\frac{2x}{t^2-x^2}\,\mathrm{d}t$ almost everywhere. Integrating against w_0 and exchanging the integrals, by Tonelli's theorem for $t \notin (a, b)$ and by the antisymmetry of the Hilbert transform between L^p and $L^{p'}$ for $t \in (a, b)$ ($\operatorname{Im} d$ is bounded and $w_0 \in L^{p'}$ for $p' < 2$), gives

$$\int_a^b w_0 \operatorname{Re} d = \frac{1}{\pi}\int_0^\infty \operatorname{Im} d(t)\, k_0(t)\,\mathrm{d}t,$$
$$k_0(t) = \mathrm{p.v.}\int_a^b \frac{2x\, w_0(x)}{t^2 - x^2}\,\mathrm{d}x.$$

In $\sigma = x^2$,

$$k_0(t) = \frac{1}{I_0}\,\mathrm{p.v.}\int_{a^2}^{b^2} \frac{\mathrm{d}\sigma}{(t^2-\sigma)\sqrt{(\sigma - a^2)(b^2 - \sigma)}},$$

which vanishes for $a < t < b$ by Glauert's integral (proof of Lemma 8.5), is negative for $t < a$ and positive for $t > b$. So $\int w_0 \operatorname{Re} d \geq 0$, with equality only if $g = g_0$ off the band. For ζ_0 the same computation, with g_0 explicit, gives

$$\int_a^b w_0 \operatorname{Re}\zeta_0 = \int_0^a k_0(t)\,\mathrm{d}t$$
$$= -\frac{\pi}{I_0}\int_0^a \frac{\mathrm{d}t}{\sqrt{(a^2 - t^2)(b^2 - t^2)}},$$

using, for $s < A$,

$$\int_A^B \frac{\mathrm{d}\sigma}{(\sigma - s)\sqrt{(\sigma - A)(B - \sigma)}} = \frac{\pi}{\sqrt{(A - s)(B - s)}};$$

the last integral is $K(a/b)/b$.[35] Adding up gives (11.16). □

12. WAVES IN PASSIVE MEDIA

12.1. A slab of a passive medium

Consider a plane wave crossing a slab of thickness L of a linear, homogeneous, isotropic and temporally dispersive medium with relative permittivity $\epsilon(\omega)$ and permeability $\mu_r(\omega)$. We consider the propagation factor of a single pass and ignore reflections at the faces, which is exact for an impedance-matched slab and otherwise describes the field inside the medium. In our convention the factor is

$$H(\omega) = \exp\Big(\frac{i\omega n(\omega) L}{c}\Big), \qquad n = \sqrt{\epsilon\mu_r} = n_r + i n_i.$$

Passivity of the medium means that $z\epsilon(z)$ and $z\mu_r(z)$ are Herglotz functions and that ϵ and μ_r tend to 1 as $z \to i\infty$ [9, 15, 16]. Reality of the fields makes ϵ and μ_r real on $i\mathbb{R}_+$, so that $z\epsilon(z)$ and $z\mu_r(z)$ lie on the closed positive imaginary axis there.[36]

Lemma 12.1. *$h(z) = zn(z)L/c$ is a Herglotz function with $h(iy) \in i\mathbb{R}$ and $h(-\bar z) = -\overline{h(z)}$, and its linear coefficient in (4.4) is $\beta = L/c$. Hence $H = e^{ih}$ is a zero-free passive system with $\sigma = 1$, phase $\varphi = \omega n_r L/c$ and attenuation $\alpha = \omega n_i L/c$.*

Proof. Put $f(z) = z\epsilon(z)$ and $g(z) = z\mu_r(z)$, so that $zn(z) = \sqrt{f(z)g(z)}$. For $z \in \mathbb{C}_+$, $\arg f$ and $\arg g$ lie in $[0, \pi]$; we take the square root with argument $\frac{1}{2}(\arg f + \arg g) \in [0, \pi]$, which is analytic and is the branch with $n \to 1$. So $zn(z)$ maps $\mathbb{C}_+$ to the closed upper half-plane. On $i\mathbb{R}_+$, f and g lie on $i\mathbb{R}_+$, so their product is negative and zn lies on $i\mathbb{R}_+$. The symmetry follows from that of f and g, and $\beta = \lim h(iy)/(iy) = L/c$ because $n(iy) \to 1$. □

The coefficient $\beta = L/c$ is the vacuum delay: no signal crosses the slab faster than light. It drops out of the bound, as every delay does.

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[35] Substitute $t = a\sin\phi$: then $\sqrt{(a^2 - t^2)(b^2 - t^2)}$ is $a\cos\phi \cdot b\sqrt{1 - (a/b)^2\sin^2\phi}$ and $\mathrm{d}t = a\cos\phi\,\mathrm{d}\phi$. Similarly $I_0 = K(k)/b$ by $t^2 = b^2 - (b^2 - a^2)\sin^2\phi$.

[36] Analyticity in $\mathbb{C}_+$ is causality of the constitutive relations; $\operatorname{Im}(\omega\epsilon(\omega)) \geq 0$ for real ω is the absence of gain; the limit expresses that every medium is transparent at very high frequency. The Herglotz property in $\mathbb{C}_+$ follows from these boundary properties by the Poisson representation of the harmonic function $\operatorname{Im}(z\epsilon(z))$.

Theorem 12.2. *Let the medium be passive, with n continuous and nonzero on a neighbourhood of $[a,b]$ in $\overline{\mathbb{C}_+}$, and suppose that on the band*

$$n_r(\omega) = n_0 + \frac{C}{\omega}, \qquad a \le \omega \le b, \tag{12.1}$$

with constants $n_0 \in \mathbb{R}$ and $C \ge 0$. Then

$$\frac{1}{b-a}\int_a^b \omega\, n_i(\omega)\,\mathrm{d}\omega \;\ge\; C\,\Lambda, \tag{12.2}$$

with equality if and only if the Herglotz measure of $zn(z)$ is carried by S.

Proof. By Lemma 12.1, H is zero-free, passive and regular on the band, with phase $\varphi = \omega n_r L/c = CL/c + (n_0 L/c)\,\omega$. Theorem 6.2 with $\theta = CL/c$ and $\tau = n_0 L/c$ gives $\frac{1}{b-a}\int \omega n_i L/c \ge (CL/c)\Lambda$, and L/c cancels. □

The thickness of the slab has dropped out: (12.2) is a property of the material. The condition (12.1) says that the phase index exceeds the group index $n_g = \mathrm{d}(\omega n_r)/\mathrm{d}\omega = n_0$ by C/ω, or equivalently that $\mathrm{d}n_r/\mathrm{d}\omega = -C/\omega^2 < 0$. So Theorem 12.2 is a quantitative form of the old observation that anomalous dispersion requires absorption [6, 7, 9]: an index falling like C/ω across a band requires the frequency-weighted extinction ωn_i to be at least $C\Lambda$ on average. In terms of the intensity absorption coefficient $a_{\mathrm{abs}} = 2\omega n_i/c$ the bound reads $\overline{a_{\mathrm{abs}}} \ge 2C\Lambda/c$.

12.2. Approximate constant phase

The sum rule is linear in the phase, so the bound is stable under perturbations. Put $P = \int_a^b |p(\omega)|\,\mathrm{d}\omega$.

Corollary 12.3. *Let H be zero-free, passive and regular on the band, and suppose $|\varphi(\omega) - \theta - \tau\omega| \le \varepsilon$ on $[a,b]$. Then*

$$\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega \;\ge\; \theta\Lambda - \frac{P}{2(b-a)}\,\varepsilon.$$

In particular a zero-free system that is lossless on the band cannot approximate a constant phase lag $\theta > 0$ to within $\varepsilon < \theta J/P$.

Proof. By Theorem 6.1 and Lemma 5.3(i), $\int_a^b \alpha \ge \frac{1}{2}\int_a^b \varphi p = \frac{1}{2}\theta J + \frac{1}{2}\int_a^b (\varphi - \theta - \tau\omega)p$, and the last term is at least $-\frac{1}{2}\varepsilon P$. If $\alpha = 0$ on the band, then $0 \ge \frac{1}{2}(\theta J - \varepsilon P)$. □

For an octave $J/P = 0.2584$, so a lossless zero-free system cannot hold its phase within $0.258\,\theta$ of any $\theta + \tau\omega$ with $\theta > 0$ across the octave; for the decade $[1, 10]$ the ratio is 0.617. The ratio J/P does not depend on the scale of the band.

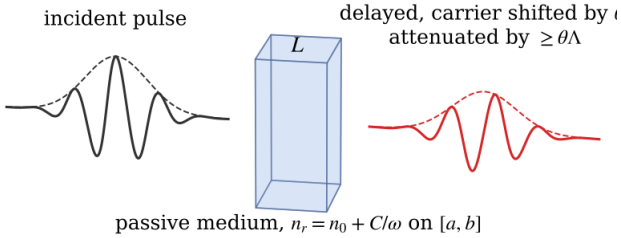


FIG. 15. Schematic of Theorem 12.2. A pulse crosses a slab of a passive medium whose index falls like $n_0 + C/\omega$ on the band. The envelope is delayed by $n_0 L/c$, the carrier is shifted under it by $\theta = CL/c$, and the pulse leaves attenuated (the factor is exaggerated). By Theorem 12.2 the mean of ωn_i over the band is at least $C\Lambda$, whatever the thickness L.

Remark 12.4 (Broadband phase shifters). Broadband "90° phase shifters" are pairs of lossless all-pass networks whose phases differ by approximately $\pi/2$ across a band [13]. Each phase is increasing (Proposition 4.5); only their difference is nearly constant. This is consistent with Corollary 12.3: what the theorem forbids is a lossless network whose own phase is nearly constant on the band, and a pair of networks never needs one. A device that shifts a single signal by θ without a reference channel must lose transmitted power.

12.3. What the bound means for energy

On the band the power transmission is $T(\omega) = |H(\omega)|^2 = e^{-2\alpha(\omega)}$.

Proposition 12.5. *Let H be as in Theorem 6.2 with $\theta > 0$.*

(i) There is $\omega_1 \in [a,b]$ with $T(\omega_1) \le e^{-2\theta\Lambda}$, and for every $\varepsilon > 0$ there is a signal with spectrum in $[-b,-a] \cup [a,b]$ that loses at least the fraction $1 - e^{-2\theta\Lambda} - \varepsilon$ of its energy.

(ii) The geometric mean of T over the band is at most $e^{-2\theta\Lambda}$.

(iii) In a quantised description, a single quantum of frequency ω_1 fails to be transmitted with probability at

least $1-e^{-2\theta\Lambda}$*, and the mean energy it removes from the transmitted channel is at least* $(1-e^{-2\theta\Lambda})\hbar\omega_1$*.*

(iv) For a narrow band of relative width $\Delta f/f$ *and* θ *in degrees,* $1-e^{-2\theta\Lambda}=\frac{\pi^2}{720}\,\theta\,\frac{\Delta f}{f}\,\big(1+O(\theta\Delta f/f)+O((\Delta f/f)^2)\big)$*.*

Proof. (i) The mean of the continuous function α is at least $\theta\Lambda$, so $\alpha(\omega_1)\ge\theta\Lambda$ at some ω_1. A signal whose spectrum is concentrated near $\pm\omega_1$ transmits a fraction of its energy arbitrarily close to $T(\omega_1)$ by Parseval's identity and the continuity of T. (ii) is Corollary 6.7. (iii) A linear lossy element acts on a single quantum as a beam splitter coupling the transmitted mode to the other modes of the device; the quantum is transmitted with probability $T(\omega_1)$. (iv) By (7.3), $\Lambda=\frac{\pi}{8}\frac{\Delta f}{f}(1+O((\Delta f/f)^2))$, and θ degrees are $\pi\theta/180$ radians, so $2\theta\Lambda=\frac{\pi^2}{720}\theta\frac{\Delta f}{f}(1+\cdots)$; finally $1-e^{-x}=x(1+O(x))$. □

The bound is on transmission. The power that is not transmitted may be absorbed, reflected or routed to other ports, and Proposition 12.5 does not say which: a scalar transfer function sees only the output channel. For the impedance-matched slab of Theorem 12.2, which reflects nothing and has no other port, all of it is absorbed. The energy that is lost is not the "energy of the phase shift". A phase factor is energy neutral (Proposition 3.1). The loss is the price that causality and passivity attach to producing that factor on a band.

12.4. The 5 MHz example done correctly

Take the example of [1]: a 1° phase shift of a wave at 5 MHz, and give it a band, 4.5 to 5.5 MHz, so $\Delta f/f=0.2$. Then $\kappa=0.1$ and

$$\Lambda=0.078836,$$
$$\theta\Lambda=\frac{\pi}{180}\Lambda=1.3760\times10^{-3}\,\mathrm{Np}=0.011951\,\mathrm{dB}.$$

A passive device that shifts the carrier–envelope phase by 1° across this band therefore fails to transmit at least 0.275% of the power at its worst frequency, $1-e^{-2\theta\Lambda}=2.748\times10^{-3}$, against 2.742×10^{-3} from Proposition 12.5(iv). A quantum of $hf=3.313\times10^{-27}$ J at that frequency removes from the output on average at least 9.10×10^{-30} J. This happens to be close to the value $hf/360=9.20\times10^{-30}$ J that a phase slipping by 1° per period would remove (Proposition 3.2), but the agreement is accidental: the ratio of the two numbers is 0.989, close to $\frac{\pi^2}{2}\frac{\Delta f}{f}=0.987$ by Proposition 12.5(iv), and it tends to zero with the bandwidth. For a single sinusoid, $\Delta f=0$ and a delay of 555.6 ps does the job without loss.

13. DISCUSSION AND OPEN PROBLEMS

13.1. What has been shown

A passive system can delay a band of frequencies for free, but it cannot shift the phase of the whole band by a constant without losing transmitted power. The least mean attenuation is $\theta\Lambda$, with $\Lambda=(K(\kappa)-E(\kappa))/\kappa$ an explicit function of the band ratio, and the bound is attained. The proof is a single sum rule, (6.1), built from the test function W, and every other statement of the paper is a consequence of it or of the explicit extremal system: the lead bound, the worst-frequency and geometric-mean bounds, the bound for anomalous dispersion in a medium, and the stability under perturbations of the phase.

For an affine phase there is more: every zero-free system is the extremal function h_g plus terms that each raise the attenuation at every point of the band (Theorem 9.2), so the extremal system, or its inversion for a lead, has the least attenuation at every frequency of the band at once. This gives the sharp worst frequency and the sharp plain mean for a lead (Corollary 9.7).

With zeros allowed the sum rule is no longer available, and the problem changes character. After the change of variables of Lemma 11.3 the zeros disappear, the phase condition becomes a curve, and a comparison argument gives a bound at every frequency, attained by one system with a single zero. At a right angle that system is algebraic, and the bound reads $\left|H(\sqrt{ab})\right|^2\le a/b$.

The ingredients are classical: the Herglotz representation, Kramers–Kronig relations, a Cauchy integral, an elliptic integral reduced by Landen's transformation, Hardy spaces and a conformal map with a free parameter. What is new, as far as we can tell, is the pair of sharp bounds, the identification of the price of a constant phase with a single number Λ for zero-free systems, and the pointwise optimality of a single system, with zeros allowed and, for an affine phase, without them.

13.2. Open problems

(1) *Delay together with zeros.* Theorem 11.1 solves Problem 10.4, and Corollary 11.8(ii) extends it to a phase $\theta-\tau\omega$ with $\tau\ge0$. For $\arg H\equiv\theta+\tau\omega$ with $\tau>0$ the system $e^{i\tau z}H_\theta^\sharp$ still has the loss $\alpha_\theta^\sharp$; is it optimal, or can a delay combined with zeros do better? The change of variables of Lemma 11.3 needs a constant phase.

(2) *General phase without zeros.* Theorem 9.2 holds for every continuously differentiable phase g, and Corollary 9.3 is sharp whenever $h_g+\max\{0,-\beta_g\}R/z$ is a Herglotz function. When v_g changes sign the cheapest measure off the band is not known; for a lead with positive slope, $-s+\tau\omega$ with $\tau>0$, this is already the case.

(3) *Several channels.* For a matrix Herglotz function F with the symmetry of (4.3), Theorem 9.2 applied to every form u^*Fu bounds $\operatorname{Im} F$ on the band from below in the Loewner order. The least compensation for a matrix phase $\Theta+\omega T$ whose coefficients do not commute is not known.

(4) *Several bands.* For a phase that is constant on each of several disjoint bands, the test function is built from a hyperelliptic square root, and the constant becomes a period of a hyperelliptic differential. We expect the sum rule to carry over with the same proof.

(5) *Reflection and matching.* For reflection coefficients, the Fano bound [12] and the present bound constrain different parts of the same function. A joint bound on the modulus and the argument of a reflection coefficient on a band would combine the two.

(6) *The optimal system in closed form.* For $\theta \neq \pi/2$, $H_\theta^\sharp$ comes from a conformal map onto the slit domain D_T of Proposition 11.6. The Joukowski map $G \mapsto \frac{1}{2}(G+1/G)$ straightens the semicircle of its boundary but bends the slit unless $\cos\theta = 0$, which is why the right angle is elementary. Is there a closed form for other θ, for instance through the Schwarz equation of the circular quadrilateral?

(7) *Formalisation.* The finite arithmetic of the paper is checked in Lean (Appendix D). The analytic part, the sum rule and the evaluation of Λ, would require a formalisation of boundary values of Cauchy integrals, of the M. Riesz theorem and of the complete elliptic integrals with Landen's transformation, and the results on zeros would need Hardy spaces, the Möbius change of variables, extremal length and the Carathéodory extension of conformal maps. We have not done this, and the analytic theorems of the paper are not machine-checked.

Appendix A: Complete elliptic integrals

This appendix derives every property of K and E used in the paper. The modulus is $k \in [0,1)$ and the complementary modulus is $k' = \sqrt{1-k^2}$; we write $K' = K(k')$ and $E' = E(k')$.

1. Power series

For $|u|<1$, $(1-u)^{-1/2} = \sum_{n\geq 0} c_n u^n$ and $(1-u)^{1/2} = \sum_{n\geq 0} c_n u^n/(1-2n)$, with $c_n = \binom{2n}{n}4^{-n}$; the first is the binomial series, $\binom{-1/2}{n}(-1)^n = c_n$, and the second follows by integrating the first term by term. The Wallis integral $\frac{2}{\pi}\int_0^{\pi/2}\sin^{2n}\phi\,\mathrm{d}\phi = c_n$ follows by induction from the reduction formula $\int_0^{\pi/2}\sin^{2n} = \frac{2n-1}{2n}\int_0^{\pi/2}\sin^{2n-2}$ and the recurrence $c_n = \frac{2n-1}{2n}c_{n-1}$. With $u = k^2\sin^2\phi$ and uniform convergence for $k<1$,

$$K(k) = \frac{\pi}{2}\sum_{n\geq 0} c_n^2 k^{2n}, \qquad E(k) = \frac{\pi}{2}\sum_{n\geq 0}\frac{c_n^2}{1-2n}k^{2n}. \tag{A.1}$$

Subtracting, $K-E = \frac{\pi}{2}\sum_{n\geq 1} c_n^2\frac{2n}{2n-1}k^{2n}$, which is the series of Proposition 7.2(i).[37]

2. Derivatives

Differentiating under the integral sign,

$$\frac{\mathrm{d}E}{\mathrm{d}k} = \frac{E-K}{k}, \qquad \frac{\mathrm{d}K}{\mathrm{d}k} = \frac{E-k'^2K}{k\,k'^2}. \tag{A.2}$$

The first is immediate: $\frac{\mathrm{d}}{\mathrm{d}k}\sqrt{1-k^2\sin^2\phi} = -k\sin^2\phi/\sqrt{\cdot}$, and $-k^2\sin^2\phi = (1-k^2\sin^2\phi)-1$. For the second, the derivative is $\int_0^{\pi/2} k\sin^2\phi\,(1-k^2\sin^2\phi)^{-3/2}\,\mathrm{d}\phi$, and the identity

$$\frac{\mathrm{d}}{\mathrm{d}\phi}\frac{\sin\phi\cos\phi}{\sqrt{1-k^2\sin^2\phi}} = \frac{1}{k^2}\Big(\sqrt{1-k^2\sin^2\phi} - \frac{k'^2}{(1-k^2\sin^2\phi)^{3/2}}\Big),$$

integrated over $[0,\pi/2]$, gives $\int(1-k^2\sin^2)^{-3/2} = E/k'^2$; then $k^2\sin^2\phi(1-k^2\sin^2\phi)^{-3/2} = (1-k^2\sin^2)^{-3/2} - (1-k^2\sin^2)^{-1/2}$ yields (A.2).[38]

3. Legendre's relation

Lemma A.1. *$E(k)K' + E'K(k) - K(k)K' = \pi/2$ for $0<k<1$.*

Proof. Let $L(k)$ be the left side. Since $\mathrm{d}k'/\mathrm{d}k = -k/k'$, (A.2) applied at k' gives

$$\frac{\mathrm{d}E'}{\mathrm{d}k} = \frac{k(K'-E')}{k'^2}, \qquad \frac{\mathrm{d}K'}{\mathrm{d}k} = -\frac{E'-k^2K'}{kk'^2}.$$

Multiply $\mathrm{d}L/\mathrm{d}k$ by kk'^2 and insert the four derivatives:

$$kk'^2L' = k'^2(E-K)K' - E(E'-k^2K') + k^2K(K'-E') + E'(E-k'^2K) - K'(E-k'^2K) + K(E'-k^2K').$$

[37] The coefficients $d_n = c_n^2\,2n/(2n-1)$ are $\frac{1}{2}, \frac{3}{16}, \frac{15}{128}, \frac{175}{2048}, \ldots$; the ratio $d_{n+1}/d_n = (2n+1)(2n-1)/((2n+2)2n)$ follows from $c_{n+1}/c_n = (2n+1)/(2n+2)$.

[38] To check the displayed identity, put $s = \sin\phi$, $c = \cos\phi$, $\Delta = 1-k^2s^2$. The derivative of $sc\,\Delta^{-1/2}$ is $(c^2-s^2)\Delta^{-1/2} + k^2s^2c^2\Delta^{-3/2} = \Delta^{-3/2}\big((c^2-s^2)\Delta + k^2s^2c^2\big)$, and with $c^2 = 1-s^2$ the bracket is $1-2s^2+k^2s^4 = \frac{1}{k^2}\big(\Delta^2 - k'^2\big)$.

Expanding, the terms in KK' and in EE' cancel in pairs, and what remains is

$$kk'^2L' = (k^2 + k'^2 - 1)\,(EK' - KE') = 0.$$

So L is constant. As $k \to 0$, $K, E \to \pi/2$ and $E' \to 1$, while $K - E = O(k^2)$ by (A.1) and $K' = O(\log(1/k))$ by (A.3) below; hence $L = E'K + (E - K)K' \to \pi/2$. □

4. The arithmetic–geometric mean and Landen's transformation

For $p, q > 0$ let

$$I(p,q) = \int_0^{\pi/2} \frac{\mathrm{d}\phi}{\sqrt{p^2\cos^2\phi + q^2\sin^2\phi}},$$

so that $K(k) = I(1, k')$.

Lemma A.2 (Gauss)**.** *$I(p,q) = I\big(\frac{p+q}{2}, \sqrt{pq}\big)$. Consequently $I(p,q) = \pi/(2M(p,q))$, with M the arithmetic–geometric mean of p and q.*

Proof. With $x = q\tan\phi$, $I(p,q) = \frac{1}{2}\int_{\mathbb{R}} \big((p^2 + x^2)(q^2 + x^2)\big)^{-1/2}\,\mathrm{d}x$. Substitute $x = \frac{1}{2}(t - pq/t)$, $t > 0$, which maps $(0,\infty)$ increasingly onto $\mathbb{R}$, with $\mathrm{d}x = \frac{1}{2}(1 + pq/t^2)\,\mathrm{d}t$. Then $4x^2 + (p+q)^2 = (t^2+p^2)(t^2+q^2)/t^2$ and $x^2 + pq = \frac{1}{4}(t + pq/t)^2$, so

$$\big((p_1^2 + x^2)(q_1^2 + x^2)\big)^{-1/2}\,\mathrm{d}x = \frac{2\;\mathrm{d}t}{\sqrt{(t^2+p^2)(t^2+q^2)}},$$

with $p_1 = \frac{p+q}{2}$, $q_1 = \sqrt{pq}$. Integrating over $t \in (0,\infty)$ and using that the integrand is even in t,

$$\begin{aligned} 2I(p_1,q_1) &= 2\int_0^\infty \frac{\mathrm{d}t}{\sqrt{(t^2+p^2)(t^2+q^2)}} \\ &= \int_{\mathbb{R}} \frac{\mathrm{d}t}{\sqrt{(t^2+p^2)(t^2+q^2)}} = 2I(p,q). \end{aligned}$$

The iterates converge to $M(p,q)$, and $I(M,M) = \pi/(2M)$. □

Proposition A.3 (Landen)**.** *Let $0 < k' < 1$ and $\kappa = (1-k')/(1+k')$. Then*

$$K(k) = (1+\kappa)K(\kappa), \qquad E(k) = (1+k')E(\kappa) - k'K(k).$$

Proof. By Lemma A.2, $K(k) = I(1,k') = I\big(\frac{1+k'}{2}, \sqrt{k'}\big) = \frac{2}{1+k'}I\big(1, \frac{2\sqrt{k'}}{1+k'}\big)$, and $\frac{2\sqrt{k'}}{1+k'} = \sqrt{1-\kappa^2} = \kappa'$, so $K(k) = \frac{2}{1+k'}K(\kappa) = (1+\kappa)K(\kappa)$. For E, regard k as a function of κ, $k = 2\sqrt{\kappa}/(1+\kappa)$, $k' = (1-\kappa)/(1+\kappa)$, and differentiate the relation for K with (A.2). Since

$$\frac{\mathrm{d}k}{\mathrm{d}\kappa} = \frac{1-\kappa}{\sqrt{\kappa}(1+\kappa)^2}, \qquad kk'^2 = \frac{2\sqrt{\kappa}(1-\kappa)^2}{(1+\kappa)^3},$$
$$\text{so} \qquad \frac{\mathrm{d}k/\,\mathrm{d}\kappa}{kk'^2} = \frac{1+\kappa}{2\kappa(1-\kappa)},$$

the derivative of $K(k)$ is $\big(E(k) - k'^2K(k)\big)\frac{1+\kappa}{2\kappa(1-\kappa)}$, and that of $(1+\kappa)K(\kappa)$ is $K(\kappa) + \big(E(\kappa) - (1-\kappa^2)K(\kappa)\big)/(\kappa(1-\kappa))$. Equating, multiplying by $2\kappa(1-\kappa)/(1+\kappa)$ and using $k'^2(1+\kappa) = (1-\kappa)^2/(1+\kappa)$,

$$\begin{aligned} E(k) &= \frac{2E(\kappa)}{1+\kappa} + \frac{K(\kappa)}{1+\kappa}\Big((1-\kappa)^2 + 2\kappa(1-\kappa) \\ &\qquad - 2(1-\kappa)(1+\kappa)\Big) \\ &= \frac{2E(\kappa)}{1+\kappa} - (1-\kappa)K(\kappa). \end{aligned}$$

Since $2/(1+\kappa) = 1 + k'$ and $(1-\kappa)K(\kappa) = k'K(k)$, this is the claim. □

5. Behaviour at the ends

As $k \to 0$, (A.1) gives $K, E = \frac{\pi}{2} + O(k^2)$. As $k \to 1$, $E(k) \to 1$ and

$$K(k) = \log\frac{4}{k'} + O\big(k'^2\log k'\big). \tag{A.3}$$

The limit of E is dominated convergence. For K, the substitution $x = k'\tan\phi$ of Lemma A.2 gives $K(k) = \int_0^\infty \big((1+x^2)(k'^2+x^2)\big)^{-1/2}\,\mathrm{d}x$. Split at $x = 1$:

$$\int_0^1 \frac{\mathrm{d}x}{\sqrt{k'^2+x^2}} = \operatorname{arsinh}\frac{1}{k'} = \log\frac{2}{k'} + o(1),$$

and, by dominated convergence,

$$\int_0^1 \frac{(1+x^2)^{-1/2} - 1}{\sqrt{k'^2+x^2}}\,\mathrm{d}x \to \log\frac{2}{1+\sqrt{2}},$$
$$\int_1^\infty \frac{\mathrm{d}x}{\sqrt{(1+x^2)(k'^2+x^2)}} \to \log(1+\sqrt{2}),$$

the first limit being $\int_0^1 \big(\frac{1}{x\sqrt{1+x^2}} - \frac{1}{x}\big)\,\mathrm{d}x = \log\frac{2}{1+\sqrt{2}}$, both from the primitive $-\operatorname{arsinh}(1/x)$ of $1/(x\sqrt{1+x^2})$. The three limits add up to $\log(4/k')$. The error term is [30, 19.12.1].

Appendix B: The elementary integrals

1. Moments of the arcsine weight

For $A < B$ put $m = \frac{A+B}{2}$, $d = \frac{B-A}{2}$ and $\sigma = m - d\cos\phi$, $0 \le \phi \le \pi$. Then $(\sigma - A)(B - \sigma) = d^2\sin^2\phi$ and

$\mathrm{d}\sigma = d\sin\phi\,\mathrm{d}\phi$, so $\mathrm{d}\sigma/\sqrt{(\sigma-A)(B-\sigma)} = \mathrm{d}\phi$, and

$$\begin{aligned}
\int_A^B \frac{\mathrm{d}\sigma}{\sqrt{(\sigma-A)(B-\sigma)}} &= \pi,\\
\int_A^B \frac{\sigma\,\mathrm{d}\sigma}{\sqrt{\cdot}} &= \pi m = \frac{\pi}{2}(A+B),\\
\int_A^B \frac{\sigma^2\,\mathrm{d}\sigma}{\sqrt{\cdot}} &= \pi\Big(m^2+\frac{d^2}{2}\Big)\\
&= \frac{\pi}{8}\big(3A^2+2AB+3B^2\big),\\
\int_A^B \frac{\mathrm{d}\sigma}{\sigma\sqrt{\cdot}} &= \int_0^\pi \frac{\mathrm{d}\phi}{m-d\cos\phi}\\
&= \frac{\pi}{\sqrt{m^2-d^2}} = \frac{\pi}{\sqrt{AB}}.
\end{aligned}$$

The last integral is the classical $\int_0^\pi \mathrm{d}\phi/(m-d\cos\phi) = \pi/\sqrt{m^2-d^2}$ for $m > |d|$, which follows from $u = \tan(\phi/2)$.[39]

These give Lemma 5.3(iii),(iv): with $x\,p(x)\,\mathrm{d}x = \frac{1}{2}\varpi(\sigma)\,\mathrm{d}\sigma$ and $p(x)\,\mathrm{d}x/x = \frac{1}{2}\varpi(\sigma)\,\mathrm{d}\sigma/\sigma$,

$$\begin{aligned}
\int_a^b \frac{p}{x}\,\mathrm{d}x &= \frac{1}{2}\Big((A+B)\frac{\pi}{\sqrt{AB}} - 2\pi\Big)\\
&= \frac{\pi(b-a)^2}{2ab},\\
\int_a^b x^3 p\,\mathrm{d}x &= \frac{1}{2}\Big(\frac{\pi}{2}(A+B)^2\\
&\qquad - \frac{\pi}{4}(3A^2+2AB+3B^2)\Big)\\
&= -\frac{\pi}{8}(B-A)^2.
\end{aligned}$$

2. The integrals I_0, I_2 and J

With $t^2 = b^2-(b^2-a^2)\sin^2\phi$ (Proposition 7.1), $t\,\mathrm{d}t = -(b^2-a^2)\sin\phi\cos\phi\,\mathrm{d}\phi$ and $r(t) = (b^2-a^2)\sin\phi\cos\phi$, so $\mathrm{d}t/r = -\,\mathrm{d}\phi/t$ and

$$\begin{aligned}
I_0 &= \int_0^{\pi/2} \frac{\mathrm{d}\phi}{b\sqrt{1-k^2\sin^2\phi}} = \frac{K(k)}{b},\\
I_2 &= \int_0^{\pi/2} b\sqrt{1-k^2\sin^2\phi}\,\mathrm{d}\phi = bE(k).
\end{aligned}$$

Hence $J = (a^2+b^2)I_0 - 2I_2 = b\big((1+k'^2)K(k) - 2E(k)\big)$. By Proposition A.3, $-2E(k) = -2(1+k')E(\kappa) + 2k'K(k)$, and $(1+k'^2)+2k' = (1+k')^2$, so

$$\begin{aligned}
J &= b(1+k')^2K(k) - 2b(1+k')E(\kappa)\\
&= b(1+k')^2(1+\kappa)K(\kappa) - 2b(1+k')E(\kappa).
\end{aligned}$$

With $1+k' = (a+b)/b$ and $1+\kappa = 2b/(a+b)$ the coefficients are $2(a+b)$ and $2(a+b)$, which is (7.1).

3. The integrals on $(0,a)$

With $s = a\sin\phi$,

$$\sqrt{(a^2-s^2)(b^2-s^2)} = a\cos\phi\,b\sqrt{1-k'^2\sin^2\phi},$$

so

$$\begin{aligned}
\int_0^a \frac{\mathrm{d}s}{\sqrt{(a^2-s^2)(b^2-s^2)}} &= \frac{K(k')}{b},\\
\int_0^a \frac{s^2\,\mathrm{d}s}{\sqrt{(a^2-s^2)(b^2-s^2)}} &= b\big(K(k')-E(k')\big),
\end{aligned}$$

the second because $a^2\sin^2\phi = b^2k'^2\sin^2\phi$ and

$$\int_0^{\pi/2} \frac{k'^2\sin^2\phi\,\mathrm{d}\phi}{\sqrt{1-k'^2\sin^2\phi}} = K(k')-E(k').$$

These are the integrals in the proof of Theorem 8.2(ii), where

$$\frac{m^2}{b}K' - b(K'-E') = \frac{b}{K}\big(EK' - KK' + E'K\big) = \frac{\pi b}{2K}$$

by $m^2 = b^2E/K$ and Lemma A.1.

Appendix C: The extremal profile in closed form

1. Three formulas for $\alpha_\star$

Lemma C.1. *For $a < x < b$,*

$$\begin{aligned}
\alpha_\star(x) &= \tau_\star\int_a^x \frac{m^2-s^2}{r(s)}\,\mathrm{d}s\\
&= \frac{2x\,r(x)\theta}{\pi}\,\mathrm{p.v.}\int_a^b \frac{\mathrm{d}t}{r(t)(x^2-t^2)}\\
&= \frac{2\theta\,r(x)}{\pi}\int_a^b \frac{\mathrm{d}t}{r(t)(x+t)}.
\end{aligned}$$

The last form shows at once that $\alpha_\star > 0$ on (a,b) and that it vanishes like $\sqrt{x-a}$ and $\sqrt{b-x}$ at the band edges.

Proof. The first equality is Theorem 8.2(ii), the second Proposition 8.4(ii),(iv) with $g = \theta$. For the third, apply Cauchy's formula to $1/R$, which is analytic off S, $O(z^{-2})$ at infinity, and has the jumps $1/R(t+i0) - 1/R(t-i0) = -2i/r(t)$ on (a,b) and $+2i/r(t)$ on $(-b,-a)$ by Lemma 8.1. Then

$$\frac{1}{R(z)} = \frac{1}{\pi}\int_a^b \frac{\mathrm{d}t}{r(t)}\Big(\frac{1}{z-t} - \frac{1}{z+t}\Big).$$

At $z = x+i0$ the left side is $-i/r(x)$, purely imaginary, so the real part of the right side vanishes: $\mathrm{p.v.}\int_a^b \frac{\mathrm{d}t}{r(t)(x-t)} = \int_a^b \frac{\mathrm{d}t}{r(t)(x+t)}$. Since $\frac{2x}{x^2-t^2} = \frac{1}{x-t} + \frac{1}{x+t}$, the second form equals the third. □

[39] With $u = \tan(\phi/2)$, $\cos\phi = (1-u^2)/(1+u^2)$ and $\mathrm{d}\phi = 2\,\mathrm{d}u/(1+u^2)$, so the integral is $\int_0^\infty 2\,\mathrm{d}u/\big((m-d)+(m+d)u^2\big) = \pi/\sqrt{(m-d)(m+d)}$.

2. The mean of $\alpha_\star$

By Fubini's theorem applied to the first form,

$$\begin{aligned}\int_a^b \alpha_\star \,\mathrm{d}x &= \tau_\star \int_a^b \frac{(m^2-s^2)(b-s)}{r(s)}\,\mathrm{d}s \\ &= \tau_\star\big(b(m^2 I_0 - I_2) - m^2 I_1 + I_3\big),\end{aligned}$$

with $I_j = \int_a^b s^j\,\mathrm{d}s/r(s)$. The first bracket vanishes by the choice of m. With $\sigma = s^2$, $I_1 = \frac{\pi}{2}$ and $I_3 = \frac{\pi}{4}(a^2+b^2)$ by Appendix B, so

$$\begin{aligned}\int_a^b \alpha_\star &= \tau_\star \frac{\pi}{4}\big(a^2+b^2-2m^2\big) \\ &= \frac{\theta K(k)}{2b}\big(a^2+b^2\big) - \theta b\, E(k) = \frac{\theta}{2} J,\end{aligned}$$

in agreement with Theorem 6.2.[40]

3. The lead extremal

The system $H_\star(-ab/z)$ of Theorem 6.6 has, in the representation (4.4), linear coefficient 0, measure on S given by its attenuation, and an atom $ab\,\tau_\star$ at the origin (Theorem 6.6, proof). Its phase on the band is $-|\theta|$. The sum rule (6.1), with the atom costing $\frac{\pi}{4}W(0)\,ab\,\tau_\star$ and $W(0) = (b-a)^2/(ab)$, gives

$$\begin{aligned}\frac{1}{b-a}\int_a^b \alpha\,\mathrm{d}\omega &= -|\theta|\,\Lambda + \frac{\pi}{4}(b-a)\tau_\star \\ &= -|\theta|\,\frac{K-E}{\kappa} + |\theta|\,\kappa K \\ &= |\theta|\,\frac{E(\kappa) - \kappa'^2 K(\kappa)}{\kappa},\end{aligned}$$

where K, E are taken at κ and $\frac{\pi}{4}(b-a)\tau_\star = |\theta|\,(b-a)K(\kappa)/(a+b) = |\theta|\,\kappa K(\kappa)$. This is Remark 6.8.

Appendix D: Verification

No statement of the paper rests on a computation: every result has a proof in the text. Two kinds of checks accompany the source files.

Numerical checks. The script `anc/checks.py` re-evaluates the identities of the paper in high-precision arithmetic: the boundary values of W and the kernel identity (5.3) at sample points; the moments of Lemma 5.3; the three forms of J and the Landen relations; the bounds on Λ and Table II; m^2, Legendre's relation and $\tau_\star$; the three formulas of Lemma C.1 and the mean of $\alpha_\star$; the octave values of Remark 8.3; the lead extremal; the single-zero reduction, its asymptotics and Table III; the ratios J/P of Corollary 12.3; for Section 11, the root $\eta_\star = \sqrt{ab}$ at a right angle, the agreement of the Cauchy integral $v_{g_\sharp}$ with both closed forms (11.5), the integral (11.6), the value at the geometric centre and the symmetry, every entry of Table IV, the positivity of $\alpha_\theta^\sharp$ on the band for three band ratios and three phases, the mean of $\alpha_\theta^\sharp$ and the pointwise comparison with $\alpha_\star$ at $\theta = 1$, the narrow-band ratios after Corollary 11.9 and the nomes and the case of equality of Corollary 11.10; for Section 9, the branch of R, both identities of Lemma 9.1 at nine points for three bands, the positivity of k_t, (9.3) for a single resonance below and above the band, Lemma 9.5 in the plane and on the band, the integral of r/x, the means Λ and Λ_- of the two profiles, the Herglotz property, the linear coefficient and the mean of the compensated optimum of Theorem 9.4, and the octave values $A(m)$, Λ_- and ϑ_c; and the numbers of Sections 2 and 12. It reports 244 checks passed and none failed.

Lean. The file `lean/PhaseShiftCost.lean` is checked by the kernel of Lean 4 (version 4.34.0, core library only). It contains 75 theorems and no `sorry`. It verifies the dimensional analysis and the arithmetic of Section 2; the coefficients d_n of the series of Λ, their identification with $\binom{2n}{n}^2 4^{-2n}\,2n/(2n-1)$ for $n \le 60$, and their monotonicity for every n; enclosures of Λ/π to nine decimals for six band ratios, from partial sums of the series and the coefficient bound behind the tail estimate; and the polynomial identities used in Sections 5 to 7, 10 and 11 and in the appendices; for Section 11 these are the identities behind (11.4), both forms of (11.5), the symmetry $\omega \mapsto ab/\omega$, the value at $\sqrt{ab}$ and (11.6), and for Section 9 the signs of the kernels, the comparison of Corollary 9.3, the step of Corollary 9.7(iv), the vanishing linear coefficients of the compensated optima and the threshold ϑ_c. Of the 75 theorems, 30 depend on no axiom, two on `propext` only, two on `propext` and `Quot.sound`, and the others on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`), through the tactics `grind` and `omega`.

The analytic theorems are not formalised. The sum rule (Theorem 6.1), the identification of Λ with elliptic integrals and with its series, the extremal system, the reconstruction of Theorem 9.2, and the results on zeros rest on boundary values of Cauchy integrals, the M. Riesz theorem, Hardy spaces, conformal maps and extremal length, Landen's transformation and the summation of a geometric series, none of which is formalised in the file. They are proved in the text and in Appendices A to C.

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[40] Here $\tau_\star\pi/4 = \theta K(k)/(2b)$ and $\tau_\star \pi m^2/2 = \theta b E(k)$; the last expression is $\frac{\theta}{2}\big((a^2+b^2)I_0 - 2I_2\big)$.

involving human participants or animals.

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Data availability. The LaTeX source, the script that re-evaluates every numerical value of the paper (244 checks) and the Lean 4 certificate of its finite and algebraic parts (75 theorems, no `sorry`) are in the repository `https://github.com/creelie/gsc`, release v1.1.0, archived on Zenodo under doi:10.5281/zenodo.23761043. No statement of the paper depends on them. Anthropic Claude assisted with exposition, figures, the numerical checks and the Lean certificate. OpenAI's ChatGPT found the positive reconstruction of Section 9 and the pointwise bounds derived from it. The authors checked every proof and are responsible for the text.

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